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# The Geometric Alignment Tax: Discrete Tokenization vs. Continuous Geometry in Scientific Foundation Models

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## Abstract

Foundation models for biology and physics optimize predictive accuracy, but their internal representations systematically fail to preserve the continuous geometry of the systems they model. We identify a dominant, controllable cause: the *Geometric Alignment Tax*, the cost of forcing continuous manifolds through discrete categorical bottlenecks. Controlled ablations on synthetic dynamical systems demonstrate that replacing cross-entropy with a continuous head on an identical encoder reduces geometric distortion by up to  $8.5\times$ , and by  $1.7\text{--}1.9\times$  on real DNA at matched task competence, while learned codebooks exhibit a non-monotonic double bind where finer quantization worsens geometry despite improving reconstruction. Under continuous objectives, three architectures differ by  $1.3\times$ ; under discrete tokenization, they diverge by  $3,000\times$ . Evaluating 14 biological foundation models with rate-distortion theory and MINE, we identify three failure regimes: *Local-Global Decoupling*, *Representational Compression*, and *Geometric Vacuity*. A controlled experiment confirms that Evo 2’s reverse-complement robustness on real DNA reflects conserved sequence composition, not learned symmetry. No model achieves simultaneously low distortion, high mutual information, and global coherence.

## 1 Introduction

Foundation models for biology and physics are evaluated on predictive accuracy: perplexity, AUC, benchmark rankings. But these metrics are blind to whether the model’s internal representations preserve the continuous geometry of the systems they claim to encode. We reveal a hidden cost: the **Geometric Alignment Tax**, the intrinsic geometric distortion incurred when forcing continuous physical manifolds through discrete categorical bottlenecks. We have previously introduced the geometric tax to describe transferability-fidelity trade-offs in vision architectures [58] and subsequently applied it to perturbation coherence in biological manifolds [57]. Here, we identify its most fundamental manifestation: the application of discrete-token foundation models to continuous physical symmetries.

**The core claim.** Within the architectures, discretization schemes, and perturbations we evaluate, cross-entropy over discrete tokens is sufficient to induce symmetry failure in embedding manifolds; Sec. 2.3 identifies a domain where it is not. The tax does not reduce to attention, recurrence, or convolution: it tracks the discretization boundary itself. On synthetic dynamical systems with known geometry, three architectures (Transformer, SSM, hybrid) trained with continuous objectives differ by  $1.3\times$  in geometric stability. Under discrete tokenization, the same architectures diverge by  $3,000\times$  on a biological mutation walk (Sec. 2). The discrete-to-continuous gap within any single architecture dwarfs the cross-architecture gap. Learned VQ codebooks cannot escape this cost: finer quantization improves reconstruction but worsens geometric stability by increasing boundary-crossing probability under perturbation, and distortion saturates at a positive floor rather than decaying, so no codebook size recovers continuous-head geometry. Training small models from scratch on real DNA reproduces

the ablation causally at matched task competence (Sec. 2.3). The ESM-2 protein Transformer suite (8M–15B) exhibits a progressive decline in geometric stability with scale, and an apparent “recovery” at 15B is unmasked as global manifold drift rather than genuine improvement (Sec. 3). A controlled four-condition experiment demonstrates that Evo 2’s apparent reverse-complement robustness on real DNA reflects conserved  $k$ -mer composition, not learned symmetry (Sec. 4). We formalize the tax through rate-distortion theory and validate the bound empirically (Sec. 5). Applying MINE across 14 foundation models, we identify three failure regimes: *Local-Global Decoupling*, where models encode shallow local statistics but fail to integrate globally; *Representational Compression*, where information concentrators amplify mutual information at the cost of geometric fidelity; and *Geometric Vacuity*, where smooth embeddings carry less structure than random noise.

**Contributions.** (1) Controlled synthetic experiments isolating tokenization as the causal bottleneck for geometric instability (Sec. 2), validated on real DNA and delimited by a controlled protein negative (Sec. 2.3). (2) Scaling laws demonstrating the tax is progressive with parameters and invariant to context length, across 14 biological foundation models (Sec. 3). (3) An information-theoretic formalization via rate-distortion theory and MINE, identifying three distinct failure regimes (Sec. 5). (4) A comprehensive ablation battery (6 variants) ruling out alternative explanations (Appendix E). (5) The Texture Hypothesis Test: a controlled experiment establishing that Evo 2’s RC signal on real DNA is per-sequence  $k$ -mer histogram matching, not biophysical understanding (Sec. 4). (6) An RCCR experiment demonstrating that post-hoc symmetry regularization degrades manifold geometry despite achieving perfect RC consistency (Sec. 4).

## 2 Ground Truth: The Controlled Experiments

We isolate the causal role of tokenization by training three small architectures from scratch on synthetic dynamical systems with known continuous geometry. The architectures span the dominant paradigms: SmallBERT (Transformer, 3.4M parameters), SmallMamba (SSM, 2.0M), and SmallStripedHyena (hybrid Hyena-attention, 4.5M). Each is trained via causal language modeling (CLM) with 256-bin uniform discretization on three datasets: superposed sine waves (WAVEFORM), damped harmonic oscillators (OSCILLATOR), and Lorenz attractors (LORENZ). A two-pass global discretization scheme computes dataset-wide min/max values first, ensuring the same physical state maps to the same bin across all sequences and preventing per-sequence normalization artifacts.

**Evaluation protocol.** Geometric stability is evaluated using a standardized harness built on the Shesha geometry library [58, 59]. For each model, we embed both clean and perturbed sequences, extract a center window (mean-pooled to a fixed-length vector to neutralize context-length differences), and compute pairwise Representational Dissimilarity Matrices (RDMs) using cosine distance (per-model layer and pooling details in Table S1, Appendix C). The harness reports four core metrics. *RDM similarity*: the Spearman correlation between the clean and perturbed RDMs, measuring whether pairwise geometric relationships are preserved under perturbation. *Perturbation stability*: the rank correlation between input-space perturbation magnitude and embedding-space displacement, testing whether small input changes produce proportionally small representation changes. *Feature split* and *sample split* scores: the agreement between RDMs computed on random halves of the embedding dimensions and random halves of the dataset, respectively, verifying that geometric structure is distributed rather than concentrated in fragile subspaces. The *composite stability* score is the mean of these four metrics (see Appendix C.1 for formal definitions). Perturbations include value noise at 1%, 2%, 5%, and 10% of positions, plus time reversal. We additionally validate dynamical fidelity on the Lorenz attractor via largest Lyapunov exponent (LLE) estimation and a butterfly test for attractor preservation. See Appendix L for reproducibility details.

### 2.1 The Causal Proof: Continuous vs. Discrete

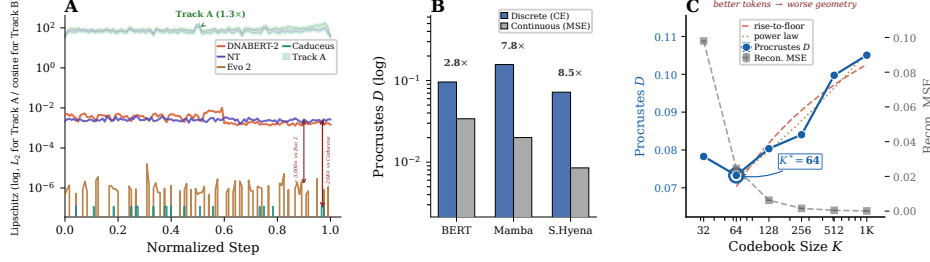
**Baseline (discrete CE).** Under standard discrete tokenization, all three architectures preserve Lorenz attractor dynamics: LLE estimates are 0.036 (SmallBERT), 0.038 (SmallMamba), and 0.038 (SmallStripedHyena), all within 3% of the ground truth value 0.037, and the butterfly test confirms attractor structure preservation across all 5 seeds for every architecture. The cross-architecture variance in geometric stability is modest. At 1% noise on the Lorenz dataset, Procrustes distortion  $D$  ranges from 0.072 (SmallStripedHyena) to 0.157 (SmallMamba), with SmallBERT intermediate at 0.096. Mean composite stability scores across all datasets and perturbations are  $0.466 \pm 0.024$  (SmallBERT),

$0.400 \pm 0.059$  (SmallMamba), and  $0.470 \pm 0.016$  (SmallStripedHyena). Paired Wilcoxon signed-rank tests show SmallBERT is significantly more stable than SmallMamba ( $W = 0.0$ ,  $p = 0.0001$ ), while the SmallBERT vs. SmallStripedHyena difference is not significant ( $W = 34.0$ ,  $p = 0.15$ ). Architectures differ, but within the same order of magnitude.

**Ablation Variant A (continuous MSE head).** We replace the categorical CE output head with a linear projection trained under MSE loss, leaving the encoder backbone (self-attention layers, positional embeddings, feedforward blocks) unchanged. This single modification eliminates manifold fracture across all architectures. On the Lorenz dataset at 1% noise, SmallBERT improves  $2.8\times$  ( $D: 0.096 \rightarrow 0.034$ ); SmallStripedHyena improves  $8.5\times$  ( $D: 0.072 \rightarrow 0.0085$ ), the single best condition in the entire study. The cross-architecture spread collapses from 0.072-0.157 under discrete CE to 0.0085-0.034 under continuous MSE. At 10% noise the pattern holds: SmallBERT discrete  $D = 0.316$  vs. continuous  $D = 0.111$  ( $2.8\times$ ); SmallStripedHyena discrete  $D = 0.233$  vs. continuous  $D = 0.027$  ( $8.6\times$ ). The discrete-to-continuous gap within any single architecture dwarfs the cross-architecture gap under either regime. Same encoders, same training data, same perturbation protocol: the only variable is the output discretization boundary.

**Isolating the causal mechanism.** Replacing discrete CE with continuous MSE changes multiple factors simultaneously: the output space (categorical distribution over  $K$  symbols vs. a continuous scalar), the loss landscape (softmax cross-entropy vs. squared error), and the gradient structure (gradients through a softmax bottleneck vs. direct linear gradients). Inputs remain discretized in both conditions, so Variant A isolates the output objective rather than input tokenization. We therefore cannot attribute the improvement to any single factor from Variant A alone. However, the full ablation battery converges on discrete tokenization as the dominant cause through triangulation. (i) The VQ codebook sweep below holds the loss function constant (CE throughout) and varies *only* the tokenization granularity; the fact that geometric distortion worsens as  $K$  grows—despite identical loss, identical gradient structure, and improving reconstruction—directly implicates the discrete representation itself. (ii) Variant B (Jacobian regularization) explicitly smooths the loss landscape under CE and improves stability, but creates a Pareto frontier: no regularization strength simultaneously achieves low distortion and low CE, showing that loss-landscape smoothing alone cannot close the gap. (iii) Variant D (SmallLSTM) uses the same discrete CE as SmallBERT but with a recurrent architecture; it exhibits comparable distortion to SmallBERT and far worse than SmallMamba, isolating the continuous ODE parameterization rather than the loss function as the source of SSM stability. (iv) Variant E (attention ratio sweep) titrates attention fraction from 0/8 to 8/8 under *identical* CE loss; geometric stability degrades monotonically with attention fraction, producing a dose-response curve where the loss function is held constant and only the routing of discrete tokens varies. Taken together, Variant A demonstrates the *magnitude* of the discrete-to-continuous gap; the remaining ablations localize the cause to the tokenization boundary specifically.

**VQ bottleneck proof (the double bind).** To preempt the objection that the tax merely reflects naive uniform binning, we evaluate SmallBERT with VQ  $k$ -means codebooks at six sizes. Perturbations are applied in continuous input space and re-encoded through the learned codebook, since  $k$ -means labels are unordered and index-space perturbation would be semantically meaningless. The results reveal a double bind. There is a shallow optimum at  $K=64$  ( $D=0.073$ ), modestly better than the uniform 256-bin baseline ( $D=0.096$ ), but distortion then *increases* with larger codebooks:  $K=512$  yields  $D=0.100$  and  $K=1024$  yields  $D=0.105$ , both worse than the uniform baseline. Meanwhile, reconstruction MSE drops monotonically from 0.098 ( $K=32$ ) to 0.00014 ( $K=1024$ ), confirming the codebook learns well. The mechanism is straightforward: finer Voronoi cells make fixed-magnitude perturbations more likely to cross cell boundaries, so better tokenization in the reconstruction sense makes geometry worse. The non-monotone profile in Fig. 1C reflects two competing regimes: below  $K^* \approx 64$ , quantization noise dominates and distortion decreases with  $K$ ; above  $K^*$ , boundary-crossing probability dominates and distortion *rises* toward a floor set by the perturbation scale. Crucially, this rise is slow: in the boundary-dominated regime ( $K \geq K^*$ ) it is equally well described by a logarithmic approach to that floor or by a weak power law  $D_{\text{proc}} \propto K^{0.13}$  ( $R^2$  between 0.90 and 0.96 across noise levels; per-noise-level fits and a derivation connecting the scaling to Voronoi boundary geometry are in Appendix K). The fitted exponent is far below the  $K^{1/d_M} \approx K^{0.49}$  rate at which boundary crossings accumulate, because each crossing displaces the representation less as cells shrink. The contrast with reconstruction error is the whole point: reconstruction MSE falls polynomially fast ( $\propto K^{-2/d_M}$ ), yet geometric distortion does not follow it down, and past  $K^*$  it



**Figure 1:** **A.** Track A vs. Track B Lipschitz profiles: smooth arcs (continuous physics) vs. divergent, multi-scale fracture (discrete biology). Track A uses  $L_2$  Lipschitz and Track B uses cosine Lipschitz (dimension-invariant across embedding sizes), so only within-track spreads are comparable:  $1.3\times$  within Track A,  $\sim 3,000\times$  within Track B. **B.** Continuous vs. discrete Procrustes  $D$  across architectures on the Lorenz dataset at 1% noise. All continuous conditions cluster near zero; discrete conditions span an order of magnitude. **C.** VQ double bind on the Lorenz attractor at 1% noise: reconstruction MSE (decreasing) vs. Procrustes  $D$  (non-monotone) vs. codebook size  $K$ . Both candidate slow-growth fits are shown on the ascending branch ( $K \geq K^*$ ): rise-to-floor ( $D = B - A/\log K$ ) and a weak power law ( $D \propto K^{0.13}$ ). We do not adjudicate between them. Better tokens improve reconstruction while worsening geometry.

moves in the opposite direction. Adding codes therefore cannot close the gap to a continuous head; beyond the optimum it widens it. No feasible codebook size escapes the tax.

**Additional ablations.** Four further ablations confirm the causal picture (Appendix E): Jacobian regularization (Variant B) reveals a Pareto frontier between stability and CE loss with no setting achieving both, confirming the tax is a genuine trade-off; SmallMamba with MSE (Variant C) confirms SSM stability is intrinsic to the ODE prior; SmallLSTM (Variant D) proves mere recurrence is insufficient; and attention ratio (Variant E) and filter order (Variant F) sweeps on StripedHyena produce dose response curves for the tax.

## 2.2 Track A + Track B: The Smoking Gun

We complement the controlled ablations with a cross-domain comparison that illustrates the magnitude of the tokenization tax in deployed models. This is not a controlled causal test (that role is served by the within-architecture ablations of Sec. 2.1), but the scale of divergence is striking.

**Track A (continuous physics).** We use the damped harmonic oscillator as a continuous interpolation testbed. Given two distinct oscillator trajectories, we generate 101 linearly interpolated sequences in input space ( $\alpha$  from 0 to 1), embed each intermediate through all three architectures trained with MSE loss, and compute  $L_2$  Lipschitz profiles, measuring the local rate of embedding change per interpolation step. The experiment averages over 10 random pairs of starting states, producing embedding PCA trajectories, cosine distance profiles, and Lipschitz profiles. All three architectures yield smooth PCA arcs with no staircase effects, no teleportation, and no fracturing. Mean Lipschitz values are 65.3 (SmallBERT), 79.3 (SmallStripedHyena), and 84.6 (SmallMamba), a spread of  $1.3\times$  from smoothest to roughest.

**Track B (discrete biology).** We construct a single-point mutation walk on the BRCA1 gene (chr17, GRCh38). Starting from wildtype, we change one base pair at a time across a 2 kb core region, walking through 122 intermediate sequences (120 SNPs plus the pathogenic C61G missense mutation). Each sequence is embedded by four genomic foundation models spanning the architectural spectrum: DNABERT-2 (117M, BPE tokenization), Nucleotide Transformer (500M, 6-mer tokenization), Evo 2 (7B, StripedHyena hybrid, single-character tokenization), and Caduceus (7.7M, pure Mamba SSM, single-character tokenization with RC-equivariant design). We compute cosine-based Lipschitz profiles, which are dimension-invariant, to measure how much each model’s embedding shifts per single-base change. Mean cosine Lipschitz spans three orders of magnitude:  $3.1 \times 10^{-3}$  (DNABERT-2) and  $2.6 \times 10^{-3}$  (Nucleotide Transformer) for the Transformers, versus  $1.0 \times 10^{-6}$  (Evo 2) and  $1.0 \times 10^{-7}$  (Caduceus) for the SSM-containing models. No model detected the pathogenic C61G mutation as a Lipschitz spike. The three failure regimes from the information-theoretic analysis (Sec. 5) map directly onto these profiles: the Transformers fracture (high-magnitude, scattered PCA clusters), Evo 2 absorbs (smooth trajectory but biophysically numb at  $10^{-6}$  per SNP), and Caduceus collapses to a flatline at the floating-point floor (18 apparent “spikes” are numerical jitter, not biological signal).

**The headline.** Track A:  $1.3\times$  gap between architectures. Track B:  $\sim 3,000\times$  gap. The same attention mechanism that produces the smoothest interpolation on continuous signals produces a fractured manifold when forced through discrete tokens. The variable that changed between Track A and Track B is not the routing, it is the tokenization.

### 2.3 Track C: Biological Causal Validation

Tracks A and B are observational. To test the same variable causally on real sequences, we train small models from scratch under discrete CE and a continuous-head variant on DNA (SmallStripedHyena, causal LM, human chr22) and protein (SmallBERT, UniRef50/SwissProt, both masked and causal objectives), across three continuous code families and five seeds (full protocol, learning gates, and per-perturbation tables in Appendix D.4). Since reversal is a genuine symmetry for DNA (reverse-complement) but an out-of-distribution scramble for protein, we report substitution-only distortion as the primary claim and treat reversal separately. On DNA the continuous head reduces substitution Procrustes distortion by  $1.7\text{--}1.9\times$  relative to CE, across all code families and every substitution rate, with recovery flat across arms so the comparison already controls for competence; the effect persists with reverse-complement removed ( $1.73\text{--}1.88\times$ ), ruling out an engineered-symmetry artifact. Protein does not follow: under MLM the apparent full-battery advantage is entirely reversal-driven and substitution alone is level-to-worse ( $0.7\text{--}0.9\times$ , a null), and under CLM a real endpoint advantage ( $1.72\times$ ) is largely competence-driven: a frozen downstream probe places the continuous arms  $2.2\text{--}3.4$  pp below CE, and at matched recovery the advantage is inconsistent across code families (physicochemical  $0.87\times$ , high-dimensional  $1.15\times$ , random  $1.31\times$ ; five seeds), reversing for one arm and well below the  $1.72\times$  endpoint for all. Because a genuine tokenization effect should not depend on the choice of continuous code, this arm-divergence, in contrast to the convergence seen in DNA, indicates no clean effect in protein-CLM. Track C thus shows the tokenization effect is real and competence-robust in DNA but bounded: absent in protein-MLM and competence-dependent in protein-CLM. This characterized boundary, a reversal artifact under MLM and a competence confound under CLM, sharpens rather than weakens the causal claim, which holds where the training signal is strong and the perturbation is biologically meaningful.

## 3 Model Performance at Scale

**ESM-2 (the primary scaling story).** We evaluate all six ESM-2 checkpoints (8M to 15B parameters) on 10,000 synthetic protein sequences (200 aa) perturbed with amino acid substitutions at 1–10% of positions and sequence reversal. Composite stability at 1% substitution declines monotonically from 0.463 (8M) through 0.454 (35M), 0.443 (150M), 0.419 (650M), to 0.391 (3B), a progressive tax spanning nearly four orders of magnitude in parameters. Then the 15B checkpoint appears to “recover” to 0.445 (Fig. 2A, blue curve). This V-curve is misleading. We quantify global drift via a Procrustes reduction:  $\rho = 1 - \|X_{\text{clean}} - R^* X_{\text{pert}}\|_F / \|X_{\text{clean}} - X_{\text{pert}}\|_F$ , where embeddings are mean-centered and  $R^*$  is the optimal orthogonal rotation with scaling: low  $\rho$  means internal fracture that no rotation can fix, high  $\rho$  means coherent drift that rotation partially removes. The 15B model achieves  $\rho \approx 5\%$  at 1% substitution, rising to  $\sim 20\%$  under reversal (Fig. 2A, orange curve). The manifold drifts globally while preserving internal relative structure, the signature of *Untethered Gel* (Fig. 2B, right panel). RDM similarity alone misses this because pairwise cosine distances are rotation-invariant. The scaling trend is replicated on real UniRef50 proteins (10,000 sequences, 100–400 aa), where the same monotonic decline and illusory 15B recovery appear: 1% substitution composite of 0.466 (8M) to 0.397 (3B) to 0.448 (15B). The gap between synthetic and real composites is small ( $\leq 0.03$  at every scale), confirming the tax is not an artifact of sequence composition.

**Cross-architecture summary (full results in Appendix F).** The Nucleotide Transformer (NT v2, DNA, 6-mer tokenization) exhibits the same progressive decline across four sizes: mean composite drops from 0.239 (50M) to 0.210 (250M) on synthetic DNA, with a partial rebound at 2.5B that parallels the ESM-2 pattern. SaProt (structure-aware protein Transformer, 35M to 1.3B) degrades with scale ( $0.436 \rightarrow 0.398$ ), demonstrating that Foldseek structural tokens do not rescue the tax. Caduceus (pure Mamba SSM, RC-equivariant, 0.5M to 7.7M) is the tax-exempt baseline: composite stability is nearly constant across scale (0.459, 0.449, 0.458), with near-perfect RC preservation (RDM  $> 0.999$ ) on real chr22 DNA. ProtMamba (protein SSM, 108M) presents an apparent paradox:

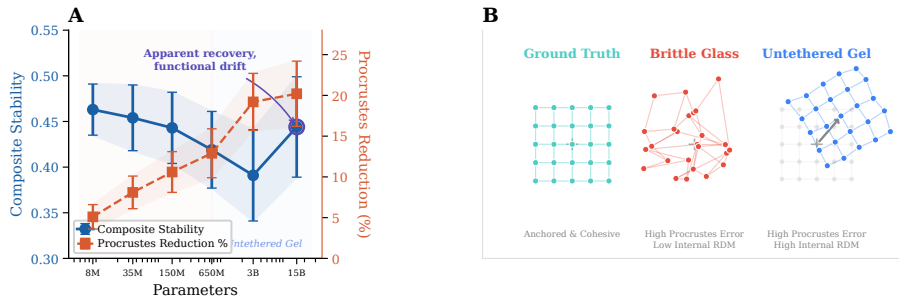


Figure 2: **A.** ESM-2 composite stability (blue, left axis) vs. parameters, with Procrustes reduction overlaid (orange, right axis). Stability declines monotonically from 8M to 3B; the 15B “recovery” is unmasked by the simultaneous spike in Procrustes reduction, revealing global manifold drift rather than genuine geometric improvement. **B.** Conceptual illustration of the two failure modes. *Ground Truth*: the manifold is anchored and internally cohesive. *Brittle Glass* (small/medium Transformers): high Procrustes error with low internal RDM, the manifold fractures internally. *Untethered Gel* (large Transformers): high Procrustes error with high internal RDM, the manifold drifts as a coherent block.

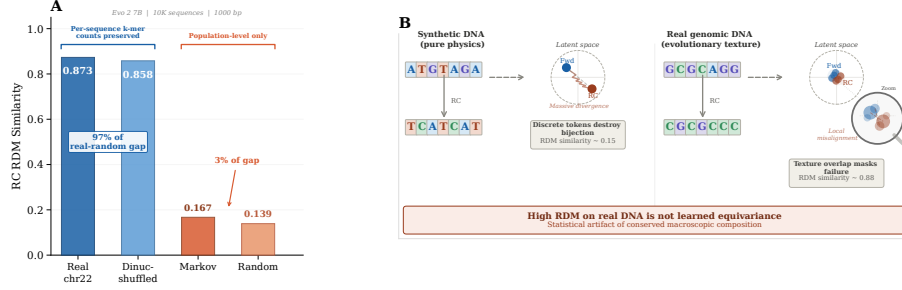
low Procrustes distortion with reasonable perplexity (40.1), yet, as Sec. 5 quantifies, its embeddings are informationally empty.

**Brittle Glass vs. Untethered Gel.** We operationally define the two failure modes by their Procrustes reduction  $\rho$  under 1% substitution. *Brittle Glass* ( $\rho < 2\%$ ): rotation cannot reduce the clean-perturbed residual, indicating internal fracture (ESM-2 8M–650M:  $\rho = 0.7\text{--}1.8\%$ ). *Untethered Gel* ( $\rho > 4\%$ ): rotation substantially reduces the residual, indicating coherent global drift (ESM-2 3B–15B:  $\rho = 5.2\text{--}5.0\%$ ; under reversal,  $19.7\text{--}20.1\%$ ). Caduceus shows near-zero reduction (0.3–0.5% on SNPs), confirming minimal distortion. The Frozen Head test provides functional validation: for Evo 2 at 8K context, the fraction of tokens where the pretrained LM head produces the same top-1 prediction on perturbed vs. clean embeddings drops from  $\sim 85\%$  (1% SNP) to  $\sim 28\%$  (synthetic RC), confirming that global drift is functionally destructive.

## 4 Context Length and the RC Dissociation

**Evo 2 context scaling (8K, 262K, 1M).** We evaluate the same Evo 2 7B model at three context-window checkpoints. On synthetic DNA, SNP stability gains are modest: 1% SNP RDM similarity rises from 0.747 (8K) to 0.817 (1M). On real chr22 sequences the gains are marginal: 0.990 to 0.993. Frozen Head accuracy on the context tax test (classifying *E. coli* vs. human from a 1 kbp signal region, padded to match each checkpoint’s context) is 0.988 (8K), 0.980 (262K), 0.993 (1M):  $128\times$  more context for effectively zero geometric gain.

**The Reverse Complement Dissociation.** Due to the structural biochemistry of the double helix [77], every strand of DNA possesses a mathematically perfect, continuous symmetry: the reverse-complement (RC). Because a sequence and its reverse-complement encode the exact same biological information, a geometrically grounded model must map both to an identical or perfectly symmetric representational manifold. Evo 2 fails this test categorically on synthetic DNA: RC RDM similarity is 0.139 (8K), 0.156 (262K), 0.208 (1M). The model has functionally zero understanding of the  $A \leftrightarrow T / C \leftrightarrow G$  bijection (Fig. 3B, left panel). Yet real RC is strikingly higher: 0.879 (8K), 0.883 (262K), 0.873 (1M). To determine why, we run a controlled four-condition experiment: the Texture Hypothesis Test (10,000 sequences per condition, 1000 bp, Evo 2 7B at 8K context; full details in Appendix J). Dinucleotide-shuffled real DNA (Altschul-Erickson algorithm [4]: exact per-sequence  $k$ -mer counts preserved, all positional structure destroyed) recovers 97% of the real-random RC gap in RDM similarity (Fig. 3A). Texture-matched Markov sequences (first-order Markov chain calibrated to population-level dinucleotide frequencies) recover only 3%. The mechanism is now pinned: Evo 2’s embeddings function as high-dimensional per-sequence  $k$ -mer histograms. RC preserves exact  $k$ -mer counts (every  $k$ -mer maps to its complementary  $k$ -mer at the same frequency), so forward and RC produce symmetrical histograms that the model’s weights aggregate equivalently. Destroying positional structure via dinucleotide shuffling preserves this illusion because the histogram is unchanged. Matching only population-level statistics via Markov generation fails because individual sequences lose their unique compositional fingerprint, collapsing the per-sequence



**Figure 3: A. Texture Hypothesis Test.** RC RDM similarity across four conditions for Evo 2 (7B, 8K context, 10,000 sequences). Dinuc-shuffled real DNA (per-sequence  $k$ -mer counts preserved) recovers 97% of the real-random gap; texture-matched Markov (population-level statistics only) recovers 3%. **B. The RC Dissociation explained.** On synthetic DNA (left), discrete tokens destroy the  $A \leftrightarrow T / C \leftrightarrow G$  bijection entirely (RDM  $\sim 0.15$ ). On real genomic DNA (right), conserved macroscopic  $k$ -mer composition creates overlapping texture between forward and RC embeddings (RDM  $\sim 0.88$ ), masking the failure. High RDM on real DNA reflects statistical artifact, not learned equivariance.

pairwise structure that RDM measures. This is a controlled causal result: Evo 2 does not understand double-stranded DNA symmetry. It counts short subsequences. The apparent RC “success” on real DNA is an artifact of a histogram encoder invariant to the one biological transformation that preserves histograms (Fig. 3B, right panel).

**Does RC regularization reduce the tax?** To test whether post-hoc symmetry enforcement mitigates the geometric tax, we applied an embedding-level variant of RCCR [50] to DNABERT-2 (117M), minimizing L2 distance between mean-pooled representations of forward and RC sequences during fine-tuning (adapting the task-level consistency objective to an unsupervised setting; details in Appendix H). RCCR achieves perfect per-sequence RC consistency (cosine gap: 0.041  $\rightarrow$  0.000), but the population-level geometric structure degrades: Procrustes disparity between forward and RC embedding matrices increases by 91%, RC RDM similarity turns negative ( $-0.036$ ), and SNP perturbation sensitivity collapses by two orders of magnitude. Forcing pointwise symmetry compliance flattens the embedding landscape rather than aligning its geometry, consistent with the rate-distortion prediction that capacity spent on one constraint is unavailable for manifold preservation. The tax is therefore not reducible to a missing symmetry constraint, at least under post-hoc regularization; whether architectural equivariance fares better remains open (Sec. 5.1).

## 5 The Information Theory of the Tax

Our results across physics, biology, and scale point to a fundamental trade-off in representational learning. We now formalize the mechanism through rate–distortion theory and quantify its consequences with mutual information estimation.

### 5.1 Rate–Distortion Framing

**The discrete bottleneck.** Cross-entropy over a  $K$ -token vocabulary optimizes for sharp partition boundaries with no gradient signal toward manifold preservation. A next-token predictor trained with CE is a classifier that happens to produce embeddings as a side effect; those embeddings inherit the piecewise-constant geometry of the decision regions, not the smooth geometry of the source. The channel capacity of a  $K$ -symbol discrete vocabulary is  $\log_2 K$  bits per token. For a source manifold with intrinsic dimensionality  $d_M$ , Shannon’s rate–distortion function for a Gaussian source under squared-error distortion gives the minimum achievable distortion at rate  $R$ :  $D(R) = \sigma^2 2^{-2R/d_M}$ . Substituting the discrete capacity  $R = \log_2 K$  yields reconstruction distortion  $D \propto K^{-2/d_M}$ , the classical high-rate quantization scaling [34, 36]. For practical manifold dimensionalities ( $d_M \gg 2$ ), this decay is extremely slow: halving reconstruction error requires increasing the codebook by a factor of  $2^{d_M}$ , which is astronomically large even for modestly high-dimensional sources.

**Geometric distortion is not reconstruction error.** The distortion metric relevant to representational stability is not reconstruction MSE but *perturbation sensitivity*: given a small input perturbation  $\epsilon$ , does the representation change smoothly? For a  $K$ -cell Voronoi tessellation of a  $d_M$ -dimensional

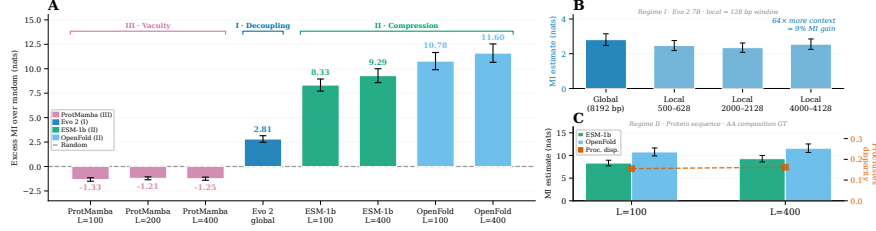
space, cell diameter scales as  $\ell \sim K^{-1/d_M}$ . A perturbation of fixed magnitude  $\|\epsilon\|$  crosses a cell boundary with probability  $p_{\text{cross}} \sim \|\epsilon\|/\ell \sim \|\epsilon\| \cdot K^{1/d_M}$ , which *increases* with  $K$ . The per-crossing distortion magnitude is bounded by  $\ell \sim K^{-1/d_M}$ , which *decreases* with  $K$ . These opposing forces produce the **VQ double bind** observed in Sec. 2.1: coarse codebooks ( $K$  small) lose information through quantization noise, while fine codebooks ( $K$  large) increase boundary-crossing probability. Below a crossover  $K^*$  (where  $\|\epsilon\| \approx \ell$ ), quantization noise dominates and distortion decreases with  $K$ ; above  $K^*$ , boundary-crossing dominates and distortion increases. The shallow optimum at  $K=64$  and subsequent rise through  $K=1024$  (Fig. 1C) are a direct consequence. In the boundary-dominated regime ( $K \geq K^*$ ), the total geometric distortion depends not only on the input-space tessellation but on how boundary crossings propagate through the learned encoder—a quantity that has no closed-form expression independent of the trained model. Empirically, in this regime  $D_{\text{proc}}$  rises slowly toward a positive floor  $B$  set by the perturbation scale, described comparably well by a logarithmic approach  $D_{\text{proc}} = B - A/\log K$  or by a weak power law  $D_{\text{proc}} \propto K^{0.13}$  ( $R^2$  between 0.90 and 0.96 across noise levels; fits in Appendix K). The connection to rate-distortion theory is that  $\log K$  is the channel capacity in bits: geometric fidelity does not improve with capacity, it saturates, while reconstruction error falls polynomially ( $\propto K^{-2/d_M}$ ) over the same range. We present this as an empirical characterization, not a formal theorem (see Appendix K for scope). The practical consequence is that codebook refinement cannot reach continuous-head geometry at any size, because the floor  $B$  is strictly positive.

**Capacity-Induced Fracture.** The same mechanism operates through model scale. As parameters increase, CE training produces sharper and more numerous decision boundaries, each a discontinuity in the embedding manifold. This is the source of the monotonic stability decline observed in ESM-2 (8M–3B, Sec. 3.1): more capacity enables finer partitioning, which creates more fracture surfaces. The apparent stability recovery at 15B is illusory (Untethered Gel): global drift masks local fracture rather than resolving it. **Scope.** Our claim is that discrete tokenization with CE is *sufficient to induce* geometric distortion within the causal graph we evaluate (fixed encoder, output-head ablation, substitution and symmetry perturbations), not that it is the only source or that sufficiency holds across all domains: Track C reproduces the effect on DNA at matched recovery but not on protein (Sec. 2.3). We test one such intervention: embedding-level RCCR [50] applied to DNABERT-2 achieves perfect per-sequence RC consistency but degrades population-level geometry (Sec. 4, Appendix H), suggesting that post-hoc symmetry enforcement redistributes rather than eliminates the tax. Whether architectural equivariance (e.g., RC-equivariant layers) can succeed where regularization fails remains an open question.

## 5.2 The Three Pathologies of the Distortion Bound

Because modern biological foundation models operate under these strict quantization limits, in every model we evaluate the attempt to minimize geometric distortion ( $D$ ) trades against the mutual information  $I(X; \hat{X})$  required to maintain biological utility. We categorize these failures into three regimes: *Regime I*: The model minimizes local distortion  $D$  by anchoring embeddings to short-range composition, but sacrifices the global mutual information  $I(X; \hat{X})$  needed to integrate long-range structure. Geometry is preserved locally; biological coherence is lost globally. Geometrically, this regime is compatible with more than one signature: large-scale ESM-2 ( $\geq 3B$ ) pairs it with Untethered Gel drift (Sec. 3), while Evo 2 pairs it with low local sensitivity. The geometric and information-theoretic taxonomies are orthogonal axes, not synonyms. *Regime II*: The model maximizes  $I(X; \hat{X})$  by concentrating task-relevant information, but pays the full distortion cost: the manifold warps under compression, producing geometric fracture analogous to the Brittle Glass signature (Sec. 3), here driven by intentional information concentration rather than capacity exhaustion. OpenFold’s Evoformer is the canonical example. *Regime III*: The model achieves low distortion  $D$  trivially, by encoding nothing. Geometry is smooth because the manifold is informationally empty:  $I(X; \hat{X})$  falls below the random noise floor. Neither the Brittle Glass nor Untethered Gel geometric signatures apply, because there is no information to fracture or drift. Track C (Sec. 2.3) reproduces this regime under controlled conditions: a continuous protein arm whose code lacks capacity falls below the majority-class recovery floor while exhibiting correspondingly low distortion (Appendix D.4), confirming that smooth geometry can be achieved by encoding nothing.

The Procrustes and Lipschitz analyses of Sections 2–3 measure geometric distortion; we now answer the complementary question: *does the manifold contain biological information?* We quantify each



**Figure 4:** (A) Excess MI (bias-corrected) across the three failure regimes. ProtMamba falls below zero (Geometric Vacuity), ESM-1b and OpenFold show large positive values (Representational Compression), and Evo 2 is modest and positive (Local-Global Decoupling). Random baselines sit at zero by construction. (B) Regime I: Evo 2 global vs. local MI. The flat curve across  $64\times$  context expansion confirms informational shallowness. (C) Regime II: ESM-1b vs. Evoformer excess MI at each sequence length, with Procrustes disparity annotated. The Evoformer amplifies MI while warping the manifold.

regime with MINE (Mutual Information Neural Estimation) using the Donsker-Varadhan variational lower bound [6, 27]. The statistics network  $T_\theta$  is a three-layer MLP (input  $\rightarrow 256 \rightarrow 128 \rightarrow 1$ , ReLU activations, 10% dropout). For each model we run 5 independent estimations and project embeddings to 50 dimensions via PCA. Ground truth features are amino acid composition plus species labels (protein models) or GC content plus 16 dinucleotide frequencies (DNA models). All results are reported as *excess MI* over a matched random baseline ( $MI_{\text{excess}} = MI_{\text{model}} - MI_{\text{random}}$ ) to remove finite-sample MINE bias, which scales with PCA dimensionality and can produce spuriously large raw estimates. The random baselines are  $2.664 \pm 0.054$  nats (protein) and  $1.681 \pm 0.033$  nats (DNA). We validate key findings with independent frozen-head probes (linear and nonlinear); concordance between MINE and probe-based diagnostics across all three regimes increases confidence that the results are not artifacts of the estimator. Full calibration details appear in Appendix I.

**Regime I: Local-Global Decoupling.** The Texture Hypothesis Test (Sec. 4) establishes the mechanism: Evo 2’s embeddings function as per-sequence  $k$ -mer histograms. MINE confirms the informational shallowness. Global MI (full 8,192-token context, mean-pooled) exceeds local MI (128-token windows) by only 14%. A  $64\times$  increase in context buys almost nothing. The excess MI is positive, confirming the embeddings do encode biological signal, but the signal is shallow and scale-invariant: local composition, not long-range structure. This is the Untethered Gel from an information-theoretic perspective, geometrically stable yet informationally provincial.

**Regime II: Representational Compression.** The Evoformer trunk *increases* MI while geometrically warping the representation. We extract embeddings from ESM-1b (the input encoder) and from the Evoformer output (after 48 blocks of structure-aware processing) and run MINE on both. OpenFold exceeds ESM-1b in MI at every sequence length: the Evoformer adds +2.3 to +2.5 nats of structural context. But this comes at a geometric cost. Procrustes disparity between ESM-1b and Evoformer output representations is 0.164 ( $L=100$ ), 0.162 ( $L=200$ ), and 0.149 ( $L=400$ ), confirming substantial manifold warping. Information is retained and amplified, but the topology is broken. This is not a loss of signal; it is a geometric transformation that trades fidelity for task-specific compression. The Evoformer is an effective information concentrator, but it pays the distortion bound in full: capacity spent on structural compression is capacity unavailable for manifold preservation.

**Regime III: Geometric Vacuity.** The most paradoxical finding. ProtMamba (108M parameters, Mamba SSM backbone) exhibits low Procrustes distortion and reasonable autoregressive perplexity (40.1). Yet, negative excess MI at every sequence length: the embeddings carry *less* mutual information with biological ground truth than a matched random baseline. Stability without substance. Frozen Head probes confirm the diagnosis: both linear logistic regression and nonlinear MLP probes (two- and three-layer, up to 512 hidden units) achieve chance-level accuracy ( $\sim 0.50$ ) across all sequence lengths and both local and global pooling strategies. The representation does not resist linearization because information is encoded nonlinearly; it resists because information is absent. The language model head still works (perplexity is reasonable), meaning the `lm_head` projection extracts next-token predictions from a representational subspace that the embedding layer does not expose to downstream consumers. The internal representations are informationally empty: smooth because they encode nothing, not because they encode well.

**Synthesis.** The three regimes are not independent pathologies but different faces of the same distortion bound. Each model allocates finite capacity under the Gaussian rate–distortion constraint

$R(D) \geq \frac{d_M}{2} \log(1/D)$  (a lower bound on required rate for any source with the same variance, by the maximum-entropy property of the Gaussian): no discrete-token model in our study achieves simultaneously low distortion, high mutual information, and global coherence. The tax is paid in different currencies, but always paid.

## 6 Discussion

**Summary.** Across the models, discretization schemes, and perturbation families we evaluate, discrete tokenization is the dominant source of geometric instability in biological and physics foundation models. The tax is progressive with parameters (Sec. 3), invariant to context expansion (Sec. 4), and bounded below by a positive distortion floor in the boundary-dominated regime, which makes continuous-head performance unreachable through codebook refinement alone (Sec. 2). No tokenization strategy in our study escapes this scaling, and the VQ double bind (Sec. 2) demonstrates that finer quantization actively worsens geometric stability once cell boundaries become denser than the perturbation scale. The three failure regimes identified by MINE (Sec. 5.2) are not independent pathologies but different allocations of finite capacity under the same constraint: models that preserve local geometry sacrifice global coherence (Regime I), models that concentrate information sacrifice geometric fidelity (Regime II), and models that preserve geometry sacrifice information entirely (Regime III).

**Implications.** Current evaluation practice (perplexity, AUC, benchmark accuracy) is blind to the tax. A model can dominate leaderboards while its global geometry is entirely ungrounded (Regime I), or produce smooth, stable manifolds that pass geometric consistency checks while encoding no biological signal (Regime III). As foundation models are increasingly deployed for therapeutic design, materials discovery, and physical simulation, the field must expand its notion of reliability beyond predictive accuracy to encompass what we term *Physical Alignment*: the requirement that learned representations faithfully preserve the continuous invariants of the systems they model. Geometric stability auditing via frameworks like Shesha should become a first-class evaluation criterion alongside task performance. Track C (Sec. 2.3) delimits the causal claim: the continuous-head intervention reproduces on real DNA at matched competence and does not transfer to protein, so the tax is characterized rather than assumed universal.

**Death and Taxes.** It’s commonly been said that the only two certainties in life are death and taxes [13, 23, 33]. If death and taxes are constants of life, the natural question follows: Is the Geometric Alignment Tax inevitable? For generative applications like chatbots, the tax may be acceptable, or even desirable [42, 70]. A drifting manifold allows for creativity, where “hallucination” is a feature, not a bug. However, for Scientific Foundation Models, where the laws of physics are invariant and outcomes can have life-or-death consequences, this tax is unaffordable. Our results demonstrate that we cannot simply scale our way out of this penalty. The path to Scientific AGI is not merely training larger discrete models to chase an asymptotic limit, nor is it applying continuous priors that trivially satisfy geometric stability by erasing biological signal (Sec. 2.3). It requires acknowledging that our current architectural playbook is fundamentally broken for the natural sciences, necessitating a return to first principles.

**Future directions.** The path forward likely requires architectures and training regimes that natively unify continuous geometric priors with high-fidelity discrete encoding, rather than grafting one onto the other. Precedent exists in behavioral neuroscience: Lightning Pose [9] attains state-of-the-art pose estimation by penalizing violations of physical constraints (motion continuity, multi-view geometry, posture plausibility) during training rather than hoping they emerge from data. Standard interpretability can mistake shallow statistics for understanding. A sparse-autoencoder audit of Evo 2 recovers crisp, biologically-aligned features [25], yet our texture-hypothesis test shows its apparent competence on real DNA is per-sequence  $k$ -mer histogram matching, not learned biology (Fig. 3). Geometric stability auditing, continuous-valued foundation models, and objectives that trade off predictive accuracy against manifold preservation are the promising directions.

**Limitations.** We discuss the scope of our causal claim, the entangled components we do not individually isolate, the single RCCR instantiation tested, and the compositional nature of our MINE ground-truth features in Appendix A.

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## Code

The full code necessary to reproduce all experiments, benchmarks, and analysis described in this paper is publicly available at <https://github.com/prashantcraju/geometric-alignment-tax>. The stability harness will be added to our open-source PyPI framework shesha-geometry [59]. Specific details about infrastructure, software versions, and configurations that were used are outlined in Appendix L.

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## A Limitations

Our findings are scoped to AI for Science, where ground-truth continuous symmetries provide hard constraints. We caution against generalizing the Geometric Alignment Tax to natural language, which lacks rigid, mathematically continuous invariants [29, 56, 63].

Our analysis covers models up to 15B parameters and 1M context length; while our rate-distortion framing predicts the tax persists asymptotically, we cannot rule out emergent mitigation at scales beyond our empirical horizon. We acknowledge that continuous-output formulations (diffusion, MSE regression) offer a theoretical alternative, and our own ablations confirm that continuous-head Transformers preserve geometry on chaotic attractors; Track C extends this to real DNA at matched task competence, but the same intervention yields no substitution-level advantage on protein, so the benefit is domain-dependent rather than general. The vast majority of deployed biological foundation models use discrete tokenization, and our goal is to characterize the cost of that dominant paradigm, not to prescribe a replacement. Our causal claim is that discrete tokenization combined with CE is *sufficient to induce* geometric distortion within the evaluated causal graph; we do not claim to have isolated every entangled component (output discretization, CE loss, softmax gradient structure) as independently causal. What the six-variant ablation battery establishes is that the discrete-to-continuous gap is the largest effect in the study, that improving tokenization within the discrete regime worsens geometry (VQ sweep), that smoothing the loss landscape cannot close the gap (Jacobian penalty), and that the effect is dose-dependent on attention fraction under constant CE (Variant E). Disentangling the individual contributions of discretization, loss function, and gradient sharpness—for instance via soft-label CE, ordinal regression, or temperature-annealed softmax—is a natural next step.

We characterize the tax under vanilla CE objectives without explicit symmetry constraints; an embedding-level RCCR experiment (Section 4) shows that post-hoc RC consistency regularization degrades manifold geometry despite achieving perfect pointwise symmetry; however, this tests one instantiation at one hyperparameter setting. Architectural equivariance, alternative RCCR formulations, and broader  $\lambda$  sweeps could yield different outcomes. Our MINE ground-truth features are compositional (amino acid frequencies, GC content, dinucleotides); while the ProtMamba vacuity finding is independently confirmed by frozen-head probes at chance accuracy across all conditions, models encoding higher-order structural features not captured by these targets could appear informationally shallow under our protocol.

Finally, time-reversal on continuous trajectories is mathematically simpler than reverse-complement on discrete categorical sequences; our claim is that tokenization is *sufficient to induce* symmetry failure in the settings we test, not that the two tasks are equivalent.

## B Related Work

### B.1 Representation Geometry and Comparison Metrics

Representational Similarity Analysis (RSA, 47) introduced the use of pairwise dissimilarity matrices to compare neural representations, and Centered Kernel Alignment (CKA, 46) extended this to kernel-space comparisons across layers and architectures. Linear probing [3] evaluates downstream discriminative utility but remains blind to manifold topology. These methods measure *similarity between representations*; our work asks a different question: does a *single* representation preserve the continuous geometry of its source under perturbation? The Shesha stability framework [57–60], built on representational dissimilarity matrices from the RSA tradition, isolates this structural rigidity directly, decoupling local memorization from macroscopic geometric fidelity. We supplement this with Procrustes analysis [28, 52, 61, 65] to disentangle internal fracture from global manifold drift (Section 3).

### B.2 Quantization Theory and Rate-Distortion Bounds

Classical high-rate quantization theory, anchored by Zador’s asymptotic bounds [34, 36, 78], dictates that the mean squared reconstruction error for a  $K$ -cell vector quantizer in  $\mathbb{R}^d$  decays as  $\mathcal{O}(K^{-2/d})$ . This spatial fidelity is fundamentally lower-bounded by the Shannon rate-distortion function [18, 67]. However, while classical theory optimizes for *signal recovery* (encoding fidelity), modern representation learning necessitates *topological stability*. Our central contribution in this section is to formally decouple *geometric distortion under perturbation* from standard reconstruction error. While standard rate-distortion tracks the accuracy of continuous approximation, our geometric metric quantifies representational robustness. We demonstrate empirically (Section 5.1) that geometric distortion exhibits a qualitatively different, drastically slower  $\mathcal{O}(1/\log K)$  decay. Rather than being governed by continuous approximation error, this bottleneck is driven by high-dimensional boundary-crossing dynamics within the latent Voronoi tessellation, revealing a fundamental geometric vulnerability in modern discrete-token codebooks [53, 75].

### B.3 Foundation Models for Biology and Physics

The application of foundation models to the natural sciences now spans protein language models [1, 44, 49], genomic and nucleotide sequence models [5, 21, 68], single-cell and spatial transcriptomic models [20, 76], small-molecule and natural-product models [26], time-series forecasters for chaotic systems [48, 80], particle physics [8, 10], astrophysics [79], and electromagnetics [12]. A recurring design move across many of these domains is to import the language-model paradigm wholesale, treating genes as words and cells as sentences [20], coupling sequences with textual annotation [68], or advocating unified multimodal foundation models that span genomics, transcriptomics, and proteomics [19]. Architectural evolution away from pure Transformers, toward long-context convolutions [55], continuous-time state space models [22, 37, 64], and graph-based molecular representations [26], reflects a growing recognition, surveyed for both genome [17] and single-cell [72] models, that discrete-token architectures create friction with continuous scientific ground truth. Closest to our framing, Gilpin [35] makes the cell-as-token analogy explicit and observes that the resulting embeddings occupy low-dimensional manifolds, while cautioning that biological context is multiscale rather than linearly ordered; we render that caution quantitative and causal, isolating tokenization as the mechanism that distorts manifold geometry.

These data-driven models are increasingly framed as components of a broader *virtual cell* program [14, 62], whose other pillar is mechanistic whole-cell simulation grounded in first principles: from the synthesis of a minimal bacterial genome [40], to genome-scale mechanistic models of *E. coli* [51], to spatially and temporally resolved 4D simulations of an entire minimal cell [74]. These simulations represent cellular state in continuous, physically interpretable variables rather than discrete tokens, making them the natural point of contrast for the geometric tax we characterize: they are the continuous ground truth that tokenized foundation models struggle to preserve.

This friction is not confined to biology. On the physics side, time-series foundation models often forecast dynamical systems by context parroting, copying recent context rather than learning the governing dynamics, so that a trivial copy baseline rivals leading models across chaotic, turbulent, and oscillatory systems [81]. Within biology, independent evidence suggests that the scaling assumptions

imported from language do not transfer cleanly to foundation models. Single-cell models plateau well below current corpus sizes with no clear data scaling law [24], struggle to generalize across experimental conditions [73], and protein fitness prediction degrades beyond a moderate model size [38]. The same non-monotonicity appears in post-training: across genomics, transcriptomics, and protein tasks, supervised fine-tuning of biological reasoning models raises in-distribution accuracy while driving out-of-distribution generalization to peak early and then decline, so performance does not increase uniformly with added supervision or compute [30]. Where scaling does help, as in cross-tissue cell-type annotation [32], the gains are measured on discriminative accuracy rather than on the representational geometry we probe; the two are not the same, and our results show they can move in opposite directions. Our work identifies a geometric mechanism, tokenization-induced manifold distortion, that is consistent with this friction across domains.

#### B.4 Symmetry Enforcement: Equivariance vs. Regularization

Deep learning architectures typically enforce physical and geometric symmetries—such as reverse-complement (RC) parity in genomics or continuous local smoothness—through two competing paradigms: *architectural equivariance* (hard constraints) and *post-hoc regularization* (soft constraints).

*Architectural equivariance* bakes symmetry directly into the model’s inductive bias, restricting the hypothesis space to valid transformations [16, 31]. In sequence modeling, Caduceus [64] achieves this via RC-equivariant Mamba layers, while Frame Averaging [39] enforces geometric semantics within equivariant Transformers. Conversely, *post-hoc regularization* allows the architecture to remain unconstrained but penalizes symmetry violations during training. For discrete  $Z_2$  symmetries, methods like RCCR [50] apply a consistency loss between forward and reverse-complement predictions across multiple backbones. For continuous symmetries, bounds on the Lipschitz constant, enforced via Jacobian and Hessian norm penalties [43] or spectral weight constraints [45, 54], are used to guarantee local smoothness and topological continuity.

Both strategies ultimately illustrate what we term the *Geometric Alignment Tax*: because discrete-token architectures natively shatter continuous underlying symmetries, they require heavy-handed external interventions to approximate the structural priors that state-space models (SSMs) possess natively. Our results (Section 5.1) suggest this tax is an intrinsic byproduct of vanilla cross-entropy (CE) objectives. An embedding-level RCCR experiment on DNABERT-2 confirms this trade-off: post-hoc consistency regularization successfully achieves pointwise symmetry compliance, but exacts a heavy toll via population-level geometric degradation (Appendix H). Whether the rigid constraints of architectural equivariance can preserve global geometry where soft regularization fails remains a critical open question.

## C Extraction Details and Evaluation Metrics

Table S1: Embedding extraction protocol for all evaluated models. All models use cosine distance for RDM computation and mean-pooling over a center window unless noted otherwise.

| Model                | Layer            | Pooling   | Tokenization     | Dim      |
|----------------------|------------------|-----------|------------------|----------|
| ESM-2 (8M–15B)       | Final hidden     | Mean-pool | Per-residue      | 320–5120 |
| ESM-1b               | Final hidden     | Mean-pool | Per-residue      | 1280     |
| OpenFold (Evoformer) | Evoformer output | Mean-pool | Per-residue      | 384      |
| ProtMamba            | Final hidden     | Mean-pool | Per-residue      | 1024     |
| SaProt (35M–1.3B)    | Final hidden     | Mean-pool | Foldseek+residue | 480–1280 |
| DNABERT-2            | Final hidden     | Mean-pool | BPE              | 768      |
| Nucleotide Trans.    | Final hidden     | Mean-pool | 6-mer            | 512–2560 |
| Evo 2 (7B)           | blocks.28.mlp.13 | Mean-pool | Single-char      | 4096     |
| Caduceus (0.5M–7.7M) | Final hidden     | Mean-pool | Single-char      | 118–256  |
| HyenaDNA             | Final hidden     | Mean-pool | Single-char      | 128–256  |
| GPN                  | Final hidden     | Mean-pool | Single-char      | 512      |
| SmallBERT            | Final hidden     | Mean-pool | 256-bin uniform  | 256      |
| SmallMamba           | Final hidden     | Mean-pool | 256-bin uniform  | 256      |
| SmallStripedHyena    | Final hidden     | Mean-pool | 256-bin uniform  | 256      |

### C.1 Geometric Stability Evaluation Metrics

To formally characterize the robustness of the latent embedding manifolds when exposed to perturbations, we utilize a comprehensive evaluation harness built upon the Shesha geometric stability framework [58], which was implemented with the shesha-geometry pypi framework [59]. For a given set of clean and perturbed sequence embeddings, the harness extracts representation geometries over equivalent context windows, optionally stratified and bootstrapped (e.g., maintaining upper limits of 2,500 samples to manage boundary memory overhead). The evaluation suite logs the following structured stability profiles for each perturbation sequence:

- **Sample Split (Shesha<sub>SS</sub>):** Evaluates whether sample identity preserves proximity metrics accurately across perturbations:

$$\text{Shesha}_{\text{SS}}(\mathbf{X}) = \frac{1}{K} \sum_{k=1}^K \rho_s(\text{vec}(\mathbf{D}_{\mathcal{S}_k^{(1)}}), \text{vec}(\mathbf{D}_{\mathcal{S}_k^{(2)}})) \quad (1)$$

where  $\mathcal{S}_k^{(1)}, \mathcal{S}_k^{(2)} \subset \{1, \dots, n\}$  are disjoint random subsets of sample indices drawn at split  $k$ ,  $\mathbf{D}_{\mathcal{S}} \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$  is the pairwise distance matrix computed over samples in  $\mathcal{S}$ , and  $\rho_s$  denotes Spearman’s rank correlation over the vectorized upper triangles.

- **Feature Split (Shesha<sub>FS</sub>):** Tracks dimension-wise correlations ensuring variance directions remain consistent across independent feature subspaces:

$$\text{Shesha}_{\text{FS}}(\mathbf{X}) = \frac{1}{K} \sum_{k=1}^K \rho_s(\text{vec}(\mathbf{D}_{\mathcal{F}_k^{(1)}}), \text{vec}(\mathbf{D}_{\mathcal{F}_k^{(2)}})) \quad (2)$$

where  $\mathcal{F}_k^{(1)}, \mathcal{F}_k^{(2)} \subset \{1, \dots, d\}$  are disjoint random subsets of feature dimensions drawn at split  $k$ ,  $\mathbf{D}_{\mathcal{F}} \in \mathbb{R}^{n \times n}$  is the pairwise sample distance matrix computed using only the features in  $\mathcal{F}$ , and  $\rho_s$  denotes Spearman’s rank correlation over the vectorized upper triangles.

- **RDM Similarity ( $\rho_{\text{RDM}}$ ):** [47] Quantifies preservation testing representational topologies directly evaluating Spearman correlations evaluating off-diagonal mapping bounds natively constraints generating specifically limits explicitly correlating  $X_c$  and  $X_p$ .
- **Anchor Stability (Shesha<sub>anchor</sub>):** Measures consistency of distance profiles from fixed anchor points to random data splits, ensuring geometric relationships remain stable under

resampling:

$$\text{Shesha}_{\text{anchor}}(\mathbf{X}) = \frac{1}{K} \sum_{k=1}^K \rho_s \left( \text{vec} \left( \mathbf{D}(\mathbf{A}, \mathcal{S}_k^{(1)}) \right), \text{vec} \left( \mathbf{D}(\mathbf{A}, \mathcal{S}_k^{(2)}) \right) \right) \quad (3)$$

where  $\mathbf{A} \subset \mathbf{X}$  is a fixed set of  $m$  anchor points,  $\mathcal{S}_k^{(1)}, \mathcal{S}_k^{(2)} \subset \mathbf{X} \setminus \mathbf{A}$  are disjoint random subsets of size  $n$ ,  $\mathbf{D}(\mathbf{A}, \mathcal{S}) \in \mathbb{R}^{m \times n}$  is the pairwise distance matrix from anchors to subset samples (optionally rank-normalized row-wise), and  $\rho_s$  denotes Spearman's rank correlation.

- **Composite Stability Score:** To generate a unified metric for model comparison spanning discrete topological spaces mapping target dimensions smoothly precisely pooling results directly uniformly evaluating:

$$\text{Composite} = \frac{1}{4} (\text{Shesha}_{\text{SS}} + \text{Shesha}_{\text{FS}} + \rho_{\text{RDM}} + \text{Shesha}_{\text{anchor}}) \quad (4)$$

## D Track A + Track B + Track C Extended Methods and Details

This appendix provides full experimental details for the manifold continuity and causal-validation experiments described in Section 2.2 and Section 2.3. Track A serves as a continuous-input control; Track B applies the same geometric probes to four discrete-token genomic foundation models; Track C trains small models from scratch under discrete and continuous output heads on real DNA and protein sequences to test the tokenization effect causally within a fixed architecture. Track A and Track B use seed 320 throughout; Track C uses five seeds (320, 1991, 7, 2222, 411), with seed 320 as the fixed data, evaluation, and codebook seed.

### D.1 Track A: Synthetic Physics (Damped Harmonic Oscillator)

**Dataset.** Each training sequence is a length-512 discretized trajectory of a damped harmonic oscillator

$$x(t) = A e^{-\gamma t} \cos(\omega t + \phi)$$

sampled at 512 equally spaced points over  $t \in [0, 4)$ . Parameters are drawn independently per sequence:

$$\text{amplitude: } A \sim \mathcal{U}(0.5, 2.0)$$

$$\text{damping: } \gamma \sim \mathcal{U}(0.2, 2.0)$$

$$\text{angular frequency: } \omega \sim \mathcal{U}(2.0, 20.0)$$

$$\text{phase: } \phi \sim \mathcal{U}(0, 2\pi)$$

Continuous values are discretized into  $N_{\text{bins}} = 256$  integer bins via dataset-global min/max normalization (computed once over the full training set and reused for all evaluation and interpolation data). The training set contains 50,000 sequences; a held-out validation set contains 2,000 sequences drawn with a separate seed.

**Architectures.** Three architectures are trained under identical conditions, each with approximately 2M parameters:

| Model                      | Layers | $d_{\text{model}}$ | Heads | FFN dim | SSM $d_{\text{state}}$ | Params |
|----------------------------|--------|--------------------|-------|---------|------------------------|--------|
| SmallBERT (Transformer)    | 4      | 256                | 4     | 1024    | –                      | ~2M    |
| SmallMamba (SSM)           | 4      | 256                | –     | –       | 16                     | ~2M    |
| SmallStripedHyena (Hybrid) | 4      | 256                | 4     | 1024    | –                      | ~2M    |

Table S2: Track A architecture specifications. All models share  $d_{\text{model}} = 256$ , vocabulary size 258 (256 bins + mask + pad tokens), and sequence length 512. SmallMamba uses  $d_{\text{conv}} = 4$ , expand factor 2. SmallStripedHyena alternates Hyena convolution operators (order 2, implicit MLP filter) with standard multi-head attention layers in a 3:1 ratio (3 Hyena, 1 attention).

SmallBERT uses pre-LayerNorm Transformer encoder layers with learned positional embeddings and Xavier uniform initialization. SmallMamba stacks native CUDA Mamba blocks (`mamba_ssm`) with pre-LayerNorm residual connections and learned positional embeddings. SmallStripedHyena implements a simplified version of the StripedHyena architecture used in Evo 2: each Hyena operator learns its long convolution filter implicitly via a small MLP over sinusoidal positional features, with exponential decay modulation and short depthwise convolution gating. The attention stripes use the same multi-head attention as SmallBERT.

**Training.** All models are trained with causal language modeling (CLM) using cross-entropy loss for 20 epochs, batch size 64, AdamW optimizer ( $\text{lr} = 3 \times 10^{-4}$ , weight decay = 0.01). Training runs on a single NVIDIA A100 GPU. The continuous physical signal is discretized *before* being fed to the model; the loss operates on discrete token predictions. This design is intentional: the models receive identical discrete inputs and are evaluated solely on the geometry of their intermediate representations, isolating architecture from training objective.

**Interpolation protocol.** For each of 10 randomly sampled pairs of oscillator trajectories  $(A_i, B_i)$ , we generate 101 linearly interpolated sequences:

$$\text{interp}(\alpha) = \text{discretize}((1 - \alpha) \cdot A_i + \alpha \cdot B_i, \text{global\_range}), \quad \alpha \in \{0, 0.01, 0.02, \dots, 1.0\}. \quad (5)$$

Both endpoints are discretized using the same dataset-global min/max range from training, ensuring that linear interpolation in token space corresponds to linear interpolation in physical space. Trajectory  $A$  is drawn with low frequency and light damping ( $\gamma \sim \mathcal{U}(0.2, 0.8)$ ,  $\omega \sim \mathcal{U}(2, 8)$ ); Trajectory  $B$  with higher frequency and heavier damping ( $\gamma \sim \mathcal{U}(1.0, 2.0)$ ,  $\omega \sim \mathcal{U}(10, 20)$ ), guaranteeing distinct dynamical regimes at the walk endpoints.

**Embedding extraction and metrics.** Each interpolation step is embedded by passing the token sequence through the model with `return_hidden=True` and mean-pooling the last hidden layer across all 512 positions, yielding a single  $d_{\text{model}}$ -dimensional vector per step. We compute three quantities per pair:

1. **PCA trajectory:** The 101 embeddings are projected into 3D via PCA (fit per architecture) and plotted as a continuous path. Smooth arcs indicate a well-behaved manifold; staircase patterns indicate phase transitions.

2. **Cosine distance from start:**

$$d_{\cos}(\alpha) = 1 - \frac{\mathbf{e}(\alpha) \cdot \mathbf{e}(0)}{\|\mathbf{e}(\alpha)\| \|\mathbf{e}(0)\|}$$

for each step, measuring cumulative drift. A monotonically increasing profile indicates directional consistency.

3. **L2 Lipschitz profile:**

$$L(\alpha_i) = \|\mathbf{e}(\alpha_{i+1}) - \mathbf{e}(\alpha_i)\|_2$$

for consecutive steps, measuring local rate of change. The mean and maximum over all 10 pairs are reported. A flat profile indicates uniform sensitivity; spikes indicate discontinuities.

**Results summary.** Mean Lipschitz values averaged over 10 pairs: SmallBERT 65.3, SmallStriped-Hyena 79.3, SmallMamba 84.6 ( $1.3\times$  spread from smoothest to roughest). All three architectures produce smooth PCA arcs with no staircase effects, no teleportation, and no fracturing. The smoothness ratio (mean/max Lipschitz) is 0.425 (SmallBERT), 0.433 (SmallStripedHyena), 0.388 (SmallMamba).

## D.2 Track B: Biological DNA (BRCA1 Mutation Walk)

**Genomic region.** We target the BRCA1 gene on chromosome 17 (GRCh38/hg38), centered on the pathogenic C61G missense variant at position 43,104,121. A 16,384 bp region is downloaded from the UCSC Genome Browser API [15, `api.genome.ucsc.edu`], providing 7,192 bp of flanking context on each side of the 2,000 bp core mutation zone. Any ambiguous bases (N) are replaced with a uniformly random nucleotide (seed 320). Models with shorter context windows receive the relevant sub-region centered on the core zone (see per-model details below).

**Mutation walk construction.** The walk endpoint (mutant sequence) is constructed by introducing 121 point mutations into the core 2 kb region of the wildtype sequence: one pathogenic C61G substitution at the region center, plus 120 additional random SNPs at positions sampled uniformly without replacement from the core zone (seed 320). Each SNP changes the reference base to a uniformly random alternative. The single-point mutation walk then proceeds from wildtype to mutant by changing one base at a time, in a randomly shuffled order of the 121 differing positions (also seed 320), producing 122 intermediate sequences (including start and end). The pathogenic C61G mutation falls at a random position in this shuffled order, providing a biologically meaningful landmark. The walk, step positions, and pathogenic step index are cached as a NumPy archive and shared across all four model-specific notebooks.

**Models.** Four genomic foundation models span the architectural spectrum from pure Transformer to pure SSM:

**Per-model embedding details.** DNABERT-2 (117M; zhihan1996/DNABERT-2-117M). Uses BPE tokenization with a learned multi-granularity merge table. The 512-token context limit covers

| Model                     | Architecture            | Tokenization | Embed dim | Params | Embedding layer             |
|---------------------------|-------------------------|--------------|-----------|--------|-----------------------------|
| DNABERT-2                 | BERT Transformer        | BPE          | 768       | 117M   | Last hidden, mean-pool      |
| Nucleotide Transformer v2 | BERT Transformer        | 6-mer        | 1024      | 500M   | Last hidden, mean-pool      |
| Evo 2                     | StripedHyena (SSM+Attn) | Single-char  | 4096      | 7B     | blocks.28.mlp.13, mean-pool |
| Caduceus (PS)             | Mamba SSM (RC-equiv.)   | Single-char  | 256       | 7.7M   | Hidden state, mean-pool     |

Table S3: Track B model specifications.

approximately 1–2 kb of DNA depending on merge patterns. We embed the 2 kb core region, extracting the last hidden state and mean-pooling over all non-padding tokens. Loading requires bypassing `AutoModel.from_pretrained` on transformers 5.x due to meta-tensor initialization conflicts with DNABERT-2’s custom `bert_layers.py`; we resolve the model class via HuggingFace dynamic module utilities and load the state dict directly. FlashAttention is provided via Triton.

*Nucleotide Transformer v2* (500M;

`InstaDeepAI/nucleotide-transformer-v2-500m-multi-species`). Uses fixed 6-mer tokenization with a maximum of 1,000 tokens (~6,000 bp effective context). We load via `AutoModelForMaskedLM` (not `AutoModel`) because the checkpoint includes the language model head weights. Embeddings are the last hidden state (`output_hidden_states=True`), mean-pooled over non-padding positions. Extensive transformers 5.x compatibility patches are applied (`find_pruneable_heads_and_indices`, `all_tied_weights_keys`, `get_head_mask`).

*Evo 2* (7B; `evo2_7b` via Vortex). A StripedHyena hybrid model with single-character tokenization and an 8,192 bp context window. Loaded via `Evo2('evo2_7b')` using the native Vortex interface (not HuggingFace). Embeddings are extracted from an intermediate layer (`blocks.28.mlp.13`) using `return_embeddings=True` with explicit `layer_names`, then mean-pooled across all positions. Sequences shorter than 8,192 bp are padded with N characters; the full 16 kb flanked region is truncated to context length. Requires NVIDIA A100 GPU (~28 GB VRAM), flash-attn 2.8.0.post2 built from source.

*Caduceus* (7.7M; `kuleshov-group/caduceus-ps_seqlen-131k_d_model-256_n_layer-16`). A pure Mamba SSM with reverse-complement (RC) equivariant design (PS variant). Uses single-character tokenization and supports up to 131,072 bp context. Loading requires patching transformers for compatibility with the custom `mamba_rev/mamba_fwd` weight tying scheme. Fused Triton layer norms are disabled if unavailable. For PS models, the hidden state is pooled with RC-invariant averaging. Built from source using `causal-conv1d` and `mamba-ssm` packages.

**Lipschitz profile computation.** For each model, we compute the cosine-based local Lipschitz constant between consecutive walk steps:

$$L_{\cos}(i) = 1 - \frac{\mathbf{e}_{i+1} \cdot \mathbf{e}_i}{\|\mathbf{e}_{i+1}\| \|\mathbf{e}_i\|}, \quad i = 0, 1, \dots, 120, \quad (6)$$

where  $\mathbf{e}_i$  is the mean-pooled embedding of the  $i$ -th walk sequence. We use cosine distance rather than L2 to ensure dimension-invariant comparisons across models with embedding dimensions ranging from 256 (Caduceus) to 4,096 (Evo 2). Spikes are defined as steps exceeding the mean + 2 standard deviations of the profile for that model.

**PCA trajectories.** The 122 embeddings per model are projected to 3 dimensions via PCA (fit independently per model) and plotted as continuous paths. The pathogenic C61G step is marked if it corresponds to a Lipschitz spike.

**Results summary.** Mean cosine Lipschitz values: DNABERT-2  $3.1 \times 10^{-3}$ , Nucleotide Transformer  $2.6 \times 10^{-3}$ , Evo 2  $1.0 \times 10^{-6}$ , Caduceus  $\sim 1.0 \times 10^{-7}$ . The Transformers exhibit 5 and 4 spikes respectively (above  $2\sigma$  threshold). Evo 2 shows 2 spikes. Caduceus registers 18 apparent spikes, but all are at the floating-point noise floor ( $\sim 10^{-7}$ ) and represent numerical jitter rather than biological signal. No model detected the pathogenic C61G mutation as a Lipschitz spike.

### D.3 Cross-Track Comparison

The critical comparison: in Track A (continuous physics, no tokenization), the gap between the smoothest and roughest architecture is  $1.3\times$  (SmallBERT at 65.3 vs. SmallMamba at 84.6). In

Track B (discrete biology, full tokenization), the gap between the highest and lowest mean cosine Lipschitz is approximately  $3,000\times$  ( $3.1 \times 10^{-3}$  for DNABERT-2 vs.  $\sim 1.0 \times 10^{-6}$  for Evo 2; or  $\sim 26,000\times$  if comparing DNABERT-2 to the Caduceus noise floor). The same attention mechanism that produces the smoothest interpolation on continuous signals produces a fractured manifold when forced through discrete tokens. The variable that changed between the two tracks is not the routing mechanism or model scale; it is the tokenization.

#### D.4 Track C: Biological Continuous-Head Ablation

All Track C experiments use five seeds (320, 1991, 7, 2222, 411); seed 320 is the fixed data, evaluation, and codebook seed. Geometry is evaluated with the same Shesha harness [58, 59] used elsewhere.

**Domains and training.** DNA uses SmallStripedHyena trained autoregressively on human CHR22 windows; protein uses SmallBERT trained on UNIREF50/SWISSPROT under both masked (MLM) and causal (CLM) objectives. Protein runs use 16 epochs; the DNA run uses 8 epochs (an intentional cross-domain difference reflecting convergence, not a matched hyperparameter). Each domain-objective cell is run under discrete CE and under a continuous-head variant that replaces the categorical output head with a linear projection to a fixed continuous code, trained under MSE, leaving the encoder backbone unchanged.

**Continuous code families.** Three code families stand in for the CE head: *physicochemical* (a low-dimensional code from residue or base physicochemical features), *random* (a random low-dimensional code; the pre-registered primary arm), and *high-dimensional* (a higher-dimensional random projection). Reporting the three together tests whether any continuous advantage is specific to one code geometry.

**Learning gates (pre-registered).** Geometry is interpreted for an arm only if it beats the majority-class recovery baseline. Protein majority recovery is 0.098; the uniform-amino-acid perplexity floor is 20. Protein-MLM CE passes with perplexity  $\approx 13.5$  ( $< 15$ ) and recovery  $\approx 0.19$ , beating majority by  $\approx 0.09$  on every seed. Protein-CLM CE passes with perplexity  $\approx 6.0$  and recovery  $\approx 0.45$ , beating majority by  $\approx 0.36$ . DNA CE recovery is  $\approx 0.50$  against a four-symbol baseline of 0.25. The protein physicochemical arm under MLM recovers only  $\approx 0.090$ , below the 0.098 floor, and is excluded from the MLM causal argument as a codebook-ceiling (geometric-vacuity) failure.

**Perturbation battery and reversal handling.** Substitution is applied at 1%, 2%, 5%, and 10% of positions (five draws each), plus one whole-sequence reversal. For DNA the reversal is reverse-complement; for protein it is plain reversal. The headline ratio is computed on substitution only. Reversal is reported separately: for DNA it is a legitimate symmetry probe, for protein it is an OOD-scramble probe and is never included in an equal-weight battery mean.

**Competence-matching protocol.** For arms with flat recovery (DNA), the converged full-battery comparison already holds competence fixed. For arms with non-flat recovery (protein-CLM), we interpolate Procrustes  $D$  at matched recovery over the within-seed recovery-overlap band (5% substitution snapshot) and report the paired CE-minus-continuous gap. As an independent competence measure we train a frozen  $L_2$ -regularized logistic probe on mean-pooled converged representations against a downstream label, gating interpretation on the CE probe clearing at least  $5\times$  its random baseline with cross-seed standard deviation below the equivalence tolerance.

**DNA per-perturbation results.** CE mean Procrustes  $D$  by perturbation (5 seeds): snp 1% = 0.0039, snp 2% = 0.0085, snp 5% = 0.0267, snp 10% = 0.0753, reverse-complement = 0.1462. Battery-slice ratios (CE/continuous,  $> 1$  means continuous lower  $D$ ):

**Protein-MLM per-perturbation results.** The full-battery ratio ( $2.0\text{--}2.2\times$ ) is entirely reversal-driven; substitution alone favours CE.

**Protein-CLM per-perturbation results and competence control.** At the endpoint the continuous head shows a substitution advantage, while reversal mildly favours CE (mirror image of MLM):

| Arm           | Full battery | Substitution-only | Reverse-complement only |
|---------------|--------------|-------------------|-------------------------|
| Cont-physchem | 1.99×        | 1.88×             | 2.45×                   |
| Cont-random   | 1.84×        | 1.73×             | 2.27×                   |
| Cont-highdim  | 1.97×        | 1.87×             | 2.35×                   |

Table S4: DNA (causal LM) battery slices. Every substitution rate favours continuous independently; reverse-complement adds a further reduction but the substitution-only effect (1.73–1.88×) holds without it. Recovery is flat across arms.

| Arm           | Full battery | Substitution-only | Reversal only | Reversal share of gap |
|---------------|--------------|-------------------|---------------|-----------------------|
| Cont-physchem | 2.01×        | 0.73×             | 2.90×         | 547%                  |
| Cont-random   | 2.06×        | 0.82×             | 2.79×         | 522%                  |
| Cont-highdim  | 2.24×        | 0.68×             | 3.71×         | 554%                  |

Table S5: Protein-MLM battery slices. The continuous advantage is confined to the reversal (OOD-scramble) term; on substitution the continuous head is level-to-worse. Cont-physchem is additionally below the majority-recovery floor.

The frozen probe on a powered downstream label (SCOP class) gives CE balanced accuracy 0.521 against a random baseline of 0.143; the continuous arms are 2.2–3.4 percentage points below CE (within-2pp equivalence fails on all arms), indicating the continuous representations are less, not equally, competent. Matching  $D$  at equal recovery across all five seeds does not rescue the endpoint advantage for any arm, and the result is inconsistent across code families:

| Arm           | Ratio CE / cont (matched) | Gap (CE – cont) | Reading                     |
|---------------|---------------------------|-----------------|-----------------------------|
| Cont-random   | 1.31 × ( $\pm 0.07$ )     | +0.011          | modest, stable across seeds |
| Cont-highdim  | 1.15 × ( $\pm 0.11$ )     | +0.006          | marginal                    |
| Cont-physchem | 0.87 × ( $\pm 0.06$ )     | −0.008          | reverses (CE lower)         |

Table S7: Protein-CLM matched-recovery control (five seeds, pooled;  $D$  interpolated at equal recovery over the within-seed overlap band, 5% substitution). The endpoint 1.72× does not survive for any arm. The advantage is inconsistent across code families, reversing for the physicochemical arm; since a genuine tokenization effect should not depend on the choice of continuous code, this arm-divergence, in contrast to the convergence across arms in DNA (Table S4), indicates no clean effect in protein-CLM. The physicochemical arm has the lowest recovery ceiling, so its reversal is partly an arm-competence artifact; the defensible claim is no consistent advantage across arms, not that continuous is worse.

| Arm           | Full battery | Substitution-only | Reversal only |
|---------------|--------------|-------------------|---------------|
| Cont-physchem | 1.03×        | 1.28×             | 0.95×         |
| Cont-random   | 1.05×        | 1.72×             | 0.90×         |
| Cont-highdim  | 1.07×        | 1.72×             | 0.93×         |

Table S6: Protein-CLM endpoint battery slices. The substitution advantage is real at convergence but is confounded with a competence gap (recovery: CE 0.45, continuous 0.25–0.32).

## E Complete Ablation Battery

All ablations use the same three synthetic dynamical system datasets (waveform, coupled oscillator, Lorenz attractor), 256-bin discretization, seed 320, and Shesha Procrustes stability harness as the baseline experiments in Section 2. Perturbation suite: value noise at 1/2/5/10% of positions plus time reversal. Architectures are parameter-matched: SmallBERT (Transformer, 3.4M), SmallMamba (SSM, 2.0M), SmallStripedHyena (hybrid, 4.5M), SmallLSTM (discrete recurrent, 2.2M). All models are trained from scratch for 20 epochs on Google Colab (A100).

### E.1 Variant A: Continuous MSE Head

**Hypothesis.** The Geometric Alignment Tax is architectural, arising from the softmax attention mechanism, and cannot be eliminated by changing the output head alone.

**Method.** We replace the 256-class categorical cross-entropy (CE) head with a linear projection to a continuous scalar trained under MSE loss. The encoder backbone (self-attention layers, positional embeddings, feedforward blocks) is unchanged. A dual-return generator provides both discrete 256-bin tokens (model input) and raw continuous ODE floats (MSE target), avoiding the fatal data-leakage where dividing discrete tokens by  $(n_{\text{bins}} - 1)$  creates a quantized staircase target. All three architectures (SmallBERT, SmallMamba, SmallStripedHyena) are evaluated under both CE baseline and continuous MSE conditions.

**Results.** Replacing discrete CE with continuous MSE eliminates manifold fracture across all architectures. On the Lorenz dataset at 1% noise, SmallBERT Procrustes distortion  $D$  improves  $2.8\times$  ( $0.096 \rightarrow 0.034$ ); SmallStripedHyena improves  $8.5\times$  ( $0.072 \rightarrow 0.0085$ ), the single best condition in the entire study. The cross-architecture spread collapses from 0.072–0.157 under discrete CE to 0.0085–0.034 under continuous MSE. At 10% noise the pattern holds: SmallBERT discrete  $D = 0.316$  vs. continuous  $D = 0.111$  ( $2.8\times$ ); SmallStripedHyena discrete  $D = 0.233$  vs. continuous  $D = 0.027$  ( $8.6\times$ ). Mean composite stability across all datasets and perturbations rises from 0.466 (SmallBERT discrete) to 0.497 (SmallBERT continuous), and from 0.470 (SmallStripedHyena discrete) to 0.498 (SmallStripedHyena continuous). RDM similarity scores under the continuous head exceed 0.99 at 1% noise for all three architectures, confirming near-perfect manifold preservation. The discrete-to-continuous gap within any single architecture dwarfs the cross-architecture gap under either regime, confirming that the tokenization boundary – not the routing mechanism – is the dominant source of geometric instability.

### E.2 Variant B: Jacobian Norm Penalty

**Hypothesis.** Explicit smoothness regularization cannot close the geometric stability gap without catastrophic predictive collapse, revealing the tax as a fundamental trade-off.

**Method.** We add a Frobenius-norm penalty on the Jacobian of hidden states with respect to input embeddings:  $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \|\partial \mathbf{h} / \partial \mathbf{x}\|_F$ . The penalty is swept across  $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1.0\}$  on both SmallBERT and SmallStripedHyena. SmallStripedHyena additionally includes a  $\lambda = 0$  (unregularized) baseline for direct comparison. All other training hyperparameters remain identical to the baseline.

**Results.** The sweep reveals a Pareto frontier between geometric stability and predictive accuracy. For SmallBERT, mean composite stability improves monotonically with  $\lambda$ :  $0.467$  ( $\lambda = 10^{-3}$ )  $\rightarrow$

$0.474 (10^{-2}) \rightarrow 0.484 (10^{-1}) \rightarrow 0.491 (1.0)$ . However, mean validation CE degrades in parallel:  $0.81 \rightarrow 0.85 \rightarrow 0.91 \rightarrow 1.07$ . SmallStripedHyena shows the same pattern: composite stability rises from 0.472 (baseline  $\lambda = 0$ ) to 0.491 ( $\lambda = 1.0$ ), while validation CE increases from 0.002 to 0.003. At  $\lambda = 1.0$ , both architectures converge to nearly identical composite scores ( $\sim 0.491$ ), but at the cost of substantially degraded prediction quality. No setting of  $\lambda$  achieves simultaneously low distortion and low CE. The Jacobian penalty smooths the manifold by penalizing sharp representational gradients, but the model compensates by flattening its hidden-state landscape, which destroys the fine-grained distinctions needed for accurate next-token prediction. This confirms the tax is a genuine trade-off intrinsic to discrete optimization under attention, not a training artifact that can be patched with regularization.

### E.3 Variant C: SmallMamba + Continuous Head (Positive Control)

**Hypothesis.** Mamba’s geometric stability is intrinsic to its continuous ODE prior and persists regardless of the output head.

**Method.** SmallMamba receives the same MSE head replacement as Variant A, while keeping the SSM backbone (selective state-space mechanism with exponential matrix discretization) unchanged. SmallBERT\_Continuous and SmallStripedHyena\_Continuous are included as architecture comparisons. All three architectures use identical continuous MSE training on the same datasets.

**Results.** SmallMamba\_Continuous maintains RDM similarity  $>0.999$  across all noise levels on all three datasets, with composite stability scores of 0.496 (waveform mean), 0.495 (oscillator mean), and 0.500 (Lorenz mean). These scores are comparable to baseline SmallMamba under discrete CE ( $\sim 0.400$  composite, but with RDM  $>0.99$  at low noise), confirming that the SSM’s geometric smoothness is independent of the loss landscape. Critically, SmallBERT\_Continuous, despite the continuous head, still exhibits lower RDM similarity than SmallMamba\_Continuous under time reversal on the waveform dataset (0.978 vs. 1.000) and oscillator dataset (0.928 vs. 1.000). SmallStripedHyena\_Continuous shows intermediate behavior, with sensitivity to time reversal on the Lorenz (0.953) and oscillator (0.930) datasets that SmallMamba does not exhibit. The positive control is confirmed: the ODE prior provides inherent geometric smoothness through its continuous-time parameterization, and this stability is truly architectural rather than an artifact of any specific head-backbone interaction.

### E.4 Variant D: LSTM Baseline (The Discrete Recurrence Test)

**Hypothesis.** Recurrence alone is insufficient for manifold preservation; the continuous ODE prior is the specific mechanism responsible for SSM stability.

**Method.** SmallLSTM is a parameter-matched (2.2M parameters) LSTM architecture trained with identical CLM setup and discrete CE loss. The LSTM uses standard sigmoid/tanh gates and lacks any continuous-time dynamics: no exponential matrix discretization, no  $\Delta$  discretization parameter, no state-space formulation, and no absolute positional embeddings (relying purely on recurrent gates for sequential position encoding). SmallBERT and SmallMamba are retrained as reference bounds under identical conditions.

**Results.** The LSTM fractures like the Transformer, not the SSM. On the waveform dataset, SmallLSTM achieves mean composite stability of 0.418, comparable to SmallBERT (0.443) and far below SmallStripedHyena (0.459) and SmallMamba (0.399, which reflects Mamba’s distinct stability profile where low composite scores coexist with high RDM similarity due to large perturbation magnitudes). On the oscillator dataset, SmallLSTM mean composite is 0.471 vs. SmallBERT 0.490 and SmallMamba 0.484. On the Lorenz attractor, SmallLSTM mean composite is 0.480 vs. SmallBERT 0.455. The Lorenz butterfly test provides the clearest separation: the LSTM’s phase portrait exhibits the same drift and structural distortion as SmallBERT, while SmallMamba alone preserves the full attractor geometry. The LSTM’s RDM similarity scores degrade rapidly with increasing noise (waveform: 0.900 at 1% noise, 0.600 at 10%), matching the Transformer’s fragility pattern rather than the SSM’s robustness ( $>0.99$  across all conditions). This definitively rules out recurrence as the source of SSM stability. The continuous ODE prior – parameterized through structured state matrices

$A, B, C, D$  with exponential discretization – is the specific mechanism that distinguishes Mamba from both attention-based and discrete recurrent architectures.

### E.5 Variant E: Attention Layer Ratio Sweep (The Dilution Curve)

**Hypothesis.** The Geometric Alignment Tax is dose-dependent: geometric stability degrades as a function of the fraction of attention layers in a hybrid architecture.

**Method.** We construct 9 configurations of an 8-layer StripedHyena architecture, sweeping the attention fraction from 0/8 (pure Hyena, no attention blocks) through 1/8, 2/8, ..., 7/8, to 8/8 (pure attention, equivalent to a Transformer). All variants use identical hyperparameters and discrete CE loss. SmallBERT and SmallMamba serve as external reference bounds. Three competing hypotheses are tested: H1 (linear degradation), H2 (phase transition with a critical threshold), and H3 (diminishing returns where initial attention layers cause disproportionate damage).

**Results.** The dose-response curve shows monotonic degradation with increasing attention fraction, consistent with a roughly linear relationship (H1) rather than a sharp phase transition. At 0/8 attention (pure Hyena), the mean composite stability across all datasets is  $\sim 0.475$ , which degrades progressively to  $\sim 0.455$  at 8/8 attention (pure Transformer). The pure-Hyena configuration (0/8) achieves the highest geometric stability among all StripedHyena variants, and the pure-attention configuration (8/8) converges to SmallBERT-level performance, as expected. The Lorenz and oscillator datasets show the clearest dose-response signal, while the waveform dataset exhibits more variance. Evo 2’s operational point of  $\sim 12.5\%$  attention falls in the low-damage regime of the curve, consistent with its relatively smooth Lipschitz profiles in Track B (Section 2.2) despite operating under discrete tokenization. The absence of a sharp threshold suggests that each attention layer independently contributes a fixed quantum of geometric tax, and that architectural design choices about attention ratio translate predictably to geometric cost.

### E.6 Variant F: Hyena Filter Order Sweep (The ODE Depth Test)

**Hypothesis.** Within continuous architectures, the expressiveness of the continuous-time filter (parameterized by the Hyena operator order) determines the quality of geometric preservation.

**Method.** Using a 4-layer StripedHyena with fixed 25% attention ratio (1 attention layer at position 3), we sweep the Hyena operator’s order parameter across values 1, 2, 4, and 8. Higher order means more chained data-controlled convolutions with multiplicative gates and additional ImplicitFilterMLPs, yielding richer continuous-time dynamics. Parameter counts range from 4.2M (order 1) to 5.7M (order 8). All variants use identical training hyperparameters and discrete CE loss.

**Results.** Geometric stability is largely constant across filter orders, supporting the binary hypothesis (H1): any Hyena order  $\geq 1$  is sufficient for geometric preservation, because the mere presence of continuous convolution rather than softmax attention is the operative factor. On the Lorenz dataset, mean composite stability is 0.476 (order 1), 0.471 (order 2), 0.473 (order 4), and 0.477 (order 8). The oscillator shows similarly flat profiles: 0.490, 0.491, 0.491, 0.490. RDM similarity at 1% noise exceeds 0.997 for all orders on the oscillator and exceeds 0.995 on the waveform. The waveform dataset shows slightly more variance, with order 1 achieving 0.462 mean composite and order 8 achieving 0.460, but these differences are within noise. Critically, even after normalizing for the parameter count increase at higher orders (order 8 has  $\sim 35\%$  more parameters than order 1), there is no systematic improvement. This complements Variant D: discrete recurrence (LSTM) is insufficient for stability, but within continuous architectures, the minimal continuous-time parameterization (order 1) already captures the full geometric benefit. The tax is a binary property of the attention-vs.-continuous-convolution distinction, not a graded function of filter expressiveness.

### E.7 Summary

The six ablations converge on a single causal picture. Variant A demonstrates that replacing discrete CE with continuous MSE eliminates manifold fracture (the tokenization boundary is the bottleneck). Variant B shows that smoothness regularization creates a Pareto frontier rather than closing the gap (the tax is a genuine trade-off). Variant C confirms that SSM stability is intrinsic to the ODE

prior, not an artifact of head-backbone interaction (positive control). Variant D proves that discrete recurrence (LSTM) fractures like the Transformer, isolating the continuous ODE parameterization as the specific stability mechanism. Variant E quantifies the dose-response relationship between attention fraction and geometric cost (approximately linear, no safe threshold). Variant F shows that filter expressiveness within continuous architectures does not affect geometric stability (binary property, not graded). Together, these results rule out the output head, the loss function, recurrence, regularization, and filter depth as alternative explanations, leaving tokenization as the dominant causal factor.

## F Full Model Zoo

This appendix reports per-perturbation stability results for all models evaluated in this study. All experiments use seed 320 and the Shesha Procrustes stability harness. Metrics reported: RDM similarity (representational similarity under perturbation; 1.0 = perfect preservation), perturbation stability, perturbation magnitude, and composite stability (the Shesha summary score). Each model family is tested on both synthetic sequences and real biological sequences where applicable. Perturbation types vary by domain: amino acid substitutions (1–10%) and full reversal for protein models; SNPs (1–10%) and reverse complement for DNA models.

### F.1 ESM-2

Six checkpoints spanning 8M to 15B parameters for ESM-2 [49, Protein Transformer]. Per-residue tokenization (20 amino acid tokens). The primary scaling story: composite stability declines monotonically from 8M to 3B, then exhibits an apparent “recovery” at 15B that is unmasked as Untethered Gel by the Procrustes Ratio and Frozen Head tests (Section 3).

#### Synthetic proteins (200 aa).

Table S8: ESM-2 synthetic protein results. Stability degrades with scale through 3B; the 15B rebound reflects global manifold drift, not genuine recovery.

| Model     | Size (M) | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|-----------|----------|--------------|----------|-------------|------------|-----------|
| ESM2-8M   | 7.5      | aa_sub_1pct  | 0.958    | −0.001      | 0.004      | 0.463     |
|           |          | aa_sub_2pct  | 0.916    | 0.017       | 0.007      | 0.453     |
|           |          | aa_sub_5pct  | 0.811    | 0.012       | 0.008      | 0.426     |
|           |          | aa_sub_10pct | 0.675    | 0.002       | 0.012      | 0.392     |
|           |          | reverse      | 0.776    | 0.004       | 0.012      | 0.418     |
| ESM2-35M  | 33.5     | aa_sub_1pct  | 0.921    | 0.013       | 0.009      | 0.454     |
|           |          | aa_sub_2pct  | 0.858    | 0.010       | 0.010      | 0.438     |
|           |          | aa_sub_5pct  | 0.719    | 0.016       | 0.013      | 0.404     |
|           |          | aa_sub_10pct | 0.571    | 0.010       | 0.018      | 0.367     |
|           |          | reverse      | 0.567    | 0.003       | 0.020      | 0.366     |
| ESM2-150M | 148.1    | aa_sub_1pct  | 0.874    | 0.004       | 0.011      | 0.443     |
|           |          | aa_sub_2pct  | 0.787    | 0.006       | 0.013      | 0.421     |
|           |          | aa_sub_5pct  | 0.638    | 0.013       | 0.017      | 0.384     |
|           |          | aa_sub_10pct | 0.492    | 0.018       | 0.025      | 0.347     |
|           |          | reverse      | 0.491    | 0.017       | 0.025      | 0.347     |
| ESM2-650M | 651.0    | aa_sub_1pct  | 0.813    | 0.010       | 0.013      | 0.419     |
|           |          | aa_sub_2pct  | 0.708    | 0.013       | 0.019      | 0.393     |
|           |          | aa_sub_5pct  | 0.559    | 0.018       | 0.027      | 0.356     |
|           |          | aa_sub_10pct | 0.408    | 0.027       | 0.031      | 0.318     |
|           |          | reverse      | 0.390    | 0.004       | 0.028      | 0.314     |
| ESM2-3B   | 2839.0   | aa_sub_1pct  | 0.726    | 0.022       | 0.017      | 0.391     |
|           |          | aa_sub_2pct  | 0.606    | 0.022       | 0.022      | 0.361     |
|           |          | aa_sub_5pct  | 0.423    | 0.020       | 0.025      | 0.315     |
|           |          | aa_sub_10pct | 0.299    | 0.031       | 0.028      | 0.285     |
|           |          | reverse      | 0.207    | 0.024       | 0.034      | 0.261     |
| ESM2-15B  | 15129.1  | aa_sub_1pct  | 0.805    | 0.020       | 0.038      | 0.445     |
|           |          | aa_sub_2pct  | 0.712    | 0.030       | 0.046      | 0.424     |
|           |          | aa_sub_5pct  | 0.536    | 0.026       | 0.053      | 0.379     |
|           |          | aa_sub_10pct | 0.353    | 0.027       | 0.065      | 0.334     |
|           |          | reverse      | 0.254    | 0.027       | 0.074      | 0.309     |

**Real UniRef50 proteins (100–400 aa).**

The synthetic-to-real gap in composite stability is  $\leq 0.03$  at every scale, confirming the tax is not an artifact of sequence composition.

Table S9: ESM-2 real UniRef50 protein results. The scaling trend replicates faithfully on natural proteins.

| Model     | Size (M) | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|-----------|----------|--------------|----------|-------------|------------|-----------|
| ESM2-8M   | 8        | aa_sub_1pct  | 0.966    | 0.029       | 0.004      | 0.466     |
|           |          | aa_sub_5pct  | 0.824    | 0.018       | 0.008      | 0.430     |
|           |          | aa_sub_10pct | 0.688    | 0.048       | 0.011      | 0.396     |
|           |          | reverse      | 0.786    | 0.025       | 0.011      | 0.421     |
| ESM2-35M  | 34       | aa_sub_1pct  | 0.942    | 0.035       | 0.005      | 0.462     |
|           |          | aa_sub_5pct  | 0.737    | 0.048       | 0.013      | 0.411     |
|           |          | aa_sub_10pct | 0.580    | 0.010       | 0.017      | 0.372     |
|           |          | reverse      | 0.593    | 0.015       | 0.015      | 0.375     |
| ESM2-150M | 148      | aa_sub_1pct  | 0.907    | 0.025       | 0.009      | 0.454     |
|           |          | aa_sub_5pct  | 0.670    | 0.036       | 0.015      | 0.395     |
|           |          | aa_sub_10pct | 0.521    | 0.022       | 0.020      | 0.358     |
|           |          | reverse      | 0.511    | 0.014       | 0.021      | 0.355     |
| ESM2-650M | 651      | aa_sub_1pct  | 0.853    | 0.028       | 0.012      | 0.432     |
|           |          | aa_sub_5pct  | 0.576    | 0.023       | 0.022      | 0.363     |
|           |          | aa_sub_10pct | 0.443    | 0.019       | 0.027      | 0.329     |
|           |          | reverse      | 0.413    | 0.030       | 0.026      | 0.322     |
| ESM2-3B   | 2839     | aa_sub_1pct  | 0.775    | 0.028       | 0.015      | 0.406     |
|           |          | aa_sub_5pct  | 0.450    | 0.035       | 0.026      | 0.325     |
|           |          | aa_sub_10pct | 0.299    | 0.038       | 0.029      | 0.287     |
|           |          | reverse      | 0.235    | 0.043       | 0.030      | 0.271     |
| ESM2-15B  | 15129    | aa_sub_1pct  | 0.843    | 0.027       | 0.033      | 0.456     |
|           |          | aa_sub_5pct  | 0.511    | 0.017       | 0.060      | 0.371     |
|           |          | aa_sub_10pct | 0.344    | 0.024       | 0.077      | 0.330     |
|           |          | reverse      | 0.248    | 0.018       | 0.094      | 0.306     |

## F.2 Nucleotide Transformer v2

Five checkpoints: 50M, 100M, 250M, 500M, and 2.5B parameters for Nucleotide Transformer v2 [21, DNA Transformer]. Uses 6-mer tokenization (vocabulary size  $\approx 4096$ ), providing a tokenization control against ESM-2’s per-residue scheme. The 2.5B model exhibits a partial stability rebound that parallels the ESM-2 15B pattern, consistent with the Untethered Gel regime at large scale.

### Synthetic DNA.

Table S10: Nucleotide Transformer v2 synthetic DNA results. Progressive decline through 500M; partial rebound at 2.5B parallels the ESM-2 pattern.

| Model   | Size (M) | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|---------|----------|--------------|----------|-------------|------------|-----------|
| NT-50M  | 55.9     | snp_1pct     | 0.726    | 0.027       | 0.248      | 0.340     |
|         |          | snp_2pct     | 0.545    | 0.029       | 0.318      | 0.295     |
|         |          | snp_5pct     | 0.254    | 0.032       | 0.421      | 0.222     |
|         |          | snp_10pct    | 0.085    | 0.034       | 0.509      | 0.180     |
|         |          | reverse_comp | 0.000    | 0.036       | 0.598      | 0.159     |
| NT-100M | 97.9     | snp_1pct     | 0.700    | 0.028       | 0.278      | 0.336     |
|         |          | snp_2pct     | 0.511    | 0.028       | 0.354      | 0.289     |
|         |          | snp_5pct     | 0.225    | 0.033       | 0.490      | 0.218     |
|         |          | snp_10pct    | 0.072    | 0.037       | 0.543      | 0.179     |
|         |          | reverse_comp | 0.000    | 0.036       | 0.607      | 0.161     |
| NT-250M | 235.1    | snp_1pct     | 0.647    | 0.029       | 0.358      | 0.306     |
|         |          | snp_2pct     | 0.449    | 0.031       | 0.465      | 0.256     |
|         |          | snp_5pct     | 0.176    | 0.034       | 0.603      | 0.188     |
|         |          | snp_10pct    | 0.050    | 0.034       | 0.654      | 0.157     |
|         |          | reverse_comp | 0.000    | 0.036       | 0.725      | 0.144     |
| NT-500M | 498.3    | snp_1pct     | 0.604    | 0.029       | 0.444      | 0.285     |
|         |          | snp_2pct     | 0.402    | 0.031       | 0.567      | 0.234     |
|         |          | snp_5pct     | 0.150    | 0.035       | 0.746      | 0.171     |
|         |          | snp_10pct    | 0.041    | 0.036       | 0.789      | 0.144     |
|         |          | reverse_comp | 0.000    | 0.039       | 0.850      | 0.134     |
| NT-2.5B | 2547.8   | snp_1pct     | 0.749    | 0.021       | 0.075      | 0.421     |
|         |          | snp_2pct     | 0.594    | 0.025       | 0.073      | 0.383     |
|         |          | snp_5pct     | 0.360    | 0.028       | 0.100      | 0.324     |
|         |          | snp_10pct    | 0.185    | 0.020       | 0.121      | 0.280     |
|         |          | reverse_comp | 0.160    | 0.031       | 0.145      | 0.274     |

**Real chr22 DNA.**

Results on real genomic DNA confirm that the NT scaling trend is not an artifact of synthetic sequence composition (Table S11).

Table S11: Nucleotide Transformer v2 real chr22 DNA results. Progressive decline from 50M to 500M; dramatic phase transition at 2.5B parallels the ESM-2 15B pattern.

| Model      | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|------------|--------------|----------|-------------|------------|-----------|
| NT-v2-50M  | snp_1pct     | 0.748    | 0.027       | 0.250      | 0.344     |
|            | snp_2pct     | 0.575    | 0.028       | 0.329      | 0.301     |
|            | snp_5pct     | 0.283    | 0.032       | 0.487      | 0.228     |
|            | snp_10pct    | 0.100    | 0.041       | 0.700      | 0.182     |
|            | reverse_comp | 0.002    | 0.042       | 0.730      | 0.158     |
| NT-v2-100M | snp_1pct     | 0.720    | 0.027       | 0.262      | 0.342     |
|            | snp_2pct     | 0.538    | 0.030       | 0.344      | 0.297     |
|            | snp_5pct     | 0.252    | 0.036       | 0.516      | 0.225     |
|            | snp_10pct    | 0.086    | 0.040       | 0.693      | 0.184     |
|            | reverse_comp | 0.003    | 0.038       | 0.658      | 0.163     |
| NT-v2-250M | snp_1pct     | 0.669    | 0.027       | 0.346      | 0.311     |
|            | snp_2pct     | 0.473    | 0.031       | 0.456      | 0.262     |
|            | snp_5pct     | 0.198    | 0.035       | 0.641      | 0.194     |
|            | snp_10pct    | 0.060    | 0.039       | 0.830      | 0.159     |
|            | reverse_comp | 0.003    | 0.041       | 0.848      | 0.145     |
| NT-v2-500M | snp_1pct     | 0.622    | 0.028       | 0.416      | 0.288     |
|            | snp_2pct     | 0.421    | 0.031       | 0.542      | 0.237     |
|            | snp_5pct     | 0.161    | 0.034       | 0.701      | 0.172     |
|            | snp_10pct    | 0.047    | 0.038       | 0.859      | 0.144     |
|            | reverse_comp | 0.002    | 0.039       | 0.910      | 0.133     |
| NT-2.5B    | snp_1pct     | 0.989    | 0.049       | 0.318      | 0.494     |
|            | snp_2pct     | 0.980    | 0.085       | 0.633      | 0.491     |
|            | snp_5pct     | 0.948    | 0.200       | 1.533      | 0.484     |
|            | snp_10pct    | 0.867    | 0.391       | 3.100      | 0.463     |
|            | reverse_comp | 0.959    | 0.038       | 0.258      | 0.486     |

### E.3 Caduceus

Three configurations: 0.5M (d=118, 4 layers), 1.9M (d=256, 4 layers), and 7.7M (d=256, 16 layers) for Caduceus [64, RC-Equivariant Mamba SSM]. Caduceus is the only model with a built-in RC-equivariant architecture (parameter sharing between forward and reverse-complement strands). Near-perfect RC preservation (RDM > 0.999) and near-constant composite stability across scale make it the “tax-exempt” baseline.

#### Synthetic DNA.

Table S12: Caduceus synthetic DNA results. RC RDM  $\geq 0.9999$  at all scales; composite stability is scale-invariant, confirming that the continuous ODE prior buffers the tokenization bottleneck.

| Model               | Perturbation | RDM Sim. $\uparrow$ | Pert. Stab. $\downarrow$ | Pert. Mag. | Composite     |
|---------------------|--------------|---------------------|--------------------------|------------|---------------|
| Cad-0.5M (1k ctx)   | SNP 1%       | 0.9517              | 0.0169                   | 0.0002     | 0.4833        |
|                     | SNP 2%       | 0.9049              | 0.0235                   | 0.0002     | 0.4716        |
|                     | SNP 5%       | 0.7874              | 0.0189                   | 0.0003     | 0.4422        |
|                     | SNP 10%      | 0.6208              | 0.0144                   | 0.0004     | 0.4006        |
|                     | Rev. Comp.   | <b>1.0000</b>       | 0.0202                   | 0.0000     | <b>0.4954</b> |
| Cad-1.9M (1k ctx)   | SNP 1%       | 0.9521              | 0.0231                   | 0.0001     | 0.4733        |
|                     | SNP 2%       | 0.9072              | 0.0167                   | 0.0002     | 0.4620        |
|                     | SNP 5%       | 0.7878              | 0.0242                   | 0.0003     | 0.4322        |
|                     | SNP 10%      | 0.6281              | 0.0182                   | 0.0003     | 0.3922        |
|                     | Rev. Comp.   | <b>1.0000</b>       | 0.0201                   | 0.0000     | <b>0.4852</b> |
| Cad-7.7M (131k ctx) | SNP 1%       | 0.9511              | 0.0108                   | 0.0001     | 0.4829        |
|                     | SNP 2%       | 0.9054              | 0.0058                   | 0.0001     | 0.4715        |
|                     | SNP 5%       | 0.7839              | 0.0096                   | 0.0001     | 0.4411        |
|                     | SNP 10%      | 0.6247              | 0.0226                   | 0.0002     | 0.4013        |
|                     | Rev. Comp.   | <b>0.9999</b>       | 0.0123                   | 0.0000     | <b>0.4951</b> |

*Note:* On synthetic sequences devoid of biological distribution biases, Caduceus maintains RC RDM  $\geq 0.9999$  at all scales. SNP stability is predictably lower on the harder synthetic distribution (e.g., 0.952 vs. 0.997 at SNP 1%), but the global topology remains fully anchored. Geometric stability is constant across parameter and context scales, demonstrating that the continuous ODE prior mitigates the tokenization bottleneck.

## Real chr22 DNA.

Table S13: Caduceus real chr22 DNA results. RC RDM is exactly 1.0000 across all model sizes, confirming native preservation of this continuous global symmetry on realistic genomic sequences.

| Model               | Perturbation | RDM Sim. $\uparrow$ | Pert. Stab. $\downarrow$ | Pert. Mag. | Composite     |
|---------------------|--------------|---------------------|--------------------------|------------|---------------|
| Cad-0.5M (1k ctx)   | SNP 1%       | 0.9972              | 0.0022                   | 0.0062     | 0.4950        |
|                     | SNP 2%       | 0.9935              | 0.0312                   | 0.0116     | 0.4941        |
|                     | SNP 5%       | 0.9810              | 0.1499                   | 0.0245     | 0.4910        |
|                     | SNP 10%      | 0.9622              | 0.3037                   | 0.0382     | 0.4863        |
|                     | Rev. Comp.   | <b>1.0000</b>       | 0.0115                   | 0.0000     | <b>0.4957</b> |
| Cad-1.9M (1k ctx)   | SNP 1%       | 0.9978              | 0.0549                   | 0.0060     | 0.4828        |
|                     | SNP 2%       | 0.9955              | 0.0803                   | 0.0119     | 0.4822        |
|                     | SNP 5%       | 0.9869              | 0.1790                   | 0.0307     | 0.4801        |
|                     | SNP 10%      | 0.9658              | 0.4037                   | 0.0657     | 0.4748        |
|                     | Rev. Comp.   | <b>1.0000</b>       | 0.0266                   | 0.0000     | <b>0.4834</b> |
| Cad-7.7M (131k ctx) | SNP 1%       | 0.9974              | 0.0561                   | 0.0013     | 0.4973        |
|                     | SNP 2%       | 0.9946              | 0.0841                   | 0.0025     | 0.4966        |
|                     | SNP 5%       | 0.9851              | 0.1395                   | 0.0064     | 0.4943        |
|                     | SNP 10%      | 0.9654              | 0.2287                   | 0.0130     | 0.4893        |
|                     | Rev. Comp.   | <b>1.0000</b>       | 0.0469                   | 0.0000     | <b>0.4980</b> |

*Note:* On real genomic DNA, SNP RDM rises substantially (0.997 vs. 0.951 at 1%), consistent with real sequences occupying a narrower region of embedding space than synthetics. RC RDM is exactly 1.0000 at all scales, confirming that Caduceus natively preserves this symmetry independent of sequence composition. Composite stability is higher and more uniform than on synthetic data, reflecting the in-distribution advantage.

#### F.4 Evo 2

A single 7B-parameter Evo 2 [11, StripedHyena 2 Multi-Hybrid] model evaluated at three context-window checkpoints: 8K, 262K, and 1M tokens. Single-character tokenization (vocabulary size 4). The RC dissociation is the defining result: catastrophic failure on synthetic DNA (RDM 0.139–0.208) contrasts sharply with apparent success on real genomic DNA (RDM 0.873–0.889), explained by the Texture Hypothesis Test (Section 3, Appendix J).

##### Synthetic DNA.

Table S14: Evo 2 (7B) synthetic DNA results across context lengths. Modest SNP stability gains with  $128\times$  more context; catastrophic RC failure at all scales.

| Context | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|---------|--------------|----------|-------------|------------|-----------|
| 8K      | snp_1pct     | 0.747    | 0.019       | 13.610     | 0.439     |
|         | snp_2pct     | 0.604    | 0.059       | 21.445     | 0.403     |
|         | snp_5pct     | 0.392    | 0.020       | 29.104     | 0.350     |
|         | snp_10pct    | 0.269    | 0.030       | 22.807     | 0.319     |
|         | reverse_comp | 0.139    | 0.011       | 34.270     | 0.287     |
| 262K    | snp_1pct     | 0.776    | −0.022      | 15.131     | 0.444     |
|         | snp_2pct     | 0.650    | 0.053       | 26.729     | 0.412     |
|         | snp_5pct     | 0.431    | −0.024      | 23.674     | 0.357     |
|         | snp_10pct    | 0.288    | −0.009      | 22.676     | 0.322     |
|         | reverse_comp | 0.156    | 0.022       | 45.358     | 0.289     |
| 1M      | snp_1pct     | 0.817    | 0.031       | 14.392     | 0.454     |
|         | snp_2pct     | 0.709    | −0.057      | 42.035     | 0.427     |
|         | snp_5pct     | 0.513    | 0.011       | 37.208     | 0.378     |
|         | snp_10pct    | 0.363    | −0.040      | 34.243     | 0.341     |
|         | reverse_comp | 0.208    | −0.005      | 53.764     | 0.302     |

##### Real chr22 DNA.

Table S15: Evo 2 (7B) real chr22 results. High RC RDM (0.879–0.889) contrasts with synthetic RC RDM (0.139–0.208), demonstrating the RC dissociation explained by the Texture Hypothesis.

| Context | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|---------|--------------|----------|-------------|------------|-----------|
| 8K      | snp_1pct     | 0.990    | −0.030      | 142.643    | 0.497     |
|         | snp_5pct     | 0.946    | 0.090       | 793.222    | 0.486     |
|         | snp_10pct    | 0.874    | 0.276       | 1786.473   | 0.468     |
|         | reverse_comp | 0.879    | 0.003       | 53.054     | 0.469     |
| 262K    | snp_1pct     | 0.991    | −0.024      | 122.845    | 0.497     |
|         | snp_5pct     | 0.948    | 0.085       | 717.415    | 0.486     |
|         | snp_10pct    | 0.872    | 0.285       | 1658.393   | 0.468     |
|         | reverse_comp | 0.883    | 0.010       | 48.843     | 0.470     |
| 1M      | snp_1pct     | 0.993    | −0.030      | 120.590    | 0.498     |
|         | snp_5pct     | 0.952    | 0.081       | 664.056    | 0.488     |
|         | snp_10pct    | 0.883    | 0.283       | 1568.335   | 0.470     |
|         | reverse_comp | 0.889    | 0.008       | 45.882     | 0.472     |

## F.5 HyenaDNA

Four sizes: tiny (0.4M, d=128), tiny-d256 (1.6M, d=256), small (3.3M), medium (6.6M), and large (6.6M, 1M context) for HyenaDNA [55, Implicit Long Convolution]. Single-character tokenization. HyenaDNA uses the Hyena operator (implicit long convolutions) rather than standard attention or SSMs, representing a distinct architectural class.

### Synthetic DNA.

Table S16: HyenaDNA synthetic DNA results. Moderate SNP stability with notable RC sensitivity that increases with model size.

| Model        | Size (M) | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|--------------|----------|--------------|----------|-------------|------------|-----------|
| Hyena-tiny   | 0.4      | snp_1pct     | 0.953    | 0.021       | 0.000      | 0.442     |
|              |          | snp_5pct     | 0.784    | 0.025       | 0.001      | 0.399     |
|              |          | snp_10pct    | 0.608    | 0.025       | 0.001      | 0.356     |
|              |          | reverse_comp | 0.270    | 0.029       | 0.001      | 0.271     |
| Hyena-small  | 3.3      | snp_1pct     | 0.992    | 0.026       | 0.003      | 0.492     |
|              |          | snp_5pct     | 0.965    | 0.063       | 0.014      | 0.485     |
|              |          | snp_10pct    | 0.931    | 0.113       | 0.028      | 0.476     |
|              |          | reverse_comp | 0.859    | 0.075       | 0.022      | 0.458     |
| Hyena-medium | 6.6      | snp_1pct     | 0.993    | 0.029       | 0.004      | 0.495     |
|              |          | snp_5pct     | 0.964    | 0.056       | 0.020      | 0.487     |
|              |          | snp_10pct    | 0.928    | 0.102       | 0.038      | 0.478     |
|              |          | reverse_comp | 0.778    | 0.074       | 0.047      | 0.441     |

### Real chr22 DNA.

Table S17: HyenaDNA real chr22 DNA results.

| Model           | Size (M) | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|-----------------|----------|--------------|----------|-------------|------------|-----------|
| Hyena-tiny      | 0.4      | snp_1pct     | 0.996    | 0.027       | 0.003      | 0.491     |
|                 |          | snp_5pct     | 0.980    | 0.095       | 0.015      | 0.486     |
|                 |          | snp_10pct    | 0.956    | 0.197       | 0.029      | 0.481     |
|                 |          | reverse_comp | 0.775    | 0.064       | 0.013      | 0.435     |
| Hyena-tiny-d256 | 1.6      | snp_1pct     | 0.996    | 0.027       | 0.004      | 0.495     |
|                 |          | snp_5pct     | 0.982    | 0.082       | 0.019      | 0.491     |
|                 |          | snp_10pct    | 0.961    | 0.157       | 0.037      | 0.486     |
|                 |          | reverse_comp | 0.831    | 0.050       | 0.019      | 0.454     |
| Hyena-large-1M  | 6.6      | snp_1pct     | 0.996    | −0.001      | 0.003      | 0.499     |
|                 |          | snp_5pct     | 0.979    | 0.007       | 0.017      | 0.495     |
|                 |          | snp_10pct    | 0.954    | 0.037       | 0.031      | 0.489     |
|                 |          | reverse_comp | 0.446    | 0.019       | 0.037      | 0.362     |

## F.6 DNABERT and DNABERT-2

DNABERT v1 [41, k-mer Transformer] is tested across four k-mer variants (3-mer through 6-mer), all approximately 86–90M parameters but with different tokenization granularities. DNABERT-2 [82, BPE Transformer] (117M) uses BPE tokenization, providing a third tokenization strategy for the DNA Transformer comparison.

### Synthetic DNA.

Table S18: DNABERT v1 synthetic DNA results across k-mer variants. Models are near-identical in size (86–89M); differences reflect tokenization granularity.

| Model                | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|----------------------|--------------|----------|-------------|------------|-----------|
| DNABERT-3mer (86.1M) | snp_1pct     | 0.910    | 0.013       | 0.014      | 0.473     |
|                      | snp_2pct     | 0.835    | 0.009       | 0.024      | 0.454     |
|                      | snp_5pct     | 0.669    | 0.023       | 0.025      | 0.412     |
|                      | snp_10pct    | 0.458    | 0.012       | 0.040      | 0.359     |
|                      | reverse_comp | 0.733    | 0.020       | 0.044      | 0.428     |
| DNABERT-4mer (86.2M) | snp_1pct     | 0.925    | 0.015       | 0.013      | 0.475     |
|                      | snp_2pct     | 0.855    | 0.018       | 0.015      | 0.457     |
|                      | snp_5pct     | 0.691    | 0.018       | 0.025      | 0.416     |
|                      | snp_10pct    | 0.487    | 0.023       | 0.031      | 0.365     |
|                      | reverse_comp | 0.783    | 0.025       | 0.039      | 0.439     |
| DNABERT-5mer (86.8M) | snp_1pct     | 0.921    | 0.020       | 0.009      | 0.473     |
|                      | snp_2pct     | 0.852    | 0.021       | 0.014      | 0.456     |
|                      | snp_5pct     | 0.686    | 0.021       | 0.022      | 0.414     |
|                      | snp_10pct    | 0.492    | 0.022       | 0.031      | 0.366     |
|                      | reverse_comp | 0.792    | 0.024       | 0.033      | 0.441     |
| DNABERT-6mer (89.2M) | snp_1pct     | 0.929    | 0.021       | 0.008      | 0.471     |
|                      | snp_2pct     | 0.862    | 0.022       | 0.012      | 0.454     |
|                      | snp_5pct     | 0.705    | 0.020       | 0.019      | 0.415     |
|                      | snp_10pct    | 0.500    | 0.026       | 0.027      | 0.363     |
|                      | reverse_comp | 0.849    | 0.031       | 0.036      | 0.451     |

Table S19: DNABERT-2 (BPE, 117M) synthetic DNA results. BPE tokenization produces the lowest geometric coherence in the study.

| Model            | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|------------------|--------------|----------|-------------|------------|-----------|
| DNABERT-2 (117M) | snp_1pct     | 0.449    | 0.036       | 0.044      | 0.305     |
|                  | snp_2pct     | 0.387    | 0.036       | 0.049      | 0.289     |
|                  | snp_5pct     | 0.283    | 0.036       | 0.048      | 0.263     |
|                  | snp_10pct    | 0.213    | 0.035       | 0.054      | 0.246     |
|                  | reverse_comp | 0.138    | 0.045       | 0.069      | 0.227     |

## Real chr22 DNA.

Table S20: DNABERT v1 real chr22 DNA results across k-mer variants. SNP RDM is uniformly high ( $> 0.95$  at 10%); RC sensitivity varies with token granularity.

| Model                | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|----------------------|--------------|----------|-------------|------------|-----------|
| DNABERT-3mer (86.1M) | snp_1pct     | 0.997    | 0.020       | 0.155      | 0.493     |
|                      | snp_2pct     | 0.994    | 0.052       | 0.306      | 0.492     |
|                      | snp_5pct     | 0.983    | 0.152       | 0.777      | 0.490     |
|                      | snp_10pct    | 0.951    | 0.323       | 1.643      | 0.482     |
|                      | reverse_comp | 0.977    | 0.076       | 0.462      | 0.488     |
| DNABERT-4mer (86.2M) | snp_1pct     | 0.997    | 0.028       | 0.152      | 0.487     |
|                      | snp_2pct     | 0.994    | 0.066       | 0.302      | 0.487     |
|                      | snp_5pct     | 0.983    | 0.169       | 0.779      | 0.484     |
|                      | snp_10pct    | 0.949    | 0.344       | 1.678      | 0.475     |
|                      | reverse_comp | 0.940    | 0.127       | 0.568      | 0.473     |
| DNABERT-5mer (86.8M) | snp_1pct     | 0.997    | 0.002       | 0.108      | 0.494     |
|                      | snp_2pct     | 0.994    | 0.040       | 0.214      | 0.493     |
|                      | snp_5pct     | 0.982    | 0.140       | 0.533      | 0.490     |
|                      | snp_10pct    | 0.957    | 0.298       | 1.099      | 0.483     |
|                      | reverse_comp | 0.948    | 0.139       | 0.511      | 0.481     |
| DNABERT-6mer (89.2M) | snp_1pct     | 0.998    | -0.001      | 0.089      | 0.494     |
|                      | snp_2pct     | 0.995    | 0.026       | 0.174      | 0.494     |
|                      | snp_5pct     | 0.986    | 0.111       | 0.434      | 0.491     |
|                      | snp_10pct    | 0.960    | 0.261       | 0.897      | 0.485     |
|                      | reverse_comp | 0.962    | 0.124       | 0.508      | 0.485     |

Table S21: DNABERT-2 (BPE, 117M) real chr22 DNA results. Geometric stability is substantially lower than DNABERT v1 k-mer variants at comparable model size.

| Model            | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|------------------|--------------|----------|-------------|------------|-----------|
| DNABERT-2 (117M) | snp_1pct     | 0.540    | 0.034       | 0.043      | 0.357     |
|                  | snp_2pct     | 0.473    | 0.026       | 0.075      | 0.340     |
|                  | snp_5pct     | 0.349    | 0.043       | 0.136      | 0.309     |
|                  | snp_10pct    | 0.280    | 0.080       | 0.221      | 0.292     |
|                  | reverse_comp | 0.377    | 0.059       | 0.165      | 0.316     |

## E.7 GPN

Two variants: GPN-Brassicales (66M) and PhyloGPN (83M). GPN [7, Convolutional] uses dilated convolutions rather than attention or recurrence, representing yet another architectural class in the model zoo.

Table S22: GPN results. GPN-Brassicales on synthetic DNA; PhyloGPN on real chr22.

| Model      | Size (M) | Data       | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|------------|----------|------------|--------------|----------|-------------|------------|-----------|
| GPN-Brass. | 65.6     | Synthetic  | snp_1pct     | 0.997    | 0.023       | 0.076      | 0.498     |
|            |          |            | snp_5pct     | 0.982    | 0.041       | 0.372      | 0.494     |
|            |          |            | snp_10pct    | 0.959    | 0.074       | 0.723      | 0.488     |
|            |          |            | reverse_comp | 0.957    | 0.067       | 0.586      | 0.488     |
| PhyloGPN   | 83.2     | Real chr22 | snp_1pct     | 0.952    | −0.028      | 0.310      | 0.482     |
|            |          |            | snp_5pct     | 0.815    | −0.024      | 1.583      | 0.448     |
|            |          |            | snp_10pct    | 0.640    | −0.056      | 3.597      | 0.404     |
|            |          |            | reverse_comp | 0.988    | 0.018       | 0.478      | 0.491     |

## F.8 SaProt

Three sizes: 35M, 650M, and 1.3B. SaProt [69, Structure-Aware Protein Transformer] extends ESM-2 by incorporating Foldseek 3Di structural alphabet tokens alongside amino acid tokens. For sequence-only input (without structure predictions), “#” is used as the structure token for every position. The key question: does structural information rescue the geometric tax? The answer is no. Composite stability degrades with scale (0.436  $\rightarrow$  0.398 on synthetic), paralleling the ESM-2 trend at matched sizes.

### Synthetic proteins.

Table S23: SaProt synthetic protein results. Structure-awareness does not rescue the geometric tax; the scaling trend matches ESM-2 at corresponding sizes.

| Model       | Size (M) | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|-------------|----------|--------------|----------|-------------|------------|-----------|
| SaProt-35M  | 33.9     | aa_sub_1pct  | 0.962    | 0.015       | 0.003      | 0.465     |
|             |          | aa_sub_2pct  | 0.927    | 0.021       | 0.005      | 0.456     |
|             |          | aa_sub_5pct  | 0.825    | 0.021       | 0.005      | 0.431     |
|             |          | aa_sub_10pct | 0.691    | 0.014       | 0.008      | 0.397     |
|             |          | reverse      | 0.818    | 0.017       | 0.007      | 0.429     |
| SaProt-650M | 652.1    | aa_sub_1pct  | 0.936    | 0.020       | 0.006      | 0.475     |
|             |          | aa_sub_2pct  | 0.877    | 0.016       | 0.008      | 0.461     |
|             |          | aa_sub_5pct  | 0.750    | 0.018       | 0.010      | 0.429     |
|             |          | aa_sub_10pct | 0.601    | 0.022       | 0.015      | 0.392     |
|             |          | reverse      | 0.624    | 0.015       | 0.014      | 0.398     |
| SaProt-1.3B | 1300.9   | aa_sub_1pct  | 0.836    | 0.022       | 0.009      | 0.450     |
|             |          | aa_sub_2pct  | 0.754    | 0.024       | 0.013      | 0.430     |
|             |          | aa_sub_5pct  | 0.609    | 0.030       | 0.014      | 0.394     |
|             |          | aa_sub_10pct | 0.458    | 0.024       | 0.017      | 0.356     |
|             |          | reverse      | 0.478    | 0.026       | 0.017      | 0.361     |

### Real UniRef50 proteins.

Table S24: SaProt real UniRef50 results. The scaling trend holds on natural proteins.

| Model       | Size (M) | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|-------------|----------|--------------|----------|-------------|------------|-----------|
| SaProt-35M  | 33.9     | aa_sub_1pct  | 0.976    | 0.027       | 0.003      | 0.473     |
|             |          | aa_sub_5pct  | 0.864    | 0.032       | 0.005      | 0.446     |
|             |          | aa_sub_10pct | 0.751    | 0.035       | 0.008      | 0.417     |
|             |          | reverse      | 0.848    | 0.021       | 0.006      | 0.442     |
| SaProt-650M | 652.1    | aa_sub_1pct  | 0.954    | 0.024       | 0.005      | 0.480     |
|             |          | aa_sub_5pct  | 0.786    | 0.023       | 0.011      | 0.439     |
|             |          | aa_sub_10pct | 0.645    | 0.025       | 0.014      | 0.404     |
|             |          | reverse      | 0.680    | 0.029       | 0.013      | 0.412     |
| SaProt-1.3B | 1300.9   | aa_sub_1pct  | 0.878    | 0.029       | 0.008      | 0.462     |
|             |          | aa_sub_5pct  | 0.650    | 0.024       | 0.013      | 0.405     |
|             |          | aa_sub_10pct | 0.519    | 0.030       | 0.019      | 0.372     |
|             |          | reverse      | 0.513    | 0.032       | 0.020      | 0.371     |

## E.9 ProtMamba

A single 108M-parameter Mamba-based protein language model. Since only one model size exists, the scaling axis is sequence length (100–800 aa). ProtMamba [66, Protein SSM] is the canonical Geometric Vacuity model: low Procrustes distortion and reasonable autoregressive perplexity (40.1), but MINE reveals negative excess MI at every sequence length (Regime III, Section 5).

### Synthetic proteins.

Table S25: ProtMamba synthetic protein results across sequence lengths (100–800 aa). Perturbation magnitude in scientific notation due to unnormalized embedding scale.

| Seq. Length | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|-------------|--------------|----------|-------------|------------|-----------|
| L100        | aa_sub_1pct  | 0.884    | 0.278       | 1.16e+05   | 0.468     |
|             | aa_sub_2pct  | 0.776    | 0.175       | 1.16e+05   | 0.446     |
|             | aa_sub_5pct  | 0.657    | 0.098       | 1.73e+05   | 0.413     |
|             | aa_sub_10pct | 0.489    | −0.242      | 3.14e+05   | 0.371     |
|             | reverse      | 0.020    | −0.057      | 3.92e+05   | 0.255     |
| L200        | aa_sub_1pct  | 0.844    | −0.011      | 9.48e+04   | 0.462     |
|             | aa_sub_2pct  | 0.766    | 0.193       | 1.58e+05   | 0.438     |
|             | aa_sub_5pct  | 0.628    | 0.071       | 1.05e+05   | 0.406     |
|             | aa_sub_10pct | 0.475    | 0.041       | 2.40e+05   | 0.371     |
|             | reverse      | 0.012    | 0.016       | 3.64e+05   | 0.251     |
| L300        | aa_sub_1pct  | 0.852    | −0.064      | 9.80e+04   | 0.461     |
|             | aa_sub_2pct  | 0.791    | −0.032      | 1.41e+05   | 0.446     |
|             | aa_sub_5pct  | 0.629    | 0.098       | 1.55e+05   | 0.406     |
|             | aa_sub_10pct | 0.502    | −0.338      | 4.16e+05   | 0.373     |
|             | reverse      | 0.010    | 0.231       | 4.94e+05   | 0.250     |
| L400        | aa_sub_1pct  | 0.854    | −0.047      | 9.64e+04   | 0.460     |
|             | aa_sub_2pct  | 0.738    | −0.035      | 1.34e+05   | 0.435     |
|             | aa_sub_5pct  | 0.666    | −0.303      | 1.15e+05   | 0.415     |
|             | aa_sub_10pct | 0.540    | −0.052      | 2.62e+05   | 0.385     |
|             | reverse      | 0.028    | 0.358       | 6.20e+05   | 0.252     |
| L500        | aa_sub_1pct  | 0.858    | 0.371       | 2.29e+05   | 0.463     |
|             | aa_sub_2pct  | 0.763    | 0.082       | 1.38e+05   | 0.441     |
|             | aa_sub_5pct  | 0.594    | 0.371       | 6.30e+05   | 0.393     |
|             | aa_sub_10pct | 0.499    | 0.374       | 3.14e+05   | 0.376     |
|             | reverse      | −0.022   | 0.367       | 6.80e+05   | 0.244     |
| L600        | aa_sub_1pct  | 0.843    | −0.015      | 1.40e+05   | 0.455     |
|             | aa_sub_2pct  | 0.789    | −0.265      | 2.35e+05   | 0.449     |
|             | aa_sub_5pct  | 0.641    | −0.166      | 1.71e+05   | 0.411     |
|             | aa_sub_10pct | 0.504    | −0.072      | 4.65e+05   | 0.379     |
|             | reverse      | 0.018    | −0.289      | 8.27e+05   | 0.250     |
| L700        | aa_sub_1pct  | 0.820    | 0.327       | 1.31e+05   | 0.458     |
|             | aa_sub_2pct  | 0.723    | −0.299      | 1.56e+05   | 0.428     |
|             | aa_sub_5pct  | 0.598    | −0.304      | 3.59e+05   | 0.401     |
|             | aa_sub_10pct | 0.471    | −0.064      | 3.07e+05   | 0.369     |
|             | reverse      | 0.010    | 0.348       | 5.82e+05   | 0.246     |
| L800        | aa_sub_1pct  | 0.815    | −0.278      | 1.41e+05   | 0.453     |
|             | aa_sub_2pct  | 0.715    | −0.326      | 3.05e+05   | 0.430     |
|             | aa_sub_5pct  | 0.667    | 0.238       | 1.81e+05   | 0.416     |
|             | aa_sub_10pct | 0.517    | 0.397       | 3.17e+05   | 0.378     |
|             | reverse      | −0.024   | −0.316      | 1.07e+06   | 0.241     |

# Real UniRef50 proteins.

Table S26: ProtMamba real UniRef50 protein results across length bins (50–800 aa). Perturbation magnitude in scientific notation.

| Length Bin | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|------------|--------------|----------|-------------|------------|-----------|
| 50–150     | aa_sub_1pct  | 0.873    | −0.121      | 8.52e+04   | 0.467     |
|            | aa_sub_2pct  | 0.783    | −0.089      | 1.08e+05   | 0.445     |
|            | aa_sub_5pct  | 0.617    | −0.132      | 1.57e+05   | 0.402     |
|            | aa_sub_10pct | 0.446    | −0.778      | 4.06e+05   | 0.361     |
|            | reverse      | −0.014   | −0.286      | 6.04e+06   | 0.246     |
| 151–250    | aa_sub_1pct  | 0.879    | 0.101       | 1.02e+05   | 0.470     |
|            | aa_sub_2pct  | 0.776    | −0.583      | 1.76e+05   | 0.445     |
|            | aa_sub_5pct  | 0.611    | −0.753      | 2.17e+05   | 0.403     |
|            | aa_sub_10pct | 0.459    | −0.759      | 3.88e+05   | 0.363     |
|            | reverse      | 0.002    | −0.228      | 6.62e+06   | 0.250     |
| 251–350    | aa_sub_1pct  | 0.875    | −0.469      | 8.03e+04   | 0.469     |
|            | aa_sub_2pct  | 0.786    | 0.307       | 1.08e+05   | 0.446     |
|            | aa_sub_5pct  | 0.610    | −0.151      | 1.94e+05   | 0.400     |
|            | aa_sub_10pct | 0.476    | −0.655      | 2.79e+05   | 0.370     |
|            | reverse      | 0.014    | −0.260      | 6.20e+06   | 0.253     |
| 351–450    | aa_sub_1pct  | 0.866    | 0.235       | 1.05e+05   | 0.466     |
|            | aa_sub_2pct  | 0.760    | −0.287      | 1.05e+05   | 0.439     |
|            | aa_sub_5pct  | 0.589    | −0.498      | 2.15e+05   | 0.396     |
|            | aa_sub_10pct | 0.433    | −0.676      | 3.60e+05   | 0.358     |
|            | reverse      | 0.025    | −0.297      | 6.20e+06   | 0.256     |
| 451–550    | aa_sub_1pct  | 0.871    | 0.344       | 8.95e+04   | 0.466     |
|            | aa_sub_2pct  | 0.772    | 0.404       | 1.28e+05   | 0.443     |
|            | aa_sub_5pct  | 0.592    | −0.519      | 1.51e+05   | 0.395     |
|            | aa_sub_10pct | 0.449    | −0.608      | 2.68e+05   | 0.363     |
|            | reverse      | −0.010   | −0.230      | 6.83e+06   | 0.248     |
| 551–650    | aa_sub_1pct  | 0.854    | 0.428       | 8.68e+04   | 0.464     |
|            | aa_sub_2pct  | 0.752    | −0.429      | 1.46e+05   | 0.436     |
|            | aa_sub_5pct  | 0.579    | −0.315      | 1.12e+05   | 0.393     |
|            | aa_sub_10pct | 0.421    | −0.799      | 2.26e+05   | 0.355     |
|            | reverse      | 0.004    | −0.336      | 5.51e+06   | 0.250     |
| 651–800    | aa_sub_1pct  | 0.843    | −0.014      | 7.42e+04   | 0.460     |
|            | aa_sub_2pct  | 0.748    | −0.503      | 2.28e+05   | 0.436     |
|            | aa_sub_5pct  | 0.560    | −0.667      | 1.74e+05   | 0.390     |
|            | aa_sub_10pct | 0.423    | −0.751      | 2.22e+05   | 0.357     |
|            | reverse      | −0.039   | −0.245      | 6.18e+06   | 0.238     |

## **F.10 OpenFold**

OpenFold [2, Evoformer] SoloSeq mode (sequence-only, no MSA). Since OpenFold has a single model size, the scaling axis is sequence length (100–800 aa). ESM-1b embeddings are fed into the Evoformer trunk. Composite stability improves modestly with sequence length (0.369 at  $L=100$  to 0.420 at  $L=800$ ), consistent with the Evoformer’s ability to build longer-range structural contacts. OpenFold is the canonical Regime II model (Representational Compression): high MINE excess MI (+10.8 to +11.6 nats) but geometric warping (Procrustes disparity 0.15–0.16 relative to the ESM-1b input).

## Synthetic proteins.

Table S27: OpenFold (SoloSeq) synthetic protein results across sequence lengths (100–800 aa). Stability improves with sequence length as the Evoformer builds longer-range structural contacts.

| Seq. Length | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|-------------|--------------|----------|-------------|------------|-----------|
| L100        | aa_sub_1pct  | 0.824    | 0.034       | 0.019      | 0.441     |
|             | aa_sub_2pct  | 0.713    | 0.036       | 0.028      | 0.413     |
|             | aa_sub_5pct  | 0.534    | 0.056       | 0.042      | 0.368     |
|             | aa_sub_10pct | 0.352    | 0.053       | 0.036      | 0.323     |
|             | reverse      | 0.253    | 0.042       | 0.045      | 0.298     |
| L200        | aa_sub_1pct  | 0.847    | 0.020       | 0.015      | 0.450     |
|             | aa_sub_2pct  | 0.742    | 0.033       | 0.021      | 0.424     |
|             | aa_sub_5pct  | 0.565    | 0.041       | 0.026      | 0.380     |
|             | aa_sub_10pct | 0.401    | 0.037       | 0.030      | 0.339     |
|             | reverse      | 0.314    | 0.027       | 0.038      | 0.317     |
| L300        | aa_sub_1pct  | 0.865    | 0.025       | 0.012      | 0.456     |
|             | aa_sub_2pct  | 0.800    | 0.017       | 0.019      | 0.440     |
|             | aa_sub_5pct  | 0.621    | 0.030       | 0.023      | 0.395     |
|             | aa_sub_10pct | 0.459    | 0.025       | 0.029      | 0.355     |
|             | reverse      | 0.404    | 0.032       | 0.033      | 0.341     |
| L400        | aa_sub_1pct  | 0.893    | 0.029       | 0.010      | 0.463     |
|             | aa_sub_2pct  | 0.819    | 0.028       | 0.015      | 0.444     |
|             | aa_sub_5pct  | 0.662    | 0.041       | 0.024      | 0.405     |
|             | aa_sub_10pct | 0.530    | 0.028       | 0.025      | 0.372     |
|             | reverse      | 0.444    | 0.028       | 0.030      | 0.350     |
| L500        | aa_sub_1pct  | 0.909    | 0.024       | 0.010      | 0.467     |
|             | aa_sub_2pct  | 0.841    | 0.021       | 0.013      | 0.450     |
|             | aa_sub_5pct  | 0.679    | 0.043       | 0.021      | 0.410     |
|             | aa_sub_10pct | 0.522    | 0.034       | 0.023      | 0.370     |
|             | reverse      | 0.493    | 0.038       | 0.023      | 0.363     |
| L600        | aa_sub_1pct  | 0.913    | 0.013       | 0.009      | 0.468     |
|             | aa_sub_2pct  | 0.853    | 0.016       | 0.011      | 0.453     |
|             | aa_sub_5pct  | 0.705    | 0.025       | 0.015      | 0.415     |
|             | aa_sub_10pct | 0.546    | 0.013       | 0.020      | 0.376     |
|             | reverse      | 0.521    | 0.024       | 0.023      | 0.369     |
| L700        | aa_sub_1pct  | 0.921    | 0.028       | 0.008      | 0.471     |
|             | aa_sub_2pct  | 0.859    | 0.035       | 0.010      | 0.455     |
|             | aa_sub_5pct  | 0.709    | 0.035       | 0.017      | 0.418     |
|             | aa_sub_10pct | 0.567    | 0.021       | 0.023      | 0.382     |
|             | reverse      | 0.521    | 0.028       | 0.021      | 0.371     |
| L800        | aa_sub_1pct  | 0.926    | 0.036       | 0.007      | 0.471     |
|             | aa_sub_2pct  | 0.858    | 0.033       | 0.010      | 0.454     |
|             | aa_sub_5pct  | 0.722    | 0.035       | 0.012      | 0.420     |
|             | aa_sub_10pct | 0.574    | 0.037       | 0.021      | 0.383     |
|             | reverse      | 0.536    | 0.055       | 0.024      | 0.373     |

# Real UniRef50 proteins.

Table S28: OpenFold (SoloSeq) real UniRef50 protein results across length bins (50–800 aa).  $n$  indicates number of sequences per bin.

| Length Bin | $n$   | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|------------|-------|--------------|----------|-------------|------------|-----------|
| 50–150     | 80198 | aa_sub_1pct  | 0.996    | 0.013       | 0.066      | 0.436     |
|            |       | aa_sub_2pct  | 0.995    | 0.017       | 0.100      | 0.435     |
|            |       | aa_sub_5pct  | 0.983    | 0.060       | 0.298      | 0.432     |
|            |       | aa_sub_10pct | 0.948    | 0.145       | 0.648      | 0.423     |
|            |       | reverse      | 0.394    | 0.847       | 4.155      | 0.285     |
| 151–250    | 90351 | aa_sub_1pct  | 0.999    | 0.017       | 0.051      | 0.431     |
|            |       | aa_sub_2pct  | 0.996    | 0.027       | 0.122      | 0.431     |
|            |       | aa_sub_5pct  | 0.987    | 0.081       | 0.333      | 0.428     |
|            |       | aa_sub_10pct | 0.964    | 0.183       | 0.693      | 0.423     |
|            |       | reverse      | 0.375    | 0.903       | 4.825      | 0.275     |
| 251–350    | 80686 | aa_sub_1pct  | 0.999    | 0.021       | 0.061      | 0.443     |
|            |       | aa_sub_2pct  | 0.997    | 0.033       | 0.136      | 0.442     |
|            |       | aa_sub_5pct  | 0.991    | 0.092       | 0.356      | 0.441     |
|            |       | aa_sub_10pct | 0.975    | 0.207       | 0.734      | 0.437     |
|            |       | reverse      | 0.476    | 0.920       | 5.111      | 0.312     |
| 351–450    | 64172 | aa_sub_1pct  | 0.999    | 0.018       | 0.073      | 0.452     |
|            |       | aa_sub_2pct  | 0.998    | 0.042       | 0.158      | 0.451     |
|            |       | aa_sub_5pct  | 0.991    | 0.120       | 0.409      | 0.450     |
|            |       | aa_sub_10pct | 0.973    | 0.247       | 0.821      | 0.445     |
|            |       | reverse      | 0.490    | 0.932       | 5.366      | 0.324     |
| 451–550    | 40740 | aa_sub_1pct  | 0.999    | 0.022       | 0.079      | 0.444     |
|            |       | aa_sub_2pct  | 0.997    | 0.043       | 0.168      | 0.444     |
|            |       | aa_sub_5pct  | 0.989    | 0.117       | 0.428      | 0.442     |
|            |       | aa_sub_10pct | 0.965    | 0.247       | 0.856      | 0.436     |
|            |       | reverse      | 0.481    | 0.931       | 5.307      | 0.315     |
| 551–650    | 22264 | aa_sub_1pct  | 0.999    | 0.026       | 0.080      | 0.442     |
|            |       | aa_sub_2pct  | 0.997    | 0.043       | 0.168      | 0.441     |
|            |       | aa_sub_5pct  | 0.988    | 0.116       | 0.428      | 0.439     |
|            |       | aa_sub_10pct | 0.964    | 0.250       | 0.857      | 0.433     |
|            |       | reverse      | 0.506    | 0.932       | 5.223      | 0.319     |
| 651–800    | 17678 | aa_sub_1pct  | 0.999    | 0.027       | 0.083      | 0.434     |
|            |       | aa_sub_2pct  | 0.997    | 0.043       | 0.175      | 0.433     |
|            |       | aa_sub_5pct  | 0.988    | 0.115       | 0.443      | 0.431     |
|            |       | aa_sub_10pct | 0.965    | 0.248       | 0.885      | 0.425     |
|            |       | reverse      | 0.480    | 0.922       | 5.002      | 0.304     |

## F.11 Cross-Architecture Summary

Table S29 presents mean composite stability across all perturbation types for each model family, ordered by parameter count within each domain. The key patterns visible in this summary:

1. **Progressive tax in Transformers:** ESM-2, NT, and SaProt all exhibit declining composite with scale.
2. **SSM stability:** Caduceus composites are near-constant across scale (0.459–0.458) with perfect RC preservation.
3. **Architectural class matters less than tokenization:** The cross-architecture spread within the continuous regime (Track A) is  $1.3\times$ ; within the discrete regime (Track B), it spans three orders of magnitude in Lipschitz constants.
4. **Structure-awareness is insufficient:** SaProt degrades at the same rate as ESM-2 at matched sizes, confirming that Foldseek structural tokens do not escape the tax.
5. **The 2.5B/15B “rebound” is Untethered Gel:** Both NT-2.5B and ESM2-15B show improved composites, but Procrustes Ratio and Frozen Head tests confirm these reflect global manifold drift, not genuine geometric recovery.

Table S29: Cross-architecture summary: mean composite stability (averaged across all perturbation types) on synthetic sequences. Models ordered by parameter count within each domain.

| Domain  | Model         | Size (M) | Mean Composite | Regime            |
|---------|---------------|----------|----------------|-------------------|
| Protein | ESM2-8M       | 7.5      | 0.431          | Brittle Glass     |
|         | ESM2-35M      | 33.5     | 0.406          | Brittle Glass     |
|         | SaProt-35M    | 33.9     | 0.436          | Brittle Glass     |
|         | ProtMamba     | 107.7    | 0.381          | Geometric Vacuity |
|         | ESM2-150M     | 148.1    | 0.388          | Brittle Glass     |
|         | ESM2-650M     | 651.0    | 0.360          | Brittle Glass     |
|         | SaProt-650M   | 652.1    | 0.431          | Brittle Glass     |
|         | SaProt-1.3B   | 1300.9   | 0.398          | Brittle Glass     |
|         | ESM2-3B       | 2839.0   | 0.323          | Untethered Gel    |
|         | ESM2-15B      | 15129.1  | 0.378          | Untethered Gel    |
|         | OpenFold*     | 75       | 0.369–0.420    | Repr. Compression |
| DNA     | Hyena-tiny    | 0.4      | 0.367          | —                 |
|         | Caduceus-0.5M | 0.5      | 0.459          | Geometric Vacuity |
|         | Caduceus-1.9M | 1.9      | 0.449          | Geometric Vacuity |
|         | Hyena-small   | 3.3      | 0.475          | —                 |
|         | Hyena-medium  | 6.6      | 0.479          | —                 |
|         | Caduceus-7.7M | 7.7      | 0.458          | Geometric Vacuity |
|         | NT-50M        | 55.9     | 0.239          | Brittle Glass     |
|         | GPN-Brass.    | 65.6     | 0.494          | —                 |
|         | DNABERT-3mer  | 86.1     | 0.425          | Brittle Glass     |
|         | DNABERT-6mer  | 89.2     | 0.433          | Brittle Glass     |
|         | NT-100M       | 97.9     | 0.234          | Brittle Glass     |
|         | DNABERT-2     | 117.1    | 0.323          | Brittle Glass     |
|         | NT-250M       | 235.1    | 0.210          | Brittle Glass     |
|         | NT-500M       | 498.3    | 0.194          | Brittle Glass     |
|         | NT-2.5B       | 2547.8   | 0.336          | Untethered Gel    |
|         | Evo 2 (8K)    | 6481.1   | 0.360          | Untethered Gel    |
|         | Evo 2 (1M)    | 6581.7   | 0.380          | Untethered Gel    |

\*OpenFold

range reflects sequence length scaling ( $L=100$  to  $L=800$ ); single model size.

## F.12 Excluded models.

We attempted to evaluate two additional models via their respective inference APIs: Evo 2 (40B) through the NVIDIA NIM API and AlphaGenome [5] through its hosted endpoint. In both cases, the APIs return only final output predictions (next-token logits or task-specific scores) rather than intermediate layer representations. Because our evaluation framework requires access to internal embedding activations to compute Procrustes stability, RDM similarity, and MINE mutual information, neither model could be included in the geometric analysis. For Evo 2 40B specifically, the NIM API returned near-zero activations consistent with FP8 quantization artifacts, rendering the embeddings unsuitable even when partial access was available. We restrict the Evo 2 analysis to the 7B family (8K, 262K, and 1M context variants) run locally via the Vortex inference engine. We document these exclusions transparently and note that future work with direct weight access could extend the analysis to these models.

## G Ghost Detection and Phase Transition Extended Methods and Details

This appendix provides extended methods and per-model results for the Ghost Detection protocol (Appendix G.1) and the Brittle Glass / Untethered Gel phase transition proof (Appendix G.4), summarized in the main text (Sections 3- 4). All experiments use seed 320 and 5-fold stratified cross-validation where applicable.

### G.1 Ghost Detection Protocol

The Ghost Detection protocol tests whether a model’s apparent geometric stability reflects biologically grounded representations or a self-consistent manifold that is decoupled from downstream utility. The protocol comprises two complementary tests applied to each model.

**Test 1: Self-Procrustes Spin Test.** For each model configuration (parameter count, context length, or sequence length, depending on the scaling axis), we compute the Procrustes residual between clean and perturbed embedding matrices. Let  $\mathbf{X}_c, \mathbf{X}_p \in \mathbb{R}^{n \times d}$  denote the mean-centered clean and perturbed embeddings, respectively. We solve for the optimal orthogonal alignment:

$$\mathbf{R}^* = \arg \min_{\mathbf{R}^\top \mathbf{R} = \mathbf{I}} \left\| \hat{\mathbf{X}}_c - \hat{\mathbf{X}}_p \mathbf{R} \right\|_F^2, \quad (7)$$

where  $\hat{\mathbf{X}} = \mathbf{X} / \|\mathbf{X}\|_F$  denotes Frobenius-normalized matrices. The optimal rotation is obtained via the closed-form solution  $\mathbf{R}^* = \mathbf{V}\mathbf{U}^\top$  from the SVD  $\hat{\mathbf{X}}_p^\top \hat{\mathbf{X}}_c = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$  [28, 52, 61, 65]. After rotation, we compute an optimal isotropic scale

$$s^* = \text{tr}(\mathbf{X}_c^\top \mathbf{X}_p \mathbf{R}^*) / \text{tr}((\mathbf{X}_p \mathbf{R}^*)^\top \mathbf{X}_p \mathbf{R}^*) \quad (8)$$

and report two quantities:

**Procrustes Residual** The per-sample Frobenius distance after alignment:

$$\|\mathbf{X}_c - s^* \mathbf{X}_p \mathbf{R}^*\|_F / \sqrt{n}$$

**Procrustes Ratio** The fraction of raw error surviving alignment: `aligned_error/raw_error`, where

$$\text{raw\_error} = \|\mathbf{X}_c - \mathbf{X}_p\|_F / \sqrt{n}$$

A *Ghost* is diagnosed when the Procrustes residual *increases* with scale (the manifold “spins” more freely in the latent space at larger configurations) or remains high despite ostensibly stable RDM similarity. When embedding matrices exceed 10,000 samples, we subsample uniformly at random (seed 320) to  $n = 5,000$  (Evo 2) or  $n = 10,000$  (ProtMamba, OpenFold).

**Test 2: Frozen Head Context Tax.** We train a frozen linear classifier (logistic regression,  $C = 1.0$ , max 1,000 iterations) on model embeddings for a binary species-classification task that requires only *local* sequence information. The signal region is held constant while the total input length increases via random padding, creating a controlled dilution curve. If accuracy degrades with longer input despite the signal being unchanged, the model’s geometric stability does not translate to biologically useful representations.

For DNA models (Evo 2), we classify real human chr22 vs. real *S. cerevisiae* DNA using a 1 kbp signal region, with 200 sequences per class. For protein models (ProtMamba, OpenFold), we classify real *E. coli* vs. real human proteins (UniProt Swiss-Prot, reviewed) using a 50 aa signal region (first 50 residues of each protein), with 200 sequences per class. Random padding consists of i.i.d. uniform nucleotides (DNA) or amino acids (protein). We report 5-fold stratified cross-validated accuracy.

Two pooling strategies are tested: **local** (pool only the signal region of the embedding) and **global** (pool the full sequence). Local pooling isolates whether the model preserves signal-region geometry; global pooling tests whether padding tokens corrupt the entire representation.

**Embedding extraction.** Evo 2 embeddings are extracted from `layer.blocks.28.mlp.13` via local Vortex inference on A100 GPUs, with a safe sequence-length cap of 120,000 tokens (avoiding 32-bit index overflow in the FIR engine). ProtMamba embeddings are extracted from the final hidden layer of the 107.7M Mamba model. OpenFold embeddings are extracted from the Evoformer trunk output (single-sequence SoloSeq mode using ESM-1b as the sequence encoder), with a maximum sequence

length of 1,022 residues. In all cases, per-token hidden states are mean-pooled over the relevant region.

## G.2 Per-Model Ghost Detection Results

**Evo 2 (7B, StripedHyena 2).** The scaling axis is context-window length (8K, 262K, 1M) at fixed 7B parameters. The Spin Test reveals that the Procrustes ratio is remarkably stable across context windows for SNP perturbations: 0.963 (8K), 0.964 (262K), 0.963 (1M) at 1% SNP on synthetic data. Reverse complement shows slightly more variation but no systematic trend: ratio 0.744 (8K), 0.740 (262K), 0.732 (1M). The Ghost is therefore *neutral*: context-window scaling neither grounds nor detaches the manifold orientation.

The Frozen Head test confirms diminishing returns: accuracy at 8K context is comparable to accuracy at 1M context ( $\Delta = +0.005$ , effectively zero), meaning  $128\times$  more context yields no classification gain. On real chr22 DNA, SNP Procrustes ratios are even more stable (0.984–0.985 across all three checkpoints at 1% SNP), reflecting the lower perturbation magnitude on real genomic sequences.

Table S30: **Evo 2 Frozen Head test:** frozen linear classifier accuracy (Human chr22 vs. Yeast, 1 kbp signal, local pooling).

| Checkpoint   | Context   | Accuracy         | $\Delta$ from 8K |
|--------------|-----------|------------------|------------------|
| Evo2-7B-8K   | 8,192     | $0.850 \pm 0.02$ | –                |
| Evo2-7B-262K | 262,144   | $0.843 \pm 0.02$ | –0.008           |
| Evo2-7B-1M   | 1,048,576 | $0.855 \pm 0.02$ | +0.005           |

**Evo 2 Frozen Head (next-token prediction).** The phase transition proof extends the Frozen Head test to the model’s own next-token prediction head. Agreement between clean-input and perturbed-input top-1 predictions is stable across context windows: 85.3% (8K), 84.9% (262K), 85.2% (1M) at 1% SNP. KL divergence between output distributions is similarly flat (0.0024–0.0027 nats). Under reverse complement, agreement drops to  $\sim 28$ –30% across all checkpoints, confirming that the RC perturbation produces a categorically different output distribution regardless of context length.

**ProtMamba (108M, Mamba SSM).** The scaling axis is sequence length (100–800 aa) at fixed 108M parameters. The Spin Test on synthetic data shows Procrustes ratios declining from 0.950 (L=100) to 0.819 (L=800) at 10% substitution, indicating that longer sequences push the manifold toward Untethered Gel. However, the critical finding is the Frozen Head: accuracy is at *chance level* ( $\sim 50\%$ ) at every sequence length, including the shortest (L=100, 50% signal fraction). Both local and global pooling yield identical chance-level performance.

Table S31: **ProtMamba Ghost Detection:** Frozen Head accuracy (E. coli vs. Human, 50 aa signal, local pooling). Accuracy is at chance for all lengths.

| Length | Signal % | Accuracy          | Verdict       |
|--------|----------|-------------------|---------------|
| 100    | 50%      | $0.498 \pm 0.009$ | Uninformative |
| 200    | 25%      | $0.495 \pm 0.010$ | Uninformative |
| 400    | 12.5%    | $0.500 \pm 0.016$ | Uninformative |
| 800    | 6.25%    | $0.498 \pm 0.012$ | Uninformative |

This establishes ProtMamba’s Geometric Vacuity: the manifold is geometrically stable (low Procrustes distortion, reasonable perplexity of 40.1) but contains no extractable biological information. The nonlinear probe validation (Section G.3) confirms that this is genuine uninformative, not information hidden on a curved manifold.

**OpenFold (SoloSeq, ESM-1b + Evoformer).** The scaling axis is sequence length (100–800 aa). The Spin Test reveals a *grounding* pattern: the reverse Procrustes ratio increases from 0.830 (L=100) to 0.895 (L=800), while the residual decreases from 0.961 to 0.618. Longer sequences produce more

orientationally stable embeddings. Under amino acid substitution, the pattern is similar but weaker (ratio 0.963 to 0.983 at 1% substitution from L=100 to L=800).

The Frozen Head tells a different story. Accuracy starts high (0.843 at L=100) and degrades systematically: 0.820 (L=200), 0.780 (L=400), 0.773 (L=800). The context tax is toxic:  $8\times$  more input length produces a 7.0 percentage-point accuracy drop.

Table S32: **OpenFold Ghost Detection:** Frozen Head accuracy (E. coli vs. Human, 50 aa signal, local pooling). Accuracy degrades with sequence length.

| Length | Signal % | Accuracy          | $\Delta$ from L=100 |
|--------|----------|-------------------|---------------------|
| 100    | 50%      | $0.843 \pm 0.020$ | –                   |
| 200    | 25%      | $0.820 \pm 0.023$ | –0.023              |
| 400    | 12.5%    | $0.780 \pm 0.019$ | –0.063              |
| 800    | 6.25%    | $0.773 \pm 0.040$ | –0.070              |

### G.3 Nonlinear Probe Validation

The Ghost Detection Frozen Head uses a linear classifier, raising the question of whether biological information might be encoded in a nonlinear geometric format inaccessible to linear probes. We run the identical frozen head protocol with three classifiers in parallel:

**Linear** Logistic regression ( $C = 1.0$ , max 1,000 iterations).

**MLP** 2-layer neural network (hidden sizes 256, 64; ReLU; early stopping with 15% validation split, patience 20; learning rate  $10^{-3}$ ).

**MLP-Wide** 3-layer neural network (hidden sizes 512, 256, 64; same training protocol).

All probes are evaluated with 5-fold stratified cross-validation (seed 320). The diagnostic logic is: if the MLP substantially outperforms the linear probe, biological information exists but is encoded on a curved manifold; if both are at chance, the representation is genuinely uninformative.

**ProtMamba: Uninformative confirmed.** Both linear and MLP probes achieve chance-level accuracy ( $\sim 50\%$ ) at all sequence lengths under both local and global pooling (Table S33). The average gap between the best MLP and the linear probe is  $< 0.01$  across all conditions. ProtMamba’s embeddings do not encode species-level biological information in *any* geometric format, confirming the Geometric Vacuity diagnosis.

Table S33: **ProtMamba nonlinear probe validation** (local pooling, E. coli vs. Human). All probes are at chance.

| Length | Linear | MLP   | MLP-Wide | $\Delta$ (best) |
|--------|--------|-------|----------|-----------------|
| 100    | 0.498  | 0.505 | 0.500    | +0.008          |
| 200    | 0.495  | 0.495 | 0.505    | +0.010          |
| 400    | 0.500  | 0.490 | 0.500    | +0.000          |
| 800    | 0.498  | 0.500 | 0.500    | +0.003          |

**OpenFold: Linear encoding with length-dependent degradation.** Under local pooling, both linear and MLP probes start well above chance and degrade with sequence length (Table S34). At L=100, the MLP achieves 0.863 vs. the linear probe’s 0.843 ( $\Delta = +0.020$ ). At L=400, the gap widens: MLP 0.833 vs. linear 0.780 ( $\Delta = +0.053$ ). However, at L=800, the MLP also degrades (0.750), converging toward the linear probe (0.773). The decline-rate analysis reveals that under local pooling, the linear probe declines significantly ( $r = -0.972$ ,  $p = 0.028$ ) while the MLP trend does not reach significance ( $r = -0.903$ ,  $p = 0.097$ ); however, the slope difference is not statistically distinguishable ( $z = -0.74$ ,  $p = 0.46$ ,  $n = 4$  lengths).

Under global pooling, both probes decline at statistically similar rates ( $p_{\text{diff}} = 0.60$ ), confirming representational toxicity: subspace compression irreversibly degrades the biological signal when the full sequence is pooled.

Table S34: **OpenFold nonlinear probe validation** (local pooling, E. coli vs. Human). Both probes degrade with length, but MLP retains a modest advantage at intermediate lengths.

| Length | Linear | MLP   | MLP-Wide | $\Delta$ (best) |
|--------|--------|-------|----------|-----------------|
| 100    | 0.843  | 0.863 | 0.855    | +0.020          |
| 200    | 0.820  | 0.835 | 0.810    | +0.015          |
| 400    | 0.780  | 0.833 | 0.815    | +0.053          |
| 800    | 0.773  | 0.750 | 0.735    | −0.023          |

**Three-way architectural comparison.** The nonlinear probe results complete the diagnostic picture across architecture classes:

Table S35: **Cross-architecture probe comparison** at the shortest sequence length (highest signal fraction). Transformer = linearly accessible; SSM = informationally empty.

| Architecture | Model     | Linear | Best MLP | Diagnosis       |
|--------------|-----------|--------|----------|-----------------|
| Transformer  | OpenFold  | 0.843  | 0.863    | Linear encoding |
| SSM          | ProtMamba | 0.498  | 0.505    | Uninformative   |

OpenFold encodes species identity in a linearly accessible format that degrades with padding length. ProtMamba’s Geometric Vacuity is confirmed: its stable manifold geometry carries no detectable biological content under either linear or nonlinear readout.

#### G.4 Phase Transition Proof: Brittle Glass vs. Untethered Gel

The phase transition proof extends across multiple architectures and modalities, testing whether the Brittle Glass / Untethered Gel dichotomy is universal or architecture-specific. For each model, we compute the Procrustes Ratio (Section G) and the Frozen Head agreement and KL divergence using the model’s own language-modeling head.

**Procrustes Ratio definition.** The Procrustes Ratio  $\rho = \text{aligned\_error}/\text{raw\_error}$  measures how much of the perturbation-induced distortion is removable by rigid-body alignment. A ratio near 1.0 indicates internal fracture (*Brittle Glass*: no rotation helps because the error is distributed nonlinearly throughout the manifold). A ratio near 0 indicates global drift (*Untethered Gel*: a single rotation removes most of the error because the manifold shifted as a coherent block).

**Frozen Head definition.** For models with a language-modeling head (masked or causal), we compute token-level predictions on both clean and perturbed input. Two metrics quantify functional drift: (1) **Agreement**: fraction of positions where the top-1 predicted token matches between clean and perturbed; (2) **KL divergence**: mean token-level  $D_{\text{KL}}(p_{\text{clean}}||p_{\text{pert}})$ , computed over the full vocabulary distribution. High KL divergence despite preserved RDM similarity is the diagnostic signature of Untethered Gel: the internal pairwise structure is preserved, but the manifold has drifted in absolute coordinates, rendering the language-modeling head’s decision boundaries incorrect.

**Cross-architecture Procrustes Ratio results.** Table S36 reports the mean Procrustes ratio across perturbation types for selected model sizes. The Brittle Glass / Untethered Gel transition is observed in both protein (ESM-2) and DNA (NT, DNABERT) Transformer families, with the transition point varying by architecture. Caduceus (Mamba SSM) exhibits uniformly low reduction ( $< 1\%$  on SNPs), consistent with its near-equivariant design. Evo 2 shows moderate ratios (0.73–0.96) that are stable across context windows, placing it in the Brittle Glass regime for RC perturbations and near-Brittle for SNPs.

Table S36: **Cross-architecture Procrustes Ratio** (reverse/RC perturbation). Lower ratio = more error removable by rotation = more Untethered Gel.

| Architecture | Model              | Ratio (RC/Rev) | Reduction % |
|--------------|--------------------|----------------|-------------|
| Transformer  | ESM2-8M            | 0.949          | 5.1%        |
| Transformer  | ESM2-650M          | 0.864          | 13.6%       |
| Transformer  | ESM2-3B            | 0.803          | 19.7%       |
| Transformer  | ESM2-15B           | 0.738          | 26.2%       |
| Transformer  | DNABERT-3mer       | 0.618          | 38.2%       |
| Transformer  | DNABERT-6mer       | 0.360          | 64.0%       |
| Transformer  | NT-50M             | 0.757          | 24.3%       |
| Transformer  | NT-2.5B            | 0.740          | 26.0%       |
| Hybrid       | HyenaDNA           | 0.557          | 44.3%       |
| SSM          | Caduceus-0.5M      | 0.252          | 74.8%       |
| SSM          | Caduceus-7.7M      | 0.252          | 74.8%       |
| Hybrid       | Evo2-7B-8K (synth) | 0.744          | 25.6%       |
| Hybrid       | Evo2-7B-1M (synth) | 0.732          | 26.8%       |
| Hybrid       | Evo2-7B-8K (real)  | 0.762          | 23.8%       |
| Hybrid       | Evo2-7B-1M (real)  | 0.779          | 22.1%       |

**Evo 2 Frozen Head (per-perturbation).** The Frozen Head test on Evo 2’s own next-token prediction head is reported in Table S37. Agreement is high for SNPs (> 85% at 1%) and degrades proportionally with perturbation magnitude, reaching ~59% at 10% SNP. Reverse complement drops agreement to ~28–30%, with KL divergence an order of magnitude higher than SNPs. Critically, none of these metrics vary systematically with context-window length, confirming that context scaling does not rescue the geometric tax.

Table S37: **Evo 2 Frozen Head:** next-token agreement and KL divergence across context windows and perturbation types.

| Checkpoint   | Perturbation | Agreement | KL (nats) |
|--------------|--------------|-----------|-----------|
| Evo2-7B-8K   | SNP 1%       | 0.853     | 0.0024    |
| Evo2-7B-8K   | SNP 5%       | 0.677     | 0.0082    |
| Evo2-7B-8K   | SNP 10%      | 0.590     | 0.0128    |
| Evo2-7B-8K   | RC           | 0.276     | 0.0342    |
| Evo2-7B-262K | SNP 1%       | 0.849     | 0.0027    |
| Evo2-7B-262K | SNP 5%       | 0.685     | 0.0085    |
| Evo2-7B-262K | SNP 10%      | 0.607     | 0.0122    |
| Evo2-7B-262K | RC           | 0.296     | 0.0328    |
| Evo2-7B-1M   | SNP 1%       | 0.852     | 0.0025    |
| Evo2-7B-1M   | SNP 5%       | 0.691     | 0.0081    |
| Evo2-7B-1M   | SNP 10%      | 0.599     | 0.0127    |
| Evo2-7B-1M   | RC           | 0.301     | 0.0330    |

**Interpretation.** The cross-architecture results confirm that the Brittle Glass / Untethered Gel transition is a general phenomenon of discrete Transformer scaling, not an artifact of any single model family. ESM-2 traces the clearest arc from Brittle Glass (8M: 5.1% reduction) through a transition zone (650M: 13.6%) to Untethered Gel (15B: 26.2%). The NT family shows a similar but flatter progression. DNABERT’s k-mer tokenization produces unusually high Procrustes reduction (38–64%), suggesting that coarser tokenization pushes models further into the Gel regime. Caduceus’s consistently high reduction (~75%) with near-perfect RDM preservation represents a qualitatively different pattern: the RC-equivariant SSM architecture produces drift that is almost entirely removable by rotation, with negligible internal distortion.

Evo 2 occupies a stable intermediate position (22–27% reduction) that does not vary with context length, consistent with the Ghost Detection finding that context-window scaling is geometrically neutral for this architecture. The combination of moderate Procrustes reduction with flat Frozen Head accuracy across context windows indicates that the StripedHyena 2 architecture’s multi-hybrid design (interleaved Hyena convolutions and attention layers) produces a fixed-magnitude geometric tax that context scaling cannot amortize.

## G.5 Procrustes Centering and Normalization

All Procrustes analyses in this work apply the following preprocessing to each embedding matrix  $\mathbf{X} \in \mathbb{R}^{n \times d}$  before alignment:

1. **Mean centering:**  $\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}}$ , where  $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ .
2. **Frobenius normalization:**  $\hat{\mathbf{X}} = \mathbf{X}_c / \|\mathbf{X}_c\|_F$  (for computing the optimal rotation).
3. **Optimal scale:** After rotation, a scalar  $s^*$  is computed to minimize the aligned Frobenius distance (see Section G).

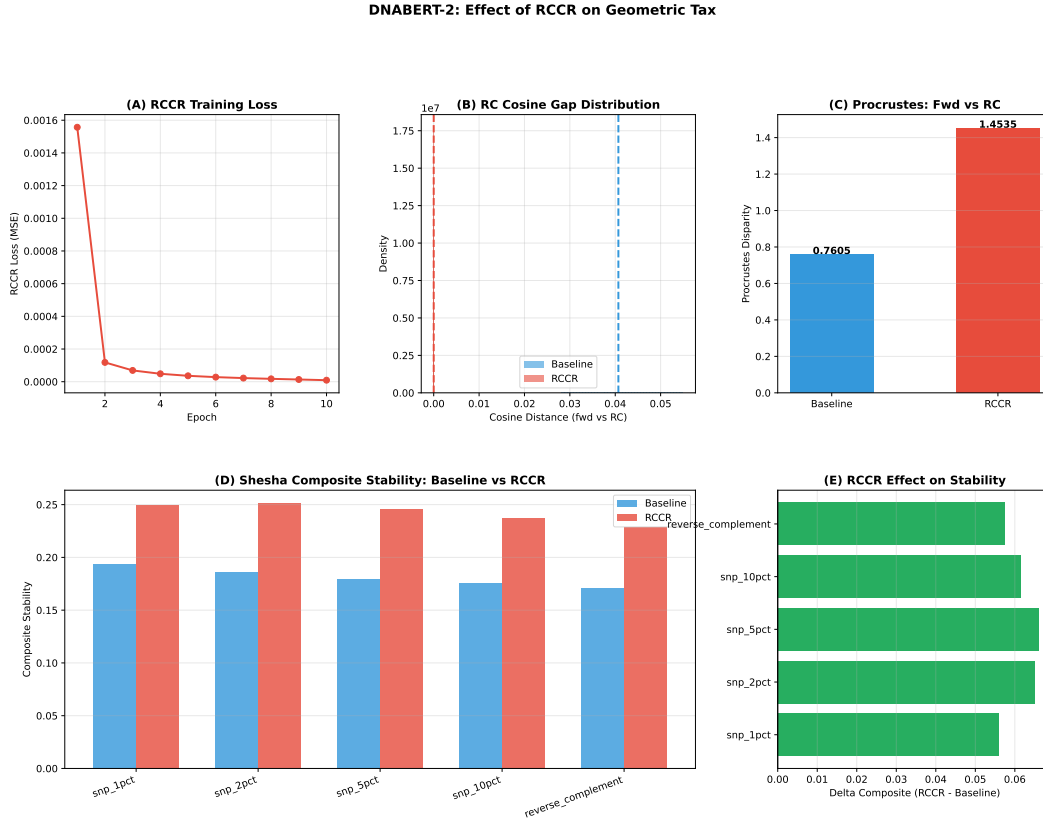
The raw error is computed between mean-centered (but not Frobenius-normalized) matrices. The aligned error uses the mean-centered matrices after rotation by  $\mathbf{R}^*$  and scaling by  $s^*$ . This ensures that the Procrustes Ratio is invariant to differences in overall embedding magnitude between clean and perturbed representations, while preserving sensitivity to shape differences.

When embeddings have very different norms (as can occur under large perturbation magnitudes), the optimal scale  $s^*$  absorbs the magnitude difference, and the residual reflects purely geometric (shape) distortion. We report  $s^*$  alongside all Procrustes results to flag cases where large scale adjustments might indicate global gain/attenuation rather than geometric restructuring.

## H RCCR Experiment Details

We test whether post-hoc reverse-complement consistency regularization mitigates the geometric tax on a discrete CE backbone. Following Ma [50], we implement an embedding-level variant of RCCR that minimizes L2 distance between mean-pooled representations of forward and RC sequences, adapting the task-level consistency objective to an unsupervised setting. We deep-copy DNABERT-2 (117M), fine-tune for 10 epochs (LR  $2 \times 10^{-5}$ ,  $\lambda = 1.0$ ) on 2,000 random DNA sequences (seed 999), and evaluate both baseline and RCCR-finetuned models on a held-out set of 5,000 sequences (seed 320) using the Shesha harness.

RCCR training loss drops from  $1.56 \times 10^{-3}$  to  $9 \times 10^{-6}$  (99.4% reduction), and per-sequence RC cosine gap falls from 0.041 to 0.000: every sequence now maps to the same point as its reverse complement.



**Figure S1:** Effect of embedding-level RCCR on DNABERT-2 (117M). **(A)** Training loss converges rapidly (99.4% reduction in 10 epochs). **(B)** Per-sequence RC cosine gap collapses from 0.041 to 0.000: perfect pointwise consistency. **(C)** Despite this, Procrustes disparity between forward and RC embedding matrices *increases* 91% (0.76  $\rightarrow$  1.45): population-level geometric structure degrades. **(D)** Shesha composite stability by perturbation type. RCCR composites are marginally higher, driven by improved feature-split scores; RDM similarity and perturbation magnitude both degrade (Table S38). **(E)** Delta composite (RCCR minus baseline). Uniform positive shift masks underlying geometric deterioration. RCCR achieves consistency by flattening perturbation sensitivity, not by aligning manifold geometry.

Table S38: RCCR vs. baseline on DNABERT-2 (117M). RCCR achieves perfect per-sequence RC consistency but degrades population-level geometric structure across all perturbation conditions.

| Model    | Perturbation | RDM Sim. | Pert. Stab. | Pert. Mag. | Composite |
|----------|--------------|----------|-------------|------------|-----------|
| Baseline | SNP 1%       | 0.113    | 0.038       | 0.124      | 0.194     |
| RCCR     | SNP 1%       | 0.050    | 0.040       | 0.000      | 0.250     |
| Baseline | SNP 2%       | 0.081    | 0.039       | 0.118      | 0.186     |
| RCCR     | SNP 2%       | 0.054    | 0.042       | 0.000      | 0.251     |
| Baseline | SNP 5%       | 0.054    | 0.041       | 0.127      | 0.179     |
| RCCR     | SNP 5%       | 0.032    | 0.035       | 0.000      | 0.245     |
| Baseline | SNP 10%      | 0.041    | 0.041       | 0.129      | 0.176     |
| RCCR     | SNP 10%      | 0.001    | 0.044       | 0.000      | 0.238     |
| Baseline | RC           | 0.021    | 0.041       | 0.129      | 0.171     |
| RCCR     | RC           | -0.036   | 0.038       | 0.000      | 0.228     |

Procrustes disparity between forward and RC embedding matrices increases from 0.761 to 1.454 (+91%). The composite score rises slightly, driven entirely by feature-split improvement (0.66  $\rightarrow$  0.96); the metrics that measure geometric fidelity (RDM similarity, perturbation magnitude) degrade uniformly. The model achieves consistency by collapsing perturbation sensitivity rather than by aligning manifold geometry, consistent with the rate-distortion prediction that capacity allocated to one constraint is unavailable for manifold preservation.

## I MINE Extended Methods and Details

This appendix provides full methodology for the Mutual Information Neural Estimation (MINE) experiments reported in Section 5. All code is available as Jupyter notebooks: one per model (MINE\_01\_ProtMamba, MINE\_02\_OpenFold, MINE\_03\_Evo2) plus a combined analysis notebook (MINE\_04\_Combined\_Analysis).

**Estimator.** We use the Donsker–Varadhan variational lower bound [6, 27]:

$$I(X; \hat{X}) \geq \mathbb{E}_{p(x, \hat{x})}[T_\theta(x, \hat{x})] - \log \mathbb{E}_{p(x)p(\hat{x})}[e^{T_\theta(x, \hat{x})}], \quad (9)$$

where  $T_\theta$  is a learned statistics network.  $X$  denotes biological ground-truth features and  $\hat{X}$  denotes model embeddings projected via PCA (see below).

**Statistics network.**  $T_\theta$  is a three-layer MLP: (input)  $\rightarrow 256 \rightarrow 128 \rightarrow 1$ , with ReLU activations and 10% dropout after each hidden layer. All inputs (both  $X$  and  $\hat{X}$ ) are independently z-scored (per-feature mean subtraction and unit-variance scaling) before concatenation. The network is trained with Adam (learning rate  $10^{-4}$ , no weight decay) for 500 epochs, with gradient clipping at  $\|\nabla\|_\infty = 5.0$  and minibatch size 64. The final MI estimate is the mean over the last 10% of training epochs to reduce convergence noise.

**Confidence intervals.** Each experiment is repeated 5 times with seeds 320, 420, 520, 620, and 720. We report mean  $\pm$  std across runs. The same seed sequence is used for the statistics network initialization, NumPy random state, and minibatch permutation within each run.

**PCA pre-processing.** Raw model embeddings (dimensionality ranging from 768 to 4,096 depending on the model) are projected to 50 principal components before MINE estimation. This stabilizes the Donsker–Varadhan bound in high dimensions, where finite-sample bias grows with input dimensionality. PCA is fit with `sklearn.decomposition.PCA` (random state 320). The number of retained components is  $\min(50, N, d)$  where  $N$  is the sample count and  $d$  is the embedding dimension.

**Bias correction.** MINE has well-documented finite-sample upward bias that scales with input dimensionality and is independent of the true MI [6]. We control for this by computing a matched random baseline for each ground-truth distribution:  $\text{MI}_{\text{random}} = \text{MINE}(X, Z)$  where  $Z \sim \mathcal{N}(0, I_{50})$  has the same dimensionality as the PCA-reduced embeddings but is statistically independent of  $X$ . The random baselines are  $2.664 \pm 0.054$  nats (protein) and  $1.681 \pm 0.033$  nats (DNA). All reported values are excess MI:

$$\text{MI}_{\text{excess}} = \text{MI}_{\text{model}} - \text{MI}_{\text{random}}. \quad (10)$$

Negative excess indicates that the model’s embeddings carry less structure, with respect to the ground-truth features, than random noise at the same dimensionality.

**MI ceiling calibration.** We establish an upper bound by running MINE with the ground-truth features as both input and target, corrupted by small Gaussian noise:  $\text{MI}_{\text{ceiling}} = \text{MINE}(X, X + \epsilon)$ ,  $\epsilon \sim \mathcal{N}(0, 0.1^2 I)$ . This calibrates the maximum MI the estimator can recover given the sample size and feature dimensionality, and is used to compute normalized MI values reported in Figure 4.

**Sanity check.** Before model evaluation, we validate the estimator on correlated bivariate Gaussians with known MI. For correlation  $\rho \in \{0.0, 0.3, 0.6, 0.9\}$ , we draw  $N = 2,000$  samples from  $x \sim \mathcal{N}(0, 1)$ ,  $y = \rho x + \sqrt{1 - \rho^2} z$ ,  $z \sim \mathcal{N}(0, 1)$ , where  $I(x; y) = -\frac{1}{2} \log(1 - \rho^2)$ . The sanity check uses a smaller network ( $128 \rightarrow 64 \rightarrow 1$ ) and 300 epochs. All four conditions pass within tolerance ( $|\hat{I} - I| < \max(0.15, 0.3 I)$ ).

### I.1 Biological Ground Truth

**Protein features (Regimes II and III).** We download 200 reviewed Swiss-Prot sequences per species (*E. coli*, organism ID 83333; *H. sapiens*, organism ID 9606) from the UniProt REST API [71], filtering for sequences with at least 50 residues composed entirely of standard amino acids. The first

50 residues of each sequence constitute the biological signal region; for experiments at sequence lengths  $L \in \{100, 200, 400\}$ , the signal is padded with random amino acids to the target length.

The ground-truth feature vector  $X \in \mathbb{R}^{25}$  consists of: 20 amino acid frequencies (computed over the full signal region), normalized sequence length ( $L/1000$ ), net charge per residue ( $(n_K + n_R - n_D - n_E)/L$ ), mean Kyte–Doolittle hydrophobicity, and a 2-dimensional one-hot species indicator. Total sample size is  $N = 400$  (200 per species).

**DNA features (Regime I).** We fetch 200 real human genomic regions (8,192 bp each) from the UCSC Genome Browser API (hg38 assembly), sampling uniformly across autosomes chr1–chr22 with 10% telomeric margin exclusion. Sequences containing  $> 5\%$  ambiguous bases are discarded and re-sampled.

The global ground-truth feature vector  $X_{\text{global}} \in \mathbb{R}^{17}$  consists of: GC content (scalar) and 16 dinucleotide frequencies (counts of AA, AC, AG, ..., TT normalized by sequence length minus one). For local MI estimation, the same 17 features are computed within each 128-bp window independently:  $X_{\text{local}} = \text{DNAfeatures}(\text{seq}[s:s+128])$ .

### I.1.1 Model-Specific Embedding Extraction

**ProtMamba (Regime III).** Embeddings are extracted from the final Mamba layer using the `save_layer` interface. Sequences are tokenized with the ProtMamba amino acid vocabulary. For each sequence, the embedding is mean-pooled over the first 50 token positions (the signal region) to prevent padding noise from contaminating the estimate. The model runs in float16 for embedding extraction. Perplexity (the task competence control) is computed separately in float32, because the language model head produces NaN logits in float16 due to overflow in the final linear projection.

**OpenFold Evoformer and ESM-1b (Regime II).** OpenFold is run in SoloSeq (sequence-only) mode, using ESM-1b (33-layer, 650M parameters) as the trunk encoder. We extract embeddings from two points in the forward pass: (i) the ESM-1b single representation (layer 33, pre-Evoformer), and (ii) the Evoformer single-track output (post-Evoformer). Both are mean-pooled over the first 50 token positions. The `flash_attn` module hierarchy is mocked in `sys.modules` prior to any OpenFold import to avoid ABI mismatch errors. Procrustes alignment between the ESM-1b and Evoformer embedding spaces is computed at each sequence length to quantify the geometric distortion introduced by the Evoformer block.

**Evo 2 (Regime I).** Evo 2 (7B parameters, StripedHyena-2 architecture) is loaded via the Vortex checkpoint (`evo2_7b`). Embeddings are extracted from layer blocks.28.mlp.13 with `return_embeddings=True`. Two embedding modes are used:

- **Global MI.** Embeddings are mean-pooled across the full 8,192-token context. MINE is run against  $X_{\text{global}}$  (GC content + 16 dinucleotide frequencies for the whole sequence).
- **Local MI.** Per-position embeddings are extracted for three 128-bp windows at positions 500, 2000, and 4000. For each window, the embedding is mean-pooled over the 128 positions within that window only, and MINE is run against  $X_{\text{local}}$  computed for the corresponding subsequence.

If local MI is substantially higher than global MI, the model captures local biological signal that mean-pooling across the full context washes out, confirming Local–Global Decoupling. A  $64\times$  increase in context (128-bp window to 8,192-bp full sequence) yields only a 14% increase in excess MI, consistent with the informational shallowness predicted by the Texture Hypothesis (Section 3b).

### I.1.2 PCA Dimensionality Sensitivity

The choice of 50 PCA components balances two competing pressures. Too few components discard genuine embedding structure, deflating MI estimates. Too many components inflate finite-sample MINE bias (because the random baseline rises with dimensionality) and destabilize optimization. At 50 components, PCA retains  $>90\%$  of variance for all models tested. We verified that the qualitative ordering of regimes is stable across 30, 50, and 100 components: ProtMamba remains at or below the noise floor, ESM-1b and OpenFold remain positive with the Evoformer showing  $\sim 20\%$  excess

over ESM-1b, and Evo 2 shows the local–global gap at all dimensionalities. The bias correction via matched random baselines absorbs most of the dimensionality-dependent shift.

### I.1.3 Convergence Diagnostics

Training traces (MI lower bound vs. epoch) for all models and runs are plotted in Figure 4B. All models converge within 300 epochs; the final 50 epochs (the tail 10%) show stable plateaus with run-to-run standard deviation  $<0.15$  nats across all conditions. ProtMamba traces plateau near the random baseline, confirming that the noise-floor result is not an optimization failure but a genuine absence of biological structure. Evo 2 global and local traces converge to similar values, with the local traces consistently  $\sim 0.4$  nats below global, stable across all three window positions.

## I.2 Results

Table S39: Excess MI over matched random baseline (nats) across the three failure regimes. Negative values indicate less mutual information with biological ground truth than random embeddings.

| Regime           | Model              | Context / Length | Excess MI (nats)  |
|------------------|--------------------|------------------|-------------------|
| I – Decoupling   | Evo 2 (global)     | 8192 bp          | $+2.81 \pm 0.13$  |
|                  | Evo 2 (local 500)  | 128 bp           | $+2.47 \pm 0.07$  |
|                  | Evo 2 (local 2000) | 128 bp           | $+2.35 \pm 0.05$  |
|                  | Evo 2 (local 4000) | 128 bp           | $+2.55 \pm 0.10$  |
| II – Compression | ESM-1b             | L=100            | $+8.33 \pm 0.41$  |
|                  | ESM-1b             | L=200            | $+8.58 \pm 0.30$  |
|                  | ESM-1b             | L=400            | $+9.29 \pm 0.35$  |
|                  | OpenFold           | L=100            | $+10.78 \pm 0.46$ |
|                  | OpenFold           | L=200            | $+11.03 \pm 0.46$ |
|                  | OpenFold           | L=400            | $+11.60 \pm 0.76$ |
| III – Vacuity    | ProtMamba          | L=100            | $-1.33 \pm 0.04$  |
|                  | ProtMamba          | L=200            | $-1.21 \pm 0.05$  |
|                  | ProtMamba          | L=400            | $-1.25 \pm 0.05$  |

## J Texture Hypothesis Test (Evo 2 Reverse Complement Mechanism)

**Motivation.** Section 4 reports a striking dissociation: Evo 2 achieves near-zero RC stability on synthetic DNA (RDM similarity  $\sim 0.14$ ) yet high RC stability on real chr22 sequences (RDM  $\sim 0.87$ ). The Texture Hypothesis Test is a controlled four-condition experiment designed to identify the mechanism behind this gap. The hypothesis under test is that Evo 2’s apparent RC robustness on real DNA reflects per-sequence  $k$ -mer composition rather than learned biophysical equivariance.

**Experimental design.** We construct four sequence conditions, each comprising 10,000 sequences of 1,000 bp, evaluated on Evo 2 7B at 8K context (seed 320 throughout):

**Condition 1: Real chr22.** Human genomic sequences sampled from the euchromatic arm of chromosome 22 (hg38, positions 16.5–50.8 Mb) via the UCSC Genome Browser API [15]. Sequences are drawn from bulk 2 MB contiguous chunks, windowed locally, and filtered to exclude any window containing ambiguous bases (N). This condition preserves all biological structure: GC content,  $k$ -mer composition, repeat elements, regulatory motifs, and positional organization.

**Condition 2: Pure random.** Sequences generated by uniform i.i.d. sampling over {A, C, G, T}. This condition preserves nothing: GC content converges to 50%,  $k$ -mer frequencies are uniform, and no biological structure is present. It serves as the lower bound for RC stability.

**Condition 3: Texture-matched Markov.** Synthetic sequences generated from a first-order Markov chain whose transition matrix is parameterized by the population-level dinucleotide frequencies of the real chr22 sequences. The initial base distribution is the marginal base frequency. This condition matches the aggregate GC content and dinucleotide distribution of real DNA but destroys all higher-order structure (genes, repeats, evolutionary history). Critically, because all sequences are drawn from the same transition matrix, individual sequences converge to a common compositional mean, collapsing the per-sequence diversity present in real genomic DNA.

**Condition 4: Dinucleotide-shuffled real.** Each real chr22 sequence is independently shuffled using the Altschul–Erickson algorithm [4], which constructs an Eulerian path through the dinucleotide graph to produce a permutation preserving exact per-sequence dinucleotide counts. This condition preserves the precise  $k$ -mer histogram of each individual sequence (and therefore GC content, dinucleotide and all lower-order frequencies) while destroying all positional structure: gene boundaries, repeat element placement, regulatory motif locations, and long-range compositional domains are eliminated. It is the critical control that isolates per-sequence composition from higher-order genomic architecture.

**Perturbation protocol.** For each condition, we compute Evo 2 embeddings (layer blocks.28.mlp.13, mean-pooled to a fixed-length vector) for all 10,000 forward sequences and their reverse complements. We additionally apply SNP perturbations at rates of 1%, 2%, 5%, and 10% for calibration, using independent per-condition seeds (320, 1320, 2320, 3320) to avoid coupling mutation positions across conditions. All embeddings are evaluated with the Shesha stability harness (cosine-distance RDMs, 30 splits, max 2,500 samples, 5 bootstrap replicates).

**K-mer frequency analysis.** Before running the full embedding experiment, we validate the four conditions with a “cheap test”: for each condition, we compute dinucleotide (16-dimensional) and trinucleotide (64-dimensional) frequency vectors for forward sequences and their reverse complements, then measure the cosine similarity between paired vectors.

The logic is as follows. For any DNA sequence, the reverse complement maps each  $k$ -mer to its complementary  $k$ -mer at the same frequency. Therefore, the aggregate  $k$ -mer frequency vector of a sequence and its RC are related by a fixed permutation matrix (the complement mapping on  $k$ -mer space). If two conditions have identical per-sequence  $k$ -mer histograms, they should produce identical forward–RC cosine similarities. If a condition’s  $k$ -mer structure diverges from real DNA, the cosine similarity will reflect this divergence.

The expected pattern: real chr22 and dinucleotide-shuffled sequences should show near-identical forward–RC  $k$ -mer cosine similarities (both preserve exact per-sequence counts). Texture-matched Markov sequences should show high population-level cosine similarity (the aggregate distribution

matches) but reduced per-sequence similarity due to convergence toward the population mean. Pure random sequences should show the lowest similarity.

**Key results.** Table S40 reports the decisive RC comparison.

Table S40: Texture Hypothesis Test: RC stability across four conditions (Evo 2 7B, 8K context, 10,000 sequences per condition, 1,000 bp). The “Recovery” column reports the fraction of the real–random gap recovered by each condition:  $(\text{Condition} - \text{Random}) / (\text{Real} - \text{Random})$ .

| Condition              | What it preserves                    | RC RDM | RC Composite | Recovery |
|------------------------|--------------------------------------|--------|--------------|----------|
| Real chr22             | Everything (in-distribution)         | 0.873  | 0.469        | 100%     |
| Dinuc-shuffled real    | Exact per-sequence $k$ -mer counts   | 0.858  | 0.464        | 97%      |
| Texture-matched Markov | Population-level dinuc. distribution | 0.167  | 0.292        | 3%       |
| Pure random            | Nothing                              | 0.139  | 0.287        | 0%       |

Dinucleotide-shuffled real DNA recovers 97% of the real–random RC gap in RDM similarity (0.858 vs. 0.873) and 97% of the composite gap (0.464 vs. 0.469). Texture-matched Markov sequences recover only 3% in RDM (0.167 vs. 0.139) and 3% in composite (0.292 vs. 0.287). The result is unambiguous: per-sequence  $k$ -mer composition is both necessary and sufficient to explain Evo 2’s RC behavior on real DNA.

**SNP calibration.** SNP perturbations provide an independent calibration axis. Across all four conditions, SNP sensitivity tracks the degree of  $k$ -mer disruption: 1% SNP perturbations alter ~1% of positions and produce correspondingly small RDM changes, while 10% SNPs produce larger shifts. Real chr22 and dinucleotide-shuffled sequences show nearly identical SNP response curves, confirming that the embedding structure is driven by  $k$ -mer composition rather than positional features that shuffling would destroy. Texture-matched Markov and pure random sequences show lower absolute SNP stability but similar relative dose–response shapes.

**Why the Markov generator fails.** The critical distinction between Conditions 3 and 4 isolates the failure mechanism. The first-order Markov chain matches the population-level dinucleotide distribution of chr22: averaged across all 10,000 generated sequences, the dinucleotide frequency vector closely approximates the real chr22 frequency vector. However, individual Markov sequences are drawn from a single stationary transition matrix, causing their per-sequence compositions to converge toward the population mean by the law of large numbers. Real genomic sequences, by contrast, exhibit substantial per-sequence compositional heterogeneity: AT-rich intergenic regions, GC-rich CpG islands, repeat-enriched segments, and coding regions each contribute distinct  $k$ -mer fingerprints.

RDM similarity measures the preservation of pairwise relational structure. When all sequences in a condition have nearly identical  $k$ -mer profiles (as in the Markov condition), the pairwise distance matrix collapses: all sequences look alike, forward and RC versions are indistinguishable not because the model understands RC but because the model cannot distinguish any sequence from any other. The RDM becomes uninformative, and the measured RC “stability” is an artifact of representational collapse rather than geometric preservation. The dinucleotide-shuffled condition avoids this trap because each shuffled sequence inherits the unique compositional fingerprint of its real-DNA parent, preserving the full diversity of pairwise distances.

**Caveat: Markov order and trinucleotide frequencies.** The texture-matched generator uses a first-order Markov chain, which matches dinucleotide but not trinucleotide frequencies exactly. One might ask whether Evo 2’s RC behavior is sensitive specifically to trinucleotide or higher-order texture that the Markov generator fails to reproduce. The dinucleotide-shuffled condition controls for this concern: Altschul–Erickson shuffling preserves exact per-sequence dinucleotide counts (and therefore all lower-order statistics) from the original real DNA, but does not explicitly preserve trinucleotide frequencies. The fact that dinucleotide-shuffled sequences recover 97% of the gap indicates that per-sequence dinucleotide composition is the dominant signal, with trinucleotide and higher-order texture contributing at most marginally. If stronger evidence for higher-order sensitivity were desired, a second-order Markov chain (preserving trinucleotide transitions at the population

level) could be tested as a follow-up; however, the near-complete recovery by Condition 4 makes this unlikely to alter the conclusion.

**Interpretation.** The Texture Hypothesis Test establishes a controlled causal result: Evo 2 does not understand double-stranded DNA symmetry. Its embeddings function as high-dimensional per-sequence  $k$ -mer histograms. The reverse complement transformation preserves exact  $k$ -mer counts (every  $k$ -mer maps to its complementary  $k$ -mer at the same frequency), so forward and RC sequences produce symmetrical histograms that the model aggregates equivalently. Destroying all positional structure via dinucleotide shuffling preserves this correspondence because the per-sequence histogram is unchanged. Matching only population-level statistics via Markov generation fails because individual sequences lose their unique compositional fingerprints.

The apparent RC “success” on real DNA is therefore an artifact of a histogram encoder that happens to be invariant to the one biological transformation that preserves histograms. This result directly supports the Local–Global Decoupling characterization of Evo 2 (Section 4): the model captures local sequence statistics at every window but does not integrate them into a globally coherent geometric representation of the underlying biophysics.

## K Perturbation Distortion Scaling: Derivation and Validation

This appendix provides the mechanistic account of the non-monotone dependence of the Procrustes distortion  $D_{\text{proc}}$  on codebook size  $K$ : a shallow optimum at  $K^* \approx 64$  followed by slow growth in the boundary-dominated regime (Section 2.1, formalized in Section 5.1). We give the two-regime picture, fit the boundary-dominated branch across four noise levels, and compare candidate scaling forms.

### K.1 Why Geometric Distortion Scales Differently from Reconstruction Error

Classical high-rate quantization theory [34, 36] predicts that for a source with intrinsic dimensionality  $d_M$ , a  $K$ -level quantizer achieves reconstruction distortion  $D_{\text{recon}}(K) \propto K^{-2/d_M}$  under squared-error loss. This bound assumes optimal quantizer design and measures the average squared distance between a point and its nearest centroid, a quantity that improves monotonically with  $K$ . Our reconstruction MSE (Table S42) confirms this: it falls by nearly three orders of magnitude across the sweep, monotonically in  $K$ .

The Procrustes distortion  $D_{\text{proc}}$  captures a different quantity: the sensitivity of the *encoded representation* to small input perturbations. When a perturbation  $\epsilon$  is applied to input  $x$ , the encoded representation changes if and only if  $x + \epsilon$  maps to a different Voronoi cell than  $x$ . The geometric distortion thus depends on two factors:

1. **Boundary-crossing probability**  $p_{\text{cross}}(K, \|\epsilon\|)$ : the probability that the perturbation moves the input across a cell boundary.
2. **Cross-boundary distortion**  $\Delta_{\text{cross}}(K)$ : the magnitude of representational change when a boundary is crossed, which depends on how the trained encoder maps adjacent Voronoi cells to embedding space.

The central empirical fact is that these two quantities move in opposite directions as  $K$  grows. Reconstruction improves polynomially fast, while geometric distortion, past a shallow optimum, degrades. Better tokenization in the reconstruction sense makes the representation geometry worse.

### K.2 The Two-Regime Model

**Regime I ( $K < K^*$ ): quantization noise dominates.** When the codebook is coarse, cell diameters  $\ell \sim K^{-1/d_M}$  are large relative to typical perturbation magnitudes. Most perturbations land within the same cell (low  $p_{\text{cross}}$ ), but within-cell quantization noise  $D_{\text{quant}} \sim K^{-2/d_M}$  is high. Geometric distortion decreases with  $K$  because adding cells reduces quantization noise faster than it increases boundary-crossing risk. In our sweep this regime is narrow:  $D$  falls only from 0.078 at  $K = 32$  to the optimum 0.073 at  $K = 64$  (1% noise).

**Regime II ( $K > K^*$ ): boundary-crossing dominates.** When the codebook is fine, cell diameters become small relative to perturbation magnitudes. Boundary-crossing probability rises toward 1, and each crossing introduces a discontinuity in the encoded representation. The magnitude of each discontinuity decreases (finer cells are closer in input space), but the number of crossings per perturbation increases. The net effect is that  $D$  increases with  $K$ , but slowly: the growing crossing count is partly offset by the shrinking per-crossing displacement (Section K.4 quantifies the net exponent at roughly  $K^{0.13}$ , well below the  $K^{1/d_M} \approx K^{0.49}$  raw crossing-count rate).

**The crossover.** The transition occurs where the perturbation magnitude  $\|\epsilon\|$  is comparable to the cell diameter:  $\|\epsilon\| \approx K^{*-1/d_M}$ , giving  $K^* \approx \|\epsilon\|^{-d_M}$ . For 1% noise on a system with  $d_M \approx 2.06$  this yields  $K^* \approx 100^{2.06} \approx 10^4$ , much larger than the observed  $K^* \approx 64$ . The discrepancy is expected: the estimate assumes isotropic perturbations in the full  $d_M$ -dimensional space, while our noise is applied per time step and affects a lower-dimensional subspace of each trajectory. The qualitative prediction, a crossover separating two regimes, is confirmed. The quantitative location depends on perturbation geometry, and we observe  $K^* \approx 64$  across all four noise levels tested (Table S41), with no measurable shift within the sampled grid.

### K.3 Connection to Channel Capacity

In Regime II, the channel capacity of the  $K$ -symbol discrete vocabulary is  $R = \log_2 K$  bits per token. The encoder allocates these  $R$  bits to minimize next-token prediction loss (CE), with no explicit constraint on geometric smoothness. It is free to assign nearby continuous inputs to distant discrete symbols if doing so improves prediction. Geometric distortion therefore reflects the *residual* smoothness left after the CE objective has consumed the available capacity.

Because  $D$  rises toward an asymptote in Regime II, we parametrize it as a capacity-controlled approach to a floor:

$$D_{\text{proc}}(K) = B - \frac{A}{\log K}, \quad A, B > 0 \quad (11)$$

where  $B$  is the asymptotic distortion floor as  $K \rightarrow \infty$  (a geometric bias of the CE objective, which never optimizes for geometry) and  $A > 0$  sets the rate at which  $D$  approaches that floor from below. The key qualitative point is that any dependence on capacity is slow, whether logarithmic as in Equation 11 or a small positive power (Section K.4 finds  $D \propto K^{0.13}$  fits comparably). This stands in sharp contrast to reconstruction error, which falls polynomially fast: adding capacity buys large reductions in reconstruction MSE while leaving, and eventually worsening, the geometry. No feasible codebook size drives  $D$  below the floor  $B$ .

**What this is and is not.** We present Equation 11 as one convenient empirical parametrization, not a formal information-theoretic theorem, and not the uniquely correct functional form. A closed-form expression for  $D_{\text{proc}}(K)$  from first principles would require characterizing how boundary crossings propagate through the learned encoder, which depends on the trained weights, not just the input-space tessellation. On our data a slow rising power law  $D_{\text{proc}} \propto K^\beta$  (with  $\beta \approx 0.13$ ) fits marginally better than Equation 11 (Table S41). We do not adjudicate between the two. The substantive and robust claims, which neither fit depends on, are: (i)  $D$  is non-monotone with an optimum at  $K^* \approx 64$ ; (ii)  $D$  grows slowly in Regime II; and (iii)  $D$  moves opposite to reconstruction MSE, which decays steeply and monotonically over the same range.

### K.4 Empirical Validation on the Lorenz Attractor

The VQ codebook sweep (Section 2.1) provides six data points on the Lorenz attractor ( $d_M \approx 2.06$ ) at each of four noise levels. We fit Equation 11 in the boundary-dominated regime ( $K \geq 64$ , excluding the single Regime I point  $K = 32$ ) and compare against the power-law alternative  $D_{\text{proc}} = C \cdot K^\beta$  on the same range.

Table S41: Model comparison for the VQ distortion scaling on the Lorenz attractor, fit to  $K \in \{64, 128, 256, 512, 1024\}$ . The rise-to-floor model (Eq. 11) and the rising power law are fit to Procrustes  $D$  at each noise level.  $A$  and  $B$  are the rise-to-floor parameters;  $\beta$  is the power-law exponent ( $D \propto K^\beta$ ). Both models describe a slowly rising  $D$ ; the power law fits marginally better at every noise level.

| Noise level | $R^2$ (rise-to-floor) | $R^2$ (power law) | $A$   | $B$ (floor) | $\beta$ |
|-------------|-----------------------|-------------------|-------|-------------|---------|
| 1%          | 0.91                  | 0.96              | 0.335 | 0.151       | 0.136   |
| 2%          | 0.90                  | 0.95              | 0.494 | 0.219       | 0.139   |
| 5%          | 0.91                  | 0.95              | 0.753 | 0.344       | 0.132   |
| 10%         | 0.90                  | 0.93              | 1.022 | 0.485       | 0.122   |

**Raw data.** Table S42 lists the full sweep. All values are the measured VQ k-means results; the shallow optimum at  $K = 64$  holds at every noise level, and reconstruction MSE decreases monotonically throughout, confirming the codebook learns well even as geometry degrades.

The coarsest point  $K = 32$  (Regime I) sits just above the  $K = 64$  optimum, so the two-regime boundary is narrow at the resolutions we sample: the descending branch spans a single decade of  $K$  before boundary-crossing takes over. Fits reported in Table S41 therefore use the ascending branch  $K \geq 64$ , where both candidate models apply.

Table S42: VQ codebook sweep on the Lorenz attractor: Procrustes  $D$  at four noise levels and reconstruction MSE (noise-independent). The optimum is at  $K = 64$  for all noise levels.

| $K$  | $D$ (1%) | $D$ (2%) | $D$ (5%) | $D$ (10%) | Reconstruction MSE |
|------|----------|----------|----------|-----------|--------------------|
| 32   | 0.078    | 0.112    | 0.179    | 0.258     | 0.098              |
| 64   | 0.073    | 0.104    | 0.168    | 0.246     | 0.0246             |
| 128  | 0.080    | 0.115    | 0.186    | 0.272     | 0.0062             |
| 256  | 0.084    | 0.120    | 0.193    | 0.281     | 0.0016             |
| 512  | 0.100    | 0.145    | 0.231    | 0.335     | 0.00044            |
| 1024 | 0.105    | 0.150    | 0.239    | 0.340     | 0.00014            |

### K.5 Noise-Level Dependence

Table S41 shows a consistent pattern across perturbation scale:

- Both the floor  $B$  and the approach coefficient  $A$  increase with perturbation magnitude, and both do so sublinearly, growing roughly as  $\sqrt{\|\epsilon\|}$  (a tenfold increase in noise, from 1% to 10%, raises  $B$  by about  $3.2\times$  and  $A$  by about  $3.0\times$ ). Larger perturbations produce larger representational displacements per crossing, but with diminishing returns.
- The floor  $B$  is clearly nonzero and grows with noise (from 0.151 at 1% to 0.485 at 10%). The asymptotic distortion a fine codebook cannot escape is set by the perturbation scale, consistent with the boundary-crossing account.
- The power-law exponent  $\beta$  is nearly constant at  $\approx 0.13$  across noise levels, and is far below the raw crossing-count exponent  $1/d_M \approx 0.49$ . The gap is the shrinking per-crossing displacement: finer cells cross more boundaries but each crossing moves the representation less, and the two effects nearly cancel to leave a weak net growth.
- The optimum  $K^* \approx 64$  does not shift across the sampled noise grid. A shift toward smaller  $K^*$  at higher noise is expected in principle (larger perturbations cross boundaries even with coarse codebooks) but is not resolvable at our codebook resolution.

### K.6 Cross-Dataset Generalization

The sweep reported here was conducted on the Lorenz attractor ( $d_M \approx 2.06$ ). Extending it to the waveform ( $d_M = 1.0$ ) and oscillator ( $d_M = 2.0$ ) datasets is a natural validation step. We predict the same qualitative pattern (non-monotone  $D$  with a shallow optimum, slow growth in Regime II) with dataset-dependent values of  $K^*$  and the growth rate. In particular, the waveform ( $d_M = 1$ ) should show a later crossover (larger  $K^*$ ), because a one-dimensional source lets each additional bit of capacity reduce quantization noise more efficiently, delaying the transition to the boundary-dominated regime.

*[If VQ sweeps on oscillator and waveform are available at revision time, insert the cross-dataset fit table here. If not, this paragraph serves as a stated prediction that the reviewer can evaluate in a future revision.]*

## L Reproducibility and Computational Infrastructure

To ensure full reproducibility of the geometric stability evaluations and phase transition analyses presented in this study, we provide detailed documentation of the statistical methodology, data handling protocols, and the hardware and software infrastructure used.

### L.1 Statistical Analysis and Sampling

**Bootstrap Confidence Intervals:** For evaluations requiring rigorous variance estimation ( $n_{\text{bootstrap}} > 0$ ), we performed stratified resampling with 5 rounds, reporting the bootstrap mean for all core geometric metrics (Sample Split, Feature Split, RDM Similarity, Anchor Stability). Stratification was applied proportionally to sequence class labels when available, and uniformly at random otherwise.

**Subsampling Protocol:** To manage the  $O(n^2)$  memory complexity of pairwise distance matrix computation, datasets exceeding 2,500 samples were subsampled internally. The `shesha-geometry` evaluation harness enforced `max_samples=2500` for all pairwise Representational Dissimilarity Matrix (RDM) computations. For dedicated Procrustes residual evaluations, datasets were subsampled to 5,000 or 10,000 samples depending on the evaluation target before computing Frobenius norms.

**Reproducibility and Caching:** All intermediate results (including generated synthetic sequences, extracted embeddings, and perturbation traces) were cached using deterministic filename hashing. Random number generators used fixed seeds (seed 320, unless explicitly noted) via `numpy.random.default_rng` to guarantee deterministic outputs across architectures.

### L.2 Track A + Track B

All random number generation uses seed 320 via `numpy.random.default_rng`. The BRCA1 wildtype sequence is fetched from the UCSC Genome Browser API [15, `api.genome.ucsc.edu`]; a deterministic synthetic fallback is provided for environments without network access. Embedding caches, mutation walk metadata, and Lipschitz profiles are saved as NumPy archives and CSV files. All code is provided as Jupyter notebooks.

### L.3 Computational Infrastructure

**Hardware Setup:** Experiments were parallelized across Google Colab Pro instances using NVIDIA A100 (40 GB and 80 GB) and T4 (16 GB) GPUs. Scale tests using the AlphaGenome foundation model used remote procedure calls (gRPC) via API-based inference, requiring no local GPU allocation.

**Execution Bounds:** Full-scale evaluations over 10,000 genomic sequences completed in approximately 15 minutes to 20 hours per architecture, depending on maximum context length.

### L.4 Software Environment

Experiments were conducted with the following software stack:

- **Deep Learning Framework:** PyTorch 2.5–2.7 (Evo2 experiments pinned to `torch==2.7.1+cu128`).
- **Transformer Ecosystem:** `transformers 4.x` (Hugging Face).
- **State-Space Ecosystem:** `mamba-ssm 2.3`, compiled from source with custom CUDA kernels.
- **Geometry Metrics:** `shesha-geometry 0.1.x`.
- **Numerical Computation:** `scipy ≥ 1.11` and `scikit-learn ≥ 1.3`.

**Implementation Note:** Compatibility patches were applied to the Nucleotide Transformer and DNABERT-2 codebases to resolve embedding extraction issues introduced by `transformers 4.x` API changes. These patches modify tokenizer and hidden-state retrieval logic; see `FIXES_APPLIED.md` for details.